Lecture 1

- book: W.D Callister, Materials Science & Engineering An Introduction 5th to 9th ed, J. W
- In tutorial Problems: - based on Assignments
  - every 2nd tutorial (first is Sep 21)
- Midterm: Friday Oct 30th
- term assignment: Nov 27th

- What is materials?

Oil, gases, pharmaceuticals: matter

Iron, copper, polymer, cement: material

materials - can be put into certain geometric shapes
- the product has some function

Property: Response of a material to an external effect, such as

- mechanical
- electrical
- optical
- thermal
- magnetic

- independent of material's shape & size

Why study Mat science: - Important to understand capabilities and limitations of materials

- An understanding of Materials science helps us to design better components, parts, devices, ect.
  - how can we make smt stronger / lighter?
  - how do elements come tg to form alloys?
  - why do sum mats have vastly different properties than others?

Info

- no tuts ts week

- no tuts switching

Classes of materials

	Polymers	Ceramics	Metals
ductility	Varies	Poor	Good
Conductivity	low	low	...
hardness/strength	low-medium	very high	fair
corrosion res	fair-good	...	fair-poor

Bonding

- Properties of materials (strength, hardness, conductivity) are determined by bonding

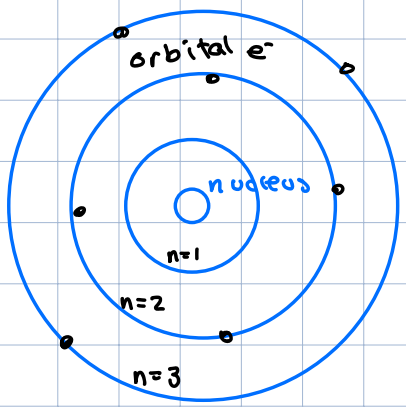
- Bonding is determined by:

  - e^- structure (distribution of e^- in atomic orbitals)

  - # of e^- and electronegativity (tendency for

an atom to attract an e^-

atomic structure



- Bohr model (1913)

- Z = # of p^+

- N = # of n^0

- Atomic mass $A \approx Z + N$

- Isotopes: same Z, diff N → same element
  diff atomic mass (^{12}C & ^{13}C)

- Reported atomic mass = abundance-weighted avg of the isotope

ex: C: $0.989(12.000) + 0.011(13.003) = 12.011 \frac{g}{mol}$

Periodic table

^{13} Al → element
26.982 → atomic mass

^{13} → Z

Electronegativity

- ↑ en = strong tendency to accept e^- (F, Cl)
- ↓ en = strong tendency to give up e^- (Li, Na, K)

terms

- Density (g/cm^3) ~ most solids range 1-23 g/cm^3 (ρ)
- Atomic # = Z
- Atomic weight = A (g/mol)
- Mole = # of particle → $N_A = 6.022 \times 10^{23}$ mol/part

Review problem

how many atoms in 6 gram of carbon

$\rho = 2.26\ g/cm^3$ $\quad 1\,mol_C = 12.011\ g/mol$

$n = \frac{m}{M} \rightarrow n = \frac{6g}{12.011\ g/mol} \approx 0.5\,mol$

$n \cdot N_A = \#\,atoms \rightarrow 0.5\,mol \cdot N_A = 3.011 \times 10^{23}\,atoms$

Calculate the Volume of 1 mol of Au

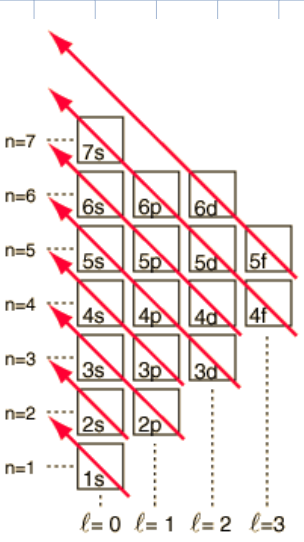
$\rho = 19.32\ g/cm^3 \qquad n = 1\ mol$

$n \cdot M = m \rightarrow 1\ mol \cdot 196.6654\ g/mol = m$

$196.665\ g = m$

$\frac{m}{\rho} = V \rightarrow \frac{196.665\ g}{19.32\ g/cm^3} = V = 10.194\ cm^3$

Energy states



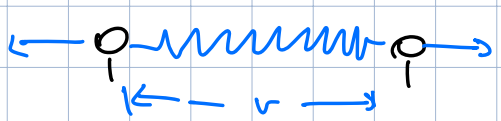
increasing energy

$s^2 \quad p^6 \quad d^{10}$

Stable e^- config

- complet s & p shells
- tend to be unreactive

Bonding Forces & Energies



r = bond length

- net force is given by the sum of an attractive and a repulsive force

- Potential E is given by the integral of the net force curve w/ respect to distance

$$E = \int F \cdot dr$$

note equilibrium seperation occurs where the net force = 0

- Bonding E : Min of the potential vs distance curve

  - Indicates how much E must be supplied to completely dissociate the 2 atoms

  - Depth well

    → Deep well = strongly bonded

    → Shallow well = weakly bonded

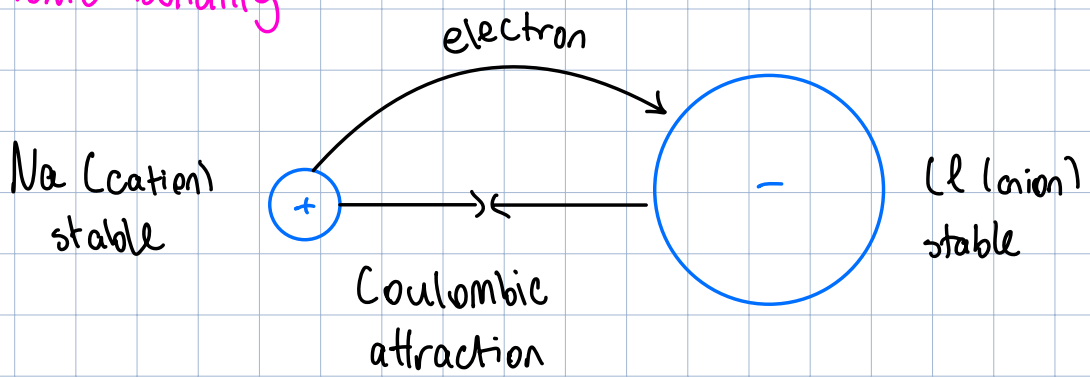
Bond E

- the ↑ the bond E
  - the ↑ the melting temp
  - the ↑ the hardness
- state as function of bonding E
  - solid (↑)
  - liquid (intermediate)
  - gas (↓ low)

Outline

- types of bonds
  - Ionic
  - Covalent
  - Metallic
  - secondary bonding
- Examples
  - relation between bond E and properties

Ionic Bonding



- between + & − ions
- large diff in electroneg is req
- requires e^- transfer
- typically between metal & non-metal

Typical bonding energies: 600 to 1500 kJ/mole (3 to 8 eV/atom)

typical characteristics of ionically-bonded mats

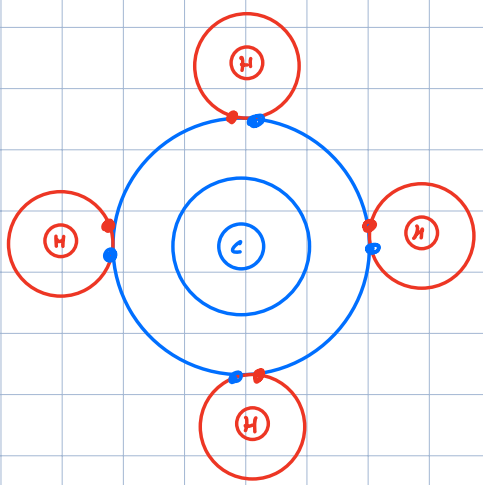
- high melting temp
- hard
- brittle
- insulators (elec & thermal)

Covalent bonding

- req shared e^- ex: CH_4

C: has 4 valence e^-, needs 4 more

H: has 1 ve^-, needs 1 more



- electronegativities are comparable

other ex:

- molecules w/ nonmetals
- molecules w/ metals & non-metal (GaAs)
- Elemental solids (RHS of Periodic table)
- Compound solids (column IV A C, Si, Ge))

summary

- each atom tries to achieve a more stable orbital filling config
- tends to be a highly directional bond
- Gives rise to a fixed orientation of the atoms
- shared e^- may be considered to belong to each atom
- bonds may be strong (diamond, $T_m > 3550°C$) or weak (Bi, $T_m = 270°C$)
- Mats may be electrically conductive (GaAs) or insulating (diamond)

Ionic Character most mats are neither 100% ionic nor 100% covalent

- % ionic character $= \left[1 - \exp\left\{-0.25 (X_A - X_B)^2\right\}\right] \times 100\%$

where X_A and X_B are the elecneg of atom A & B

Metallic Bonding

- Arises from a sea of donated valence e^-

(they're not bounded / free to drift throughout the mat)

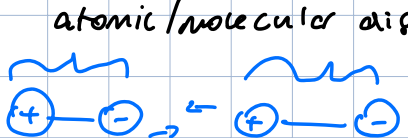
- primary bond for metals & their alloys
- it can be strong or weak
- it gives rise to ↑ elec & thermal conductivity, and ductibility

↓ why?

the $\bar{e}$ are loosely held tg since each atom has many unoccupied valence orbitals; it's relatively easy for the $\bar{e}$ to move about. which allows atoms to slide past eachother

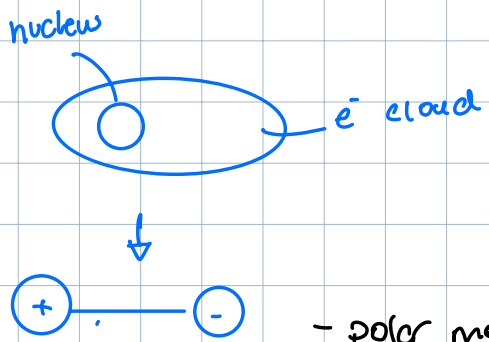
Secondary Bonding

- Van der Waals bonding
- weak compared to others
- exist between almost all atoms / molecules
- Arises from atomic / molecular dipoles



physical bonds, not chemical

↳ (sharing of e^-)

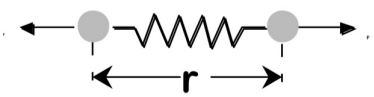


- polar molecule induced dipoles & Hydrogen bonding

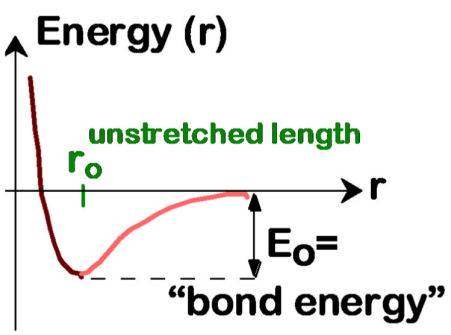
↳ the H bond is stronger than the Van der Waals

PROPERTIES FROM BONDING: T_M

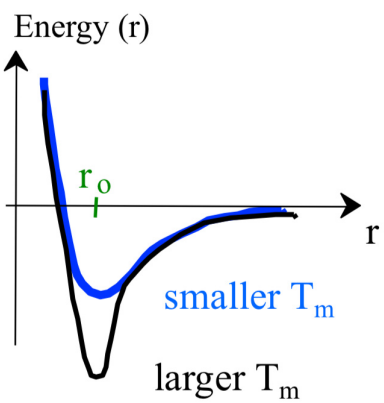
- Bond length, r



- Bond energy, E_o



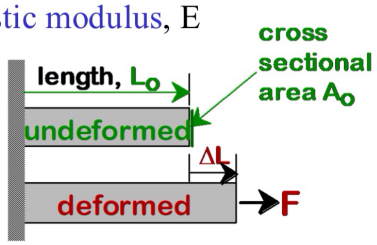
- Melting Temperature, T_m



T_m is larger, if E_o is larger.

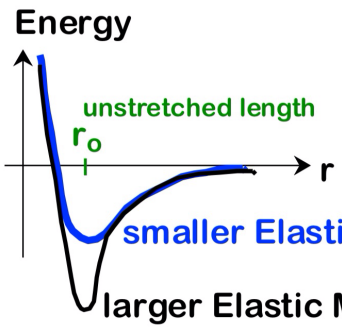
PROPERTIES FROM BONDING: *E*

- Elastic modulus, E



Elastic modulus

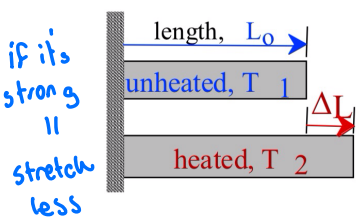
$$\frac{F}{A_o} = E\frac{\Delta L}{L_o}$$



E is larger if E_o is larger.

PROPERTIES FROM BONDING: α

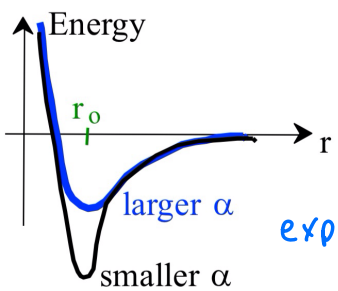
- Coefficient of thermal expansion, α



if it's strong it stretch less

coeff. thermal expansion

$$\frac{\Delta L}{L_o} = \alpha\,(T_2 - T_1)$$



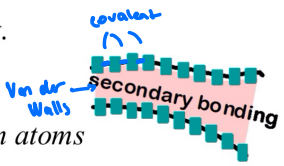
α is larger if E_o is smaller.

expands more when α is larger

Bonding type summary

A comparison of the type of bonding found in different materials:

- For brass, the bonding is metallic since it is a metal alloy.
- For rubber, the bonding is covalent with some V.d.W...
  Rubber is composed primarily of carbon and hydrogen atoms



- For BaS, the bonding is predominantly ionic. (*but with some covalent character*) - *on the basis of the relative positions of Ba and S in the periodic table.*
- For solid xenon, the bonding is Van der Waals since xenon is an inert gas.
- For nylon, the bonding is covalent with perhaps some V.d.W...
  Nylon is composed primarily of carbon and hydrogen
- For AlP the bonding is predominantly covalent (but with some ionic character)
  on the basis of the relative positions of Al and P in the periodic table.

secondary

ex:

Brass (copper + zinc) Cu Zn

2 metals, metallic bonding, alloy

Rubber (C_5H_8)

covalent, N-M, comparable elecneg

Solid Xenon (Xe)

Van der Waals, one element and noble gas

AlP (AlP)

Ionic, M-NM X covalent due to diff in electneg

Lecture 4

Outline

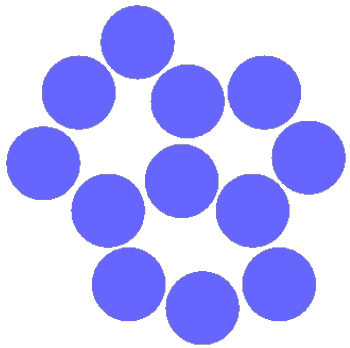
- Crystalline vs amorphous structure
- Crystal struct
  - unit cell
  - Coordination #
  - Atomic packing factor
- Crystal systems

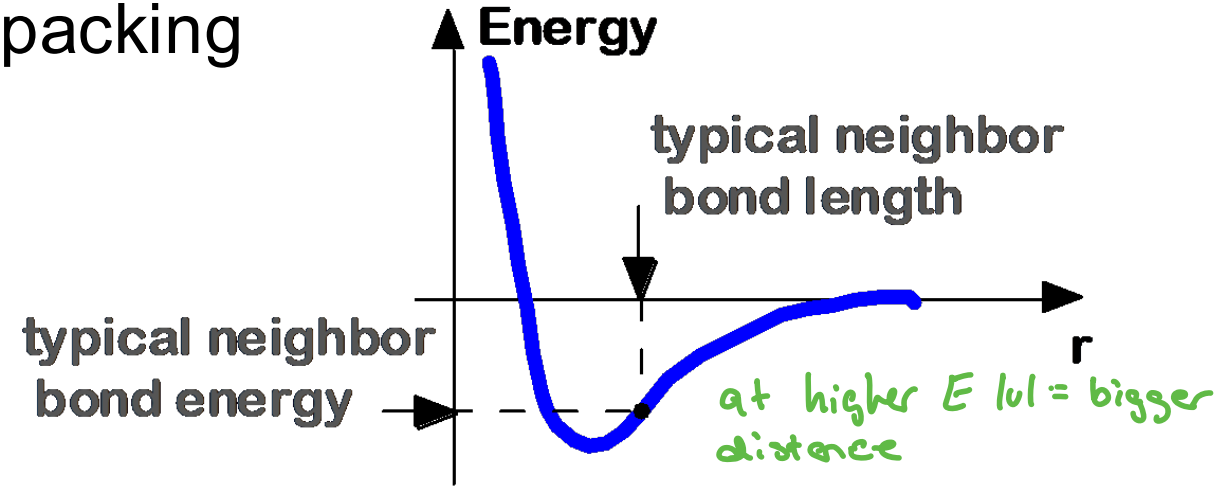
Outline:

- Crystalline versus amorphous structures
- Crystal structure
  - Unit cell
  - Coordination number
  - Atomic packing factor
- Crystal systems

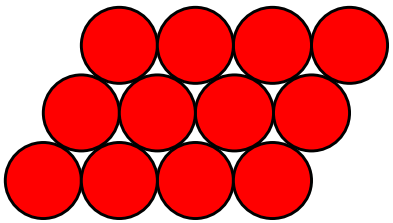
ENERGY AND PACKING

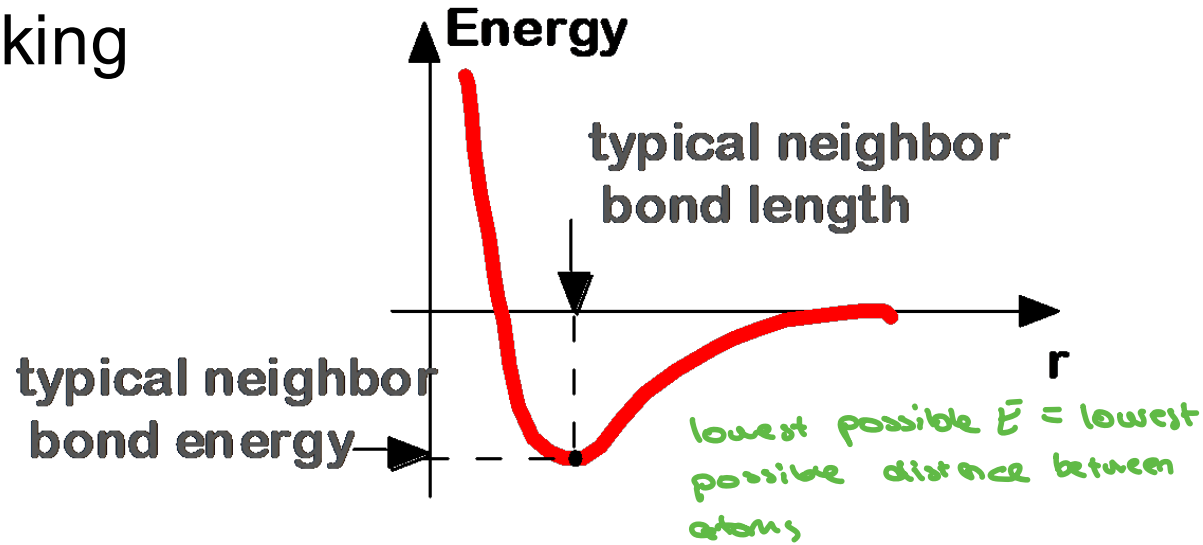
- Non dense, ... packing





- Dense, ... packing





Dense, regular-packed structures tend to have lower energy.

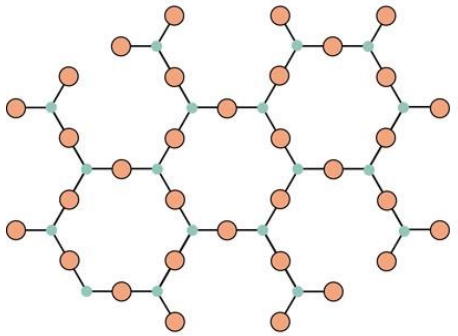
MATERIALS AND PACKING

order = crystalline disorder = Amorphous

Crystalline materials...

- atoms pack in periodic, 3D arrays
- typical of:
  - -metals
  - -many ceramics
  - -some polymers

↳ polymer crystal = metal / ceramic crystals

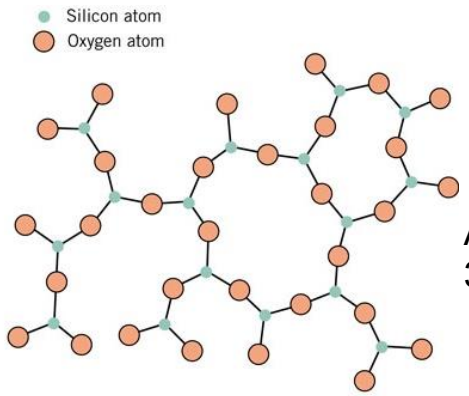


crystalline SiO_2

Adapted from Fig. 3.23(a), *Callister 8e.*

Noncrystalline materials...

- atoms have no periodic packing
- occurs for:
  - -complex structures
  - -rapid cooling



noncrystalline SiO_2

Adapted from Fig. 3.18(b), *Callister 6e.*

Amorphous..... = Noncrystalline

(like frozen liquids)

Atoms can be arranged either in a regular, periodic array (i.e., long-range order) or completely disordered (amorphous).

Crystal Structure

- Motivation: Many of the properties of materials (*especially mechanical*) are determined by the arrangement of the constituent atoms.

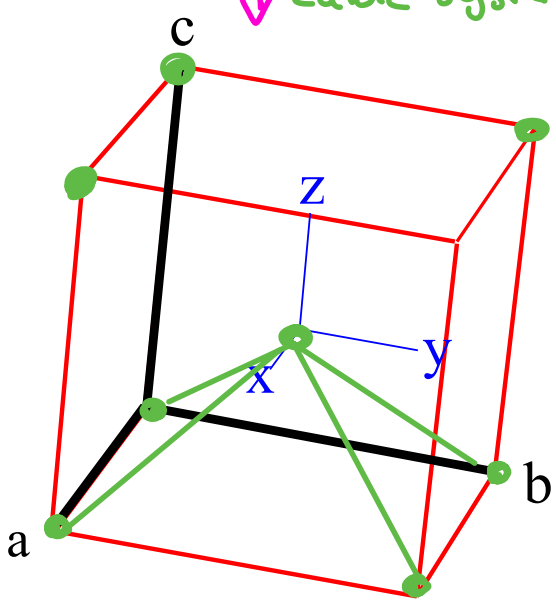
  This arrangement is called the material's crystal structure.

- An important distinction…
  - Atomic structure relates to the number of protons and neutrons in the nucleus of an atom, as well as the number and probability distributions of the constituent electrons.
  - On the other hand, crystal structure pertains to the arrangement of atoms in the crystalline solid material.

Crystal Structure

- We need a way to specify crystallographic directions and planes.
- Let's start with the hard sphere model (in which nearest neighbor atoms "touch" each other)…

cube form
cubic system



To illustrate the concept of crystal structure and lattice systems, we first identify a coordinate system (x, y, z):

a b c

We can't specify directions or planes without knowing what the reference system is.

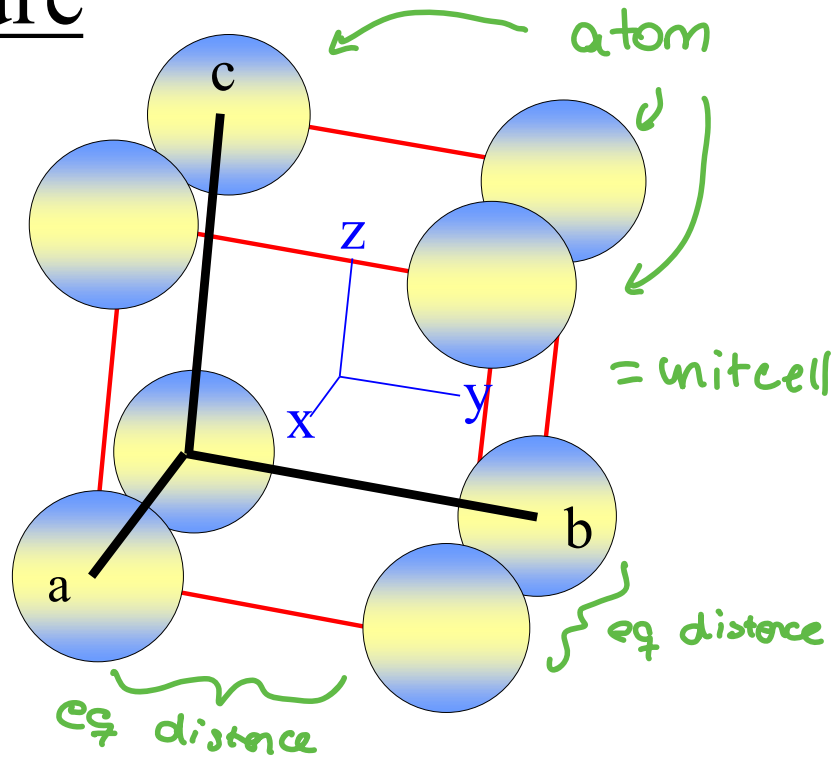
building block

unit cell < lattice < crystal

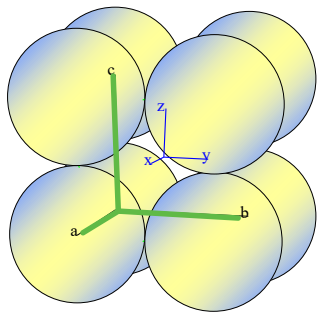
grain?

Crystal Structure

Now place an atom at each corner…



This represents the "hard sphere" model of a ...Simple cubic (SC)... *crystal system*



- Atoms "touch" along the crystal axes
- These directions are referred to as "close-packed" in the simple cubic system

only case for SC

not many examples of simple cubic systems in nature, except for α-Po

The above diagram represents a simple cubic ...unit cell...

What is Unit Cell?

A unit cell is the smallest entity that exhibits the chemical and crystallographic properties of the material.

– *Unit cells are the most elementary arrangement of atoms which can generate the entire crystal upon application of suitable translation, rotation, mirror, or inversion operations.*

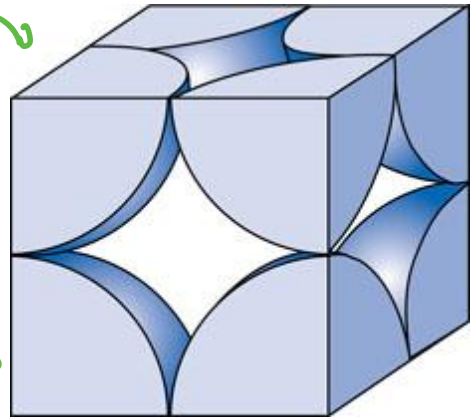
Definition:

the length of each unit cell axis is called a lattice parameter

– *In cubic systems, all three orthogonal lattice parameters are equal*

– Lattice parameters are typically on the order of a few Angstroms (or a few tenths of a nanometer)

Simple Cubic Unit Cell



- volume of the unit cell = a^3
  - where *a* is the lattice parameter
- coordination # = 6
  - for simple cubic structures

How many atoms does the simple cubic unit cell contain?

A simple cubic structure contains 1 *atom/unit cell.*

- In general, the number of atoms/unit cell, N, is given by

$$N = N_i + \frac{N_f}{2} + \frac{N_c}{8}$$

Where: N_i = # interior atoms,
N_f = # face atoms,
N_c = # corner atoms

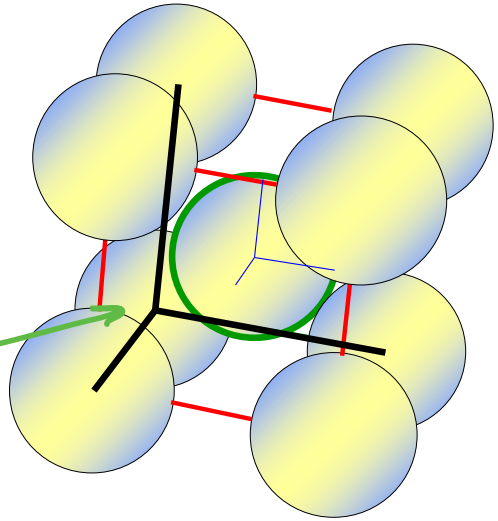
✓CN is the number of nearest-neighbor atoms

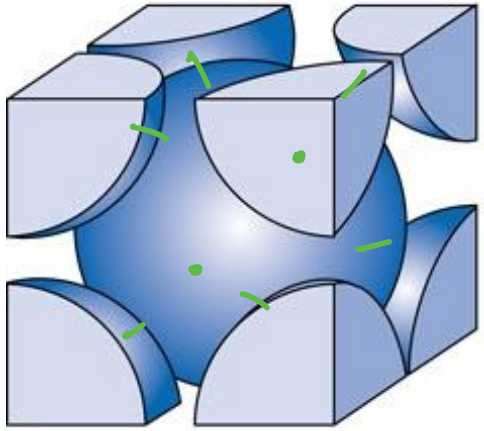
✓ Coordination number is important in determining the structure of crystalline materials.

Body Centered Cubic Unit Cell

2 atoms total

Now, suppose we add another atom at the **center** of the cube





We no longer have a simple cubic structure but instead, a ..Body-.. Centered Cubic.. (BCC) structure

Examples of BCC systems: Cr, W, Mo, Ta, Fe (αFe stable below 912°C)

Notice that in the BCC structure, atoms touch along the body diagonals. These are the close-packed directions in the body-centered cubic structure.

atoms are touching only on diagonal now

BCC unit cells have a CN = ...8..

Atomic Packing Factor

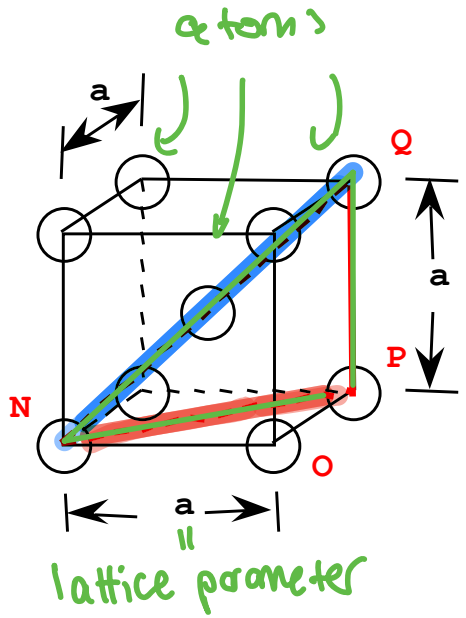
$$\text{Atomic Packing Factor (APF or APE)} = \frac{\text{total "sphere" volume}}{\text{total cell volume}}$$

"total sphere volume" is just the volume per atomic "sphere" multiplied by the number of atoms in the unit cell

Example: Calculate the APF for a BCC unit cell:

$$APF = \frac{N * \frac{4}{3}\pi R^3}{a^3}$$

where
N = # atoms/unit cell
R = atomic radius
a = lattice param



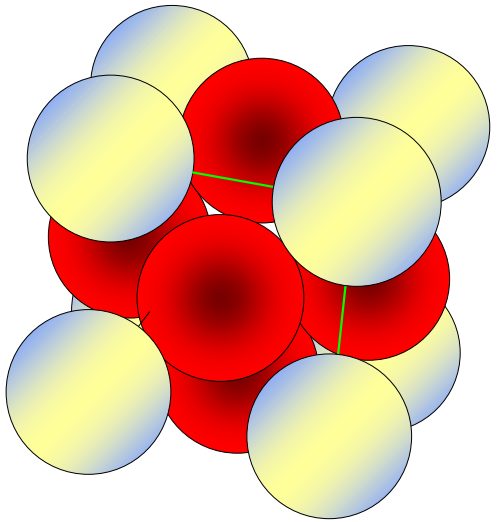
Using the triangle NOP, $(NP)^2 = a^2 + a^2 = 2a^2$

And then for triangle NPQ, $(NQ)^2 = (NP)^2 + a^2$

the packing efficiency of a BCC structure is 68. %

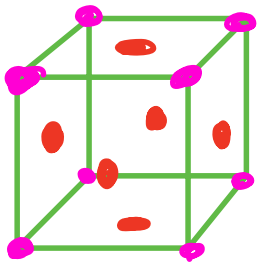
Face Centered Cubic Unit Cell

all Al atoms



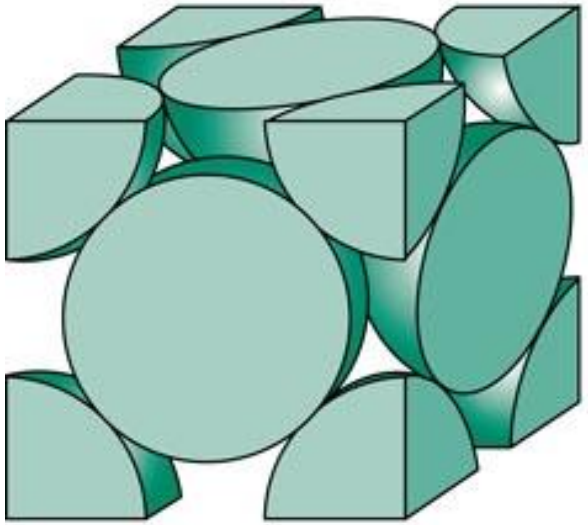
Now suppose we place equivalent atoms at the corners of the unit cell, AND in the center of each face:

This is a face centered cubic (FCC) crystal structure



- Examples of FCC metals: Cu, Ni, Au, Ag, Fe (γFe stable above 912°C)
- Close-packed directions in FCC metals are along face diagonals

Face Centered Cubic Unit Cell



- Q: How many atoms per unit cell in the FCC structure? =4

$$\text{recall, } N = N_i + \frac{N_f}{2} + \frac{N_c}{8} = 4$$

- FCC unit cells have a CN = 12 and an APF = 0.75
  - *maximum packing efficiency for* ***monosized spheres***

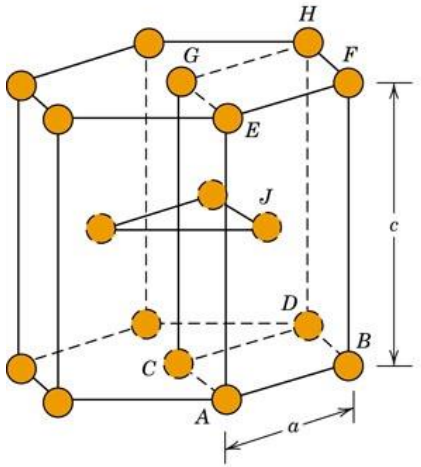
simple, body, face = 1 type of crystal system

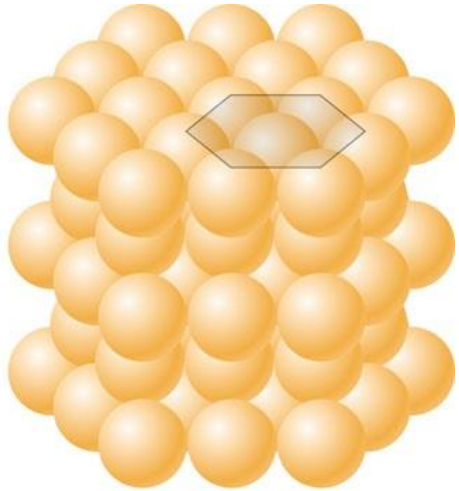
Hexagonal Unit Cell

Cubic crystal system

- There are other ways in which atoms can be arranged to form unit cells:
  - For example…

This represents an HCP (hexagonal close packed) structure





➢ Examples of HCP systems ***(Mg, Co, Ti, Zn, Zr, RE)***

➢ The HCP unit cell consists of 6 atoms forming the corners of a hexagon in the basal planes + 1 atom in the center. In addition, there are 3 interior atoms midway between basal planes along the c-axis.

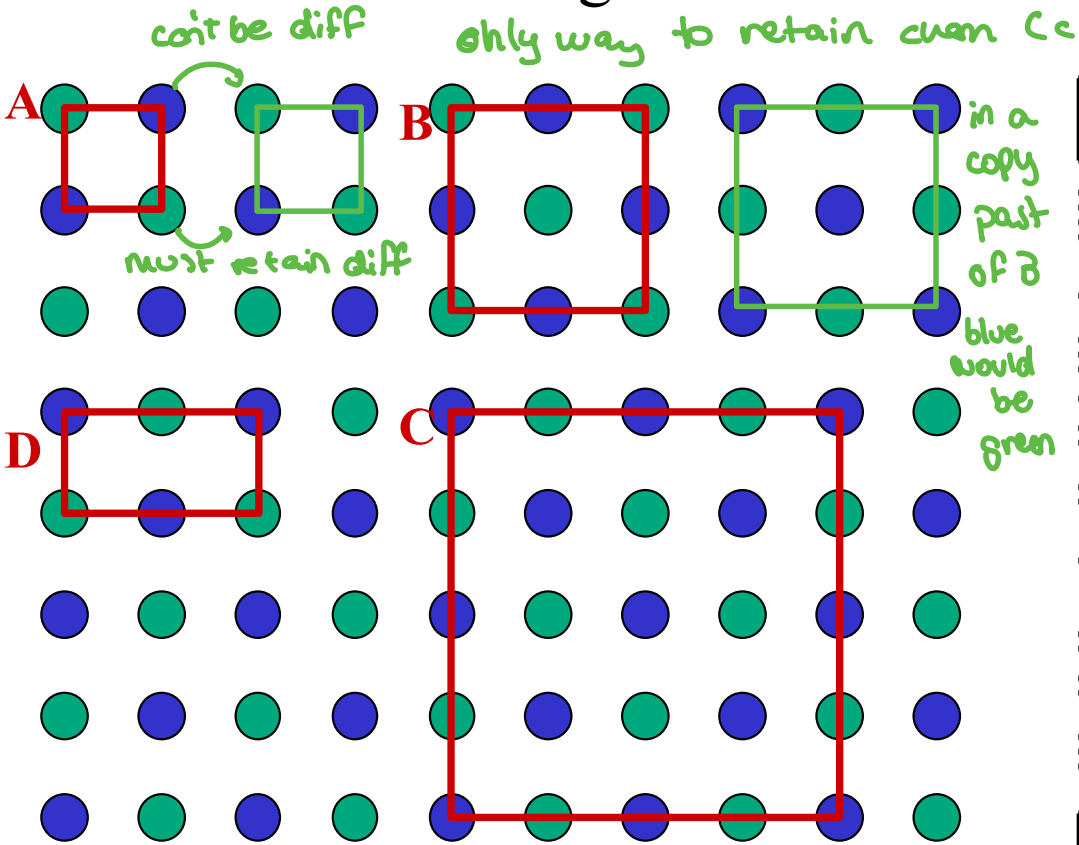
atoms/unit cell = (1/6)*12 (corner atoms) + (1/2)*2 (center face atoms) + 3 (interior atoms) = 6

➢ Coordination # = 12.

Quick Check: Can You Spot the Unit Cell?

Which outlined region is the unit cell?

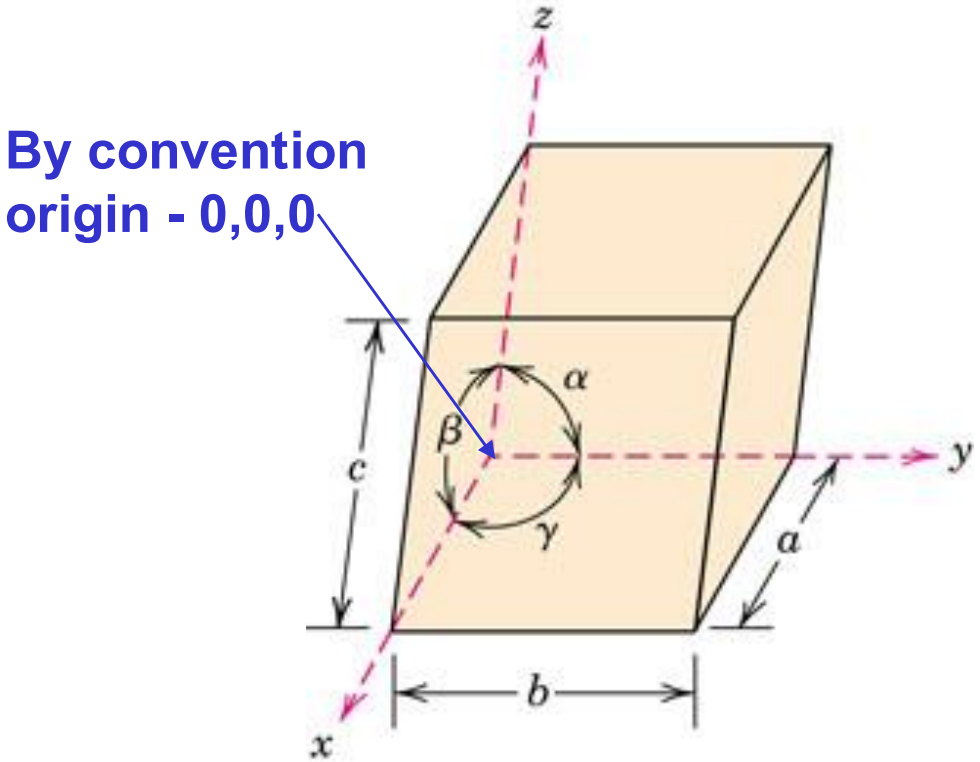
answer = B



Pattern continues infinitely in all directions.

Use the QR code shown in the classroom. This one won't work.

General convention for unit cell axis and angle notation:

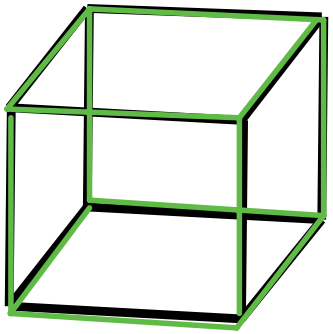


In total, there are 7 distinct and unique crystal systems:

cubic represents only one of the 7

Simple, body, face

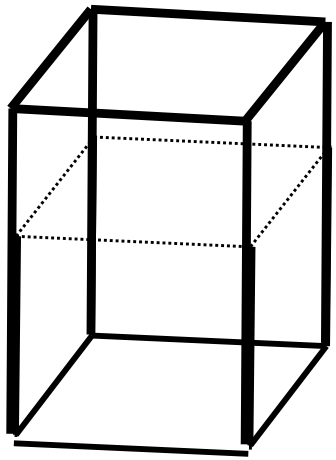
Crystal Systems : only for knowledge (exc cubic)



Cubic

$a = b = c$

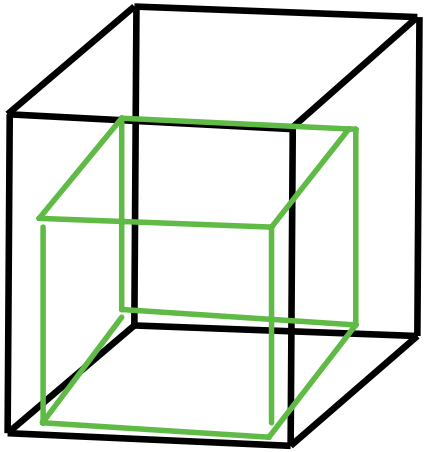
$\alpha = \beta = \gamma = 90$



Tetragonal

$a = b \neq c$

$\alpha = \beta = \gamma = 90$

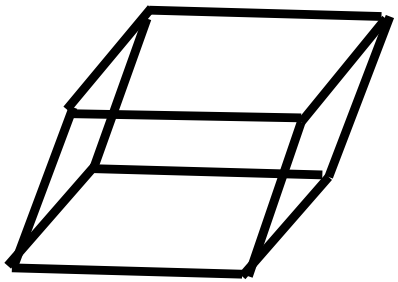


Orthorhombic

$a \neq b \neq c$

$\alpha = \beta = \gamma = 90$

Crystal Systems

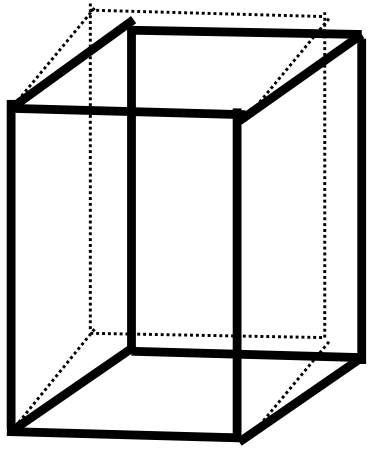


Rhombohedral

$a = b = c$

$\alpha = \beta = \gamma \neq 90$

"Pushed over" cube



Hexagonal

$a = b \neq c$

$\alpha = \beta = 90 \quad \gamma = 120$

"Squished" tetragonal

Monoclinic

$a \neq b \neq c$

$\alpha = \gamma = 90, \beta \neq 90$

"Pushed over" orthorhombic (in one direction)

Triclinic

$a \neq b \neq c$

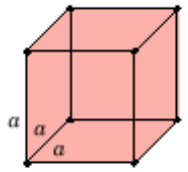
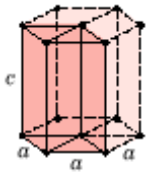
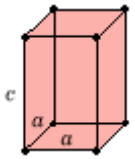
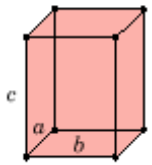
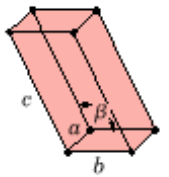
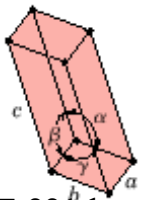
$\alpha \neq \beta \neq \gamma \neq 90$

"Pushed over" orthorhombic (in two directions)

Crystal Systems

Note that these 7 crystal systems do not account for all the possible lattice types

for example, the cubic system contains SC, FCC, and BCC as subsets

Crystal System	Axial Relationships	Interaxial Angles	Unit Cell Geometry
Cubic	$a = b = c$	$\alpha = \beta = \gamma = 90°$	
Hexagonal	$a = b \neq c$	$\alpha = \beta = 90°, \gamma = 120°$	
Tetragonal	$a = b \neq c$	$\alpha = \beta = \gamma = 90°$	
Rhombohedral	$a = b = c$	$\alpha = \beta = \gamma \neq 90°$	
Orthorhombic	$a \neq b \neq c$	$\alpha = \beta = \gamma = 90°$	
Monoclinic	$a \neq b \neq c$	$\alpha = \gamma = 90° \neq \beta$	
Triclinic	$a \neq b \neq c$	$\alpha \neq \beta \neq \gamma \neq 90°$	

Crystal Systems

There are 14 unique lattice types from this framework of 7 crystal systems (called Bravais lattices):

Crystal system	Types of possible lattice arrangements
Cubic	SC, BCC, FCC
Hexagonal	HCP
Tetragonal	Simple, body-centered
Orthorhombic	Simple, base-centered, BC, FC
Rhombohedral	Simple
Monoclin	Simple, base-centered
Triclinic	Simple

7 14

We will mainly be concerned with cubic and hexagonal systems in this class. But you need to realize that many other types of symmetries exist!

Next time:

Crystallographic Directions and Planes

Outline

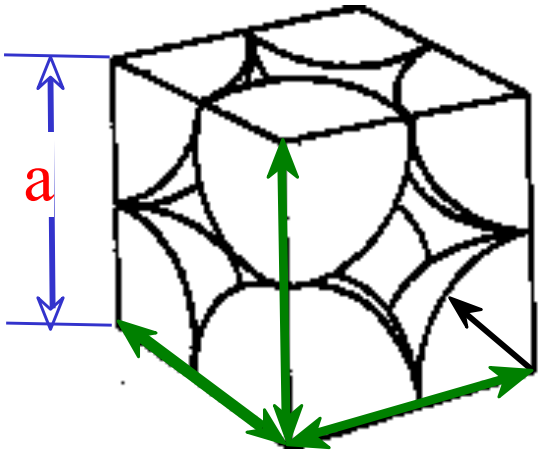
- Review Crystal structure
- Stacking sequence
- theoretical Density
- Crystallographic Directions & Planes
  - example

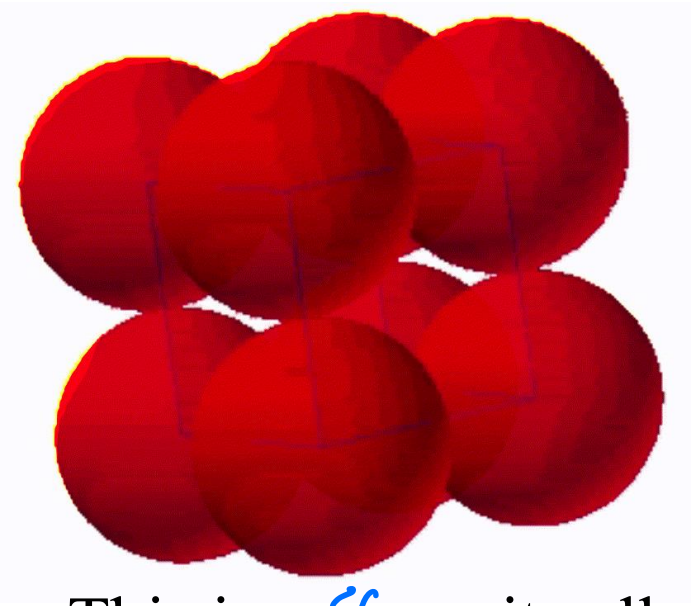
Outline:

- Review of Crystal structure
- Stacking Sequence
- Theoretical Density
- Crystallographic Directions and Planes
  - *examples*

Review: *Crystal Structure*

- Close-packed directions are ..cube..edges





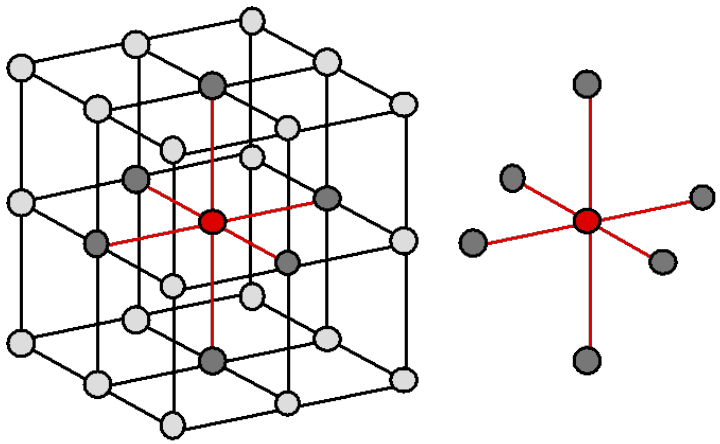
- Contains = ..1 atom/unit cell...

This is a ..SC.. unit cell

Simple cubic

- Coordination # = ..6.. *(# nearest neighbors)*

$$APF = \frac{1 \cdot \frac{4}{3}\pi (0.5a)^3}{a^3}$$

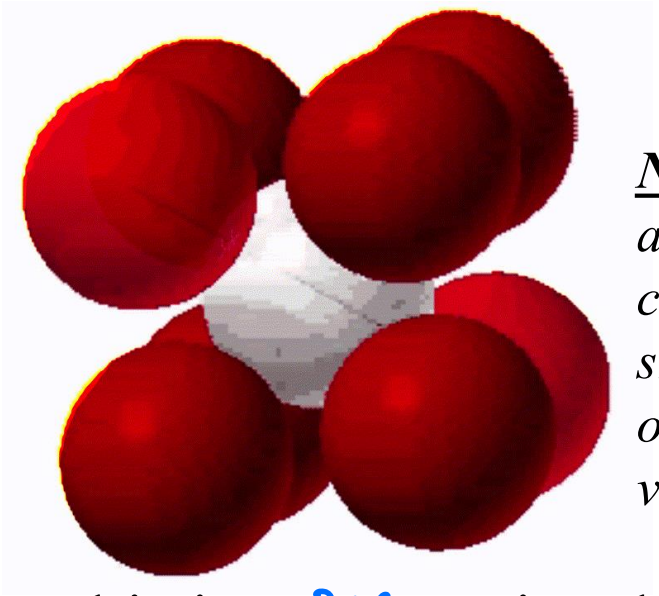


- APF for a simple cubic structure = ...

Review: *Crystal Structure*

- Close packed directions are ……Cube diagonals……

- Coordination # = ……



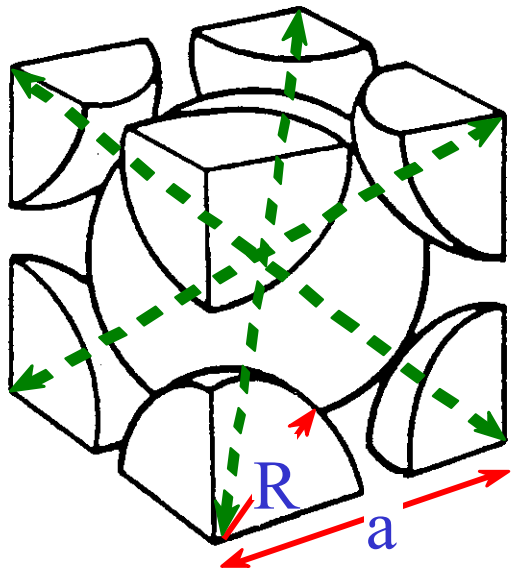
Note: *All atoms are identical; the center atom is shaded differently only for ease of viewing.*

This is a ……BCC…… unit cell

Body centric

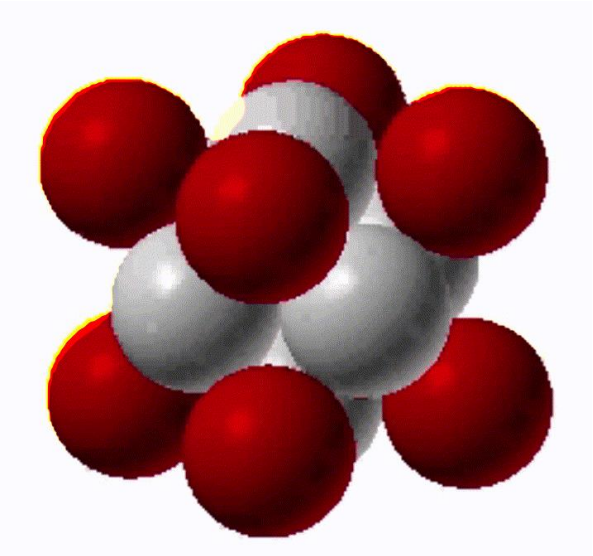
$$APF = \frac{2\left(\frac{4}{3}\pi\left(\sqrt{3}a/4\right)^3\right)}{a^3}$$

- APF for a BCC structure = …….



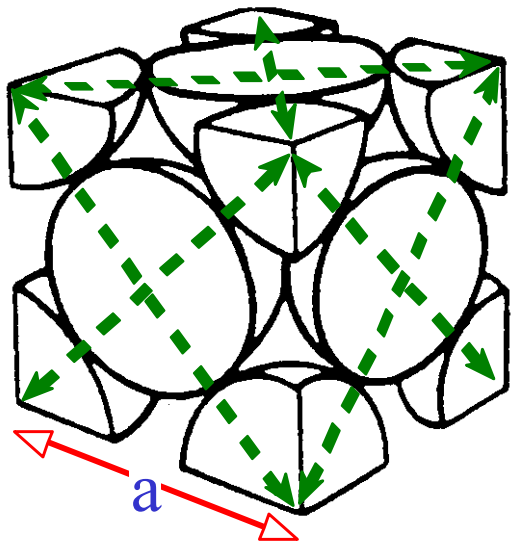
Review: *Crystal Structure*

- Close packed directions are ……face diagonals……
- Coordination # = ……12……



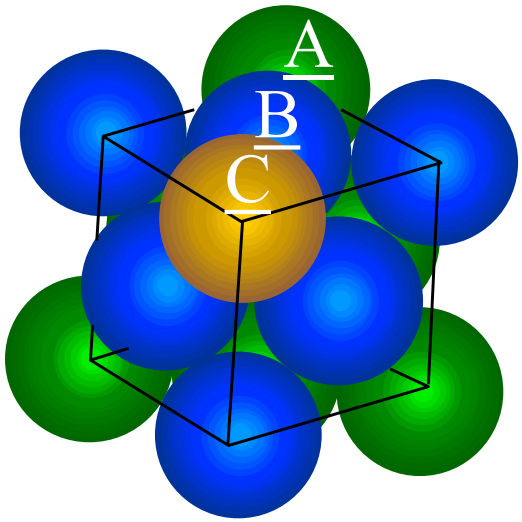
This is a ……FCC…… unit cell

$$APF = \frac{4\left(\frac{4}{3}\pi\left(\sqrt{2}a/4\right)^3\right)}{a^3}$$



- APF for a FCC structure = ……0.74……

FCC Stacking Sequence



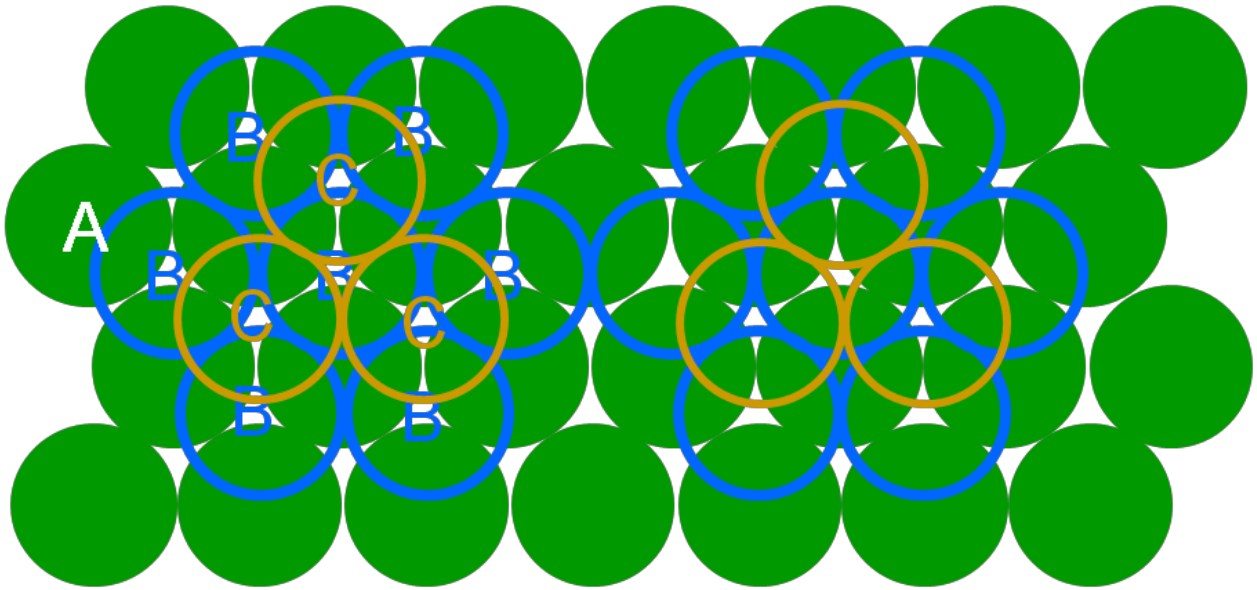
- FCC Unit Cell

3 layers

A sites

B sites

C sites



2D Projection

- ABCABC...etc Stacking Sequence

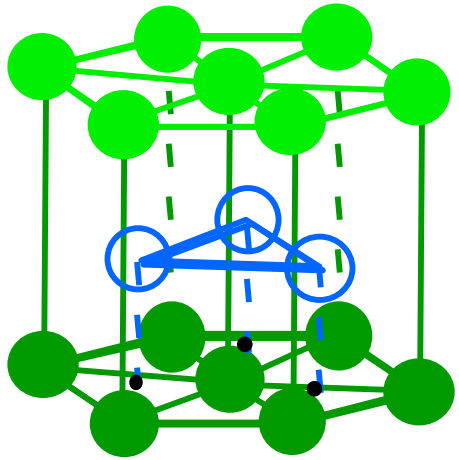
Review: *Crystal Structure*

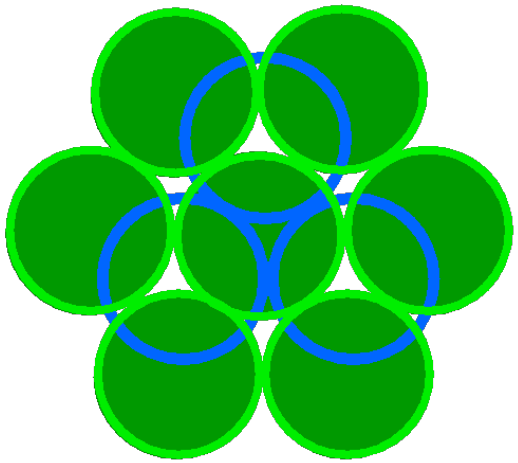
- Coordination # = 12
- APF = 0.74

A sites

B sites

A sites





Top layer

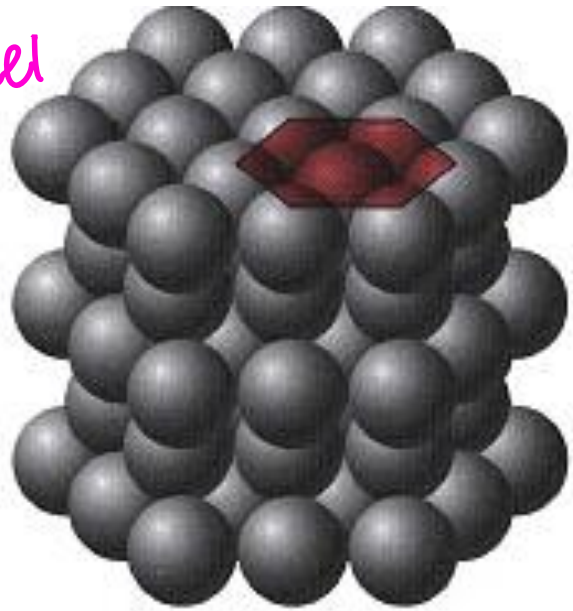
Middle layer

Bottom layer

parallel

2D Projection

This is a HCP unit cell

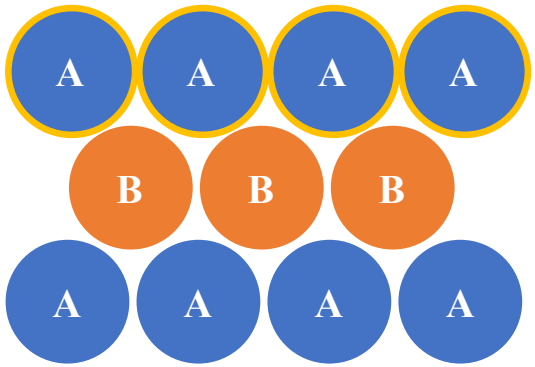


- Stacking Sequence: ABA

HCP vs. FCC: *the Third Layer Decides*

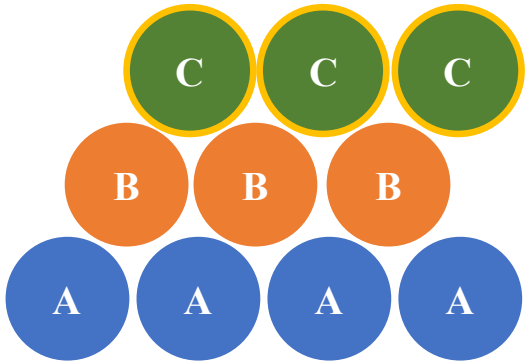
Both start the same way: a close-packed A layer, then a B layer in the hollows.

HCP — ABAB



Third layer sits directly above the A atoms. The C hollows stay empty, so the pattern repeats every two layers: ABAB…

FCC — ABCABC



Third layer shifts into the empty C hollows — a new position. Only the fourth layer returns to A: ABCABC…

Same packing, different order: CN = 12 and APF = 0.74 for both. The stacking sequence is what differs — and it is why FCC and HCP deform so differently.

Example Problem

If you know

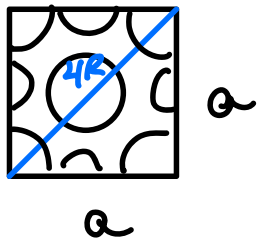
- the crystal structure,
- the atomic radius
- the atomic weight,

you can calculate the density of a particular material

Example:

Copper has an atomic radius 0.128 nm an FCC crystal structure and an atomic weight of 63.5 g/mol. Calculate its density.

4 atoms



$N_A = 6.02 \times 10^{23}$ atoms/mol

$a = \sqrt{8}R$

$V = a^3 = 0.04745\ nm^3$

$\rho = \frac{m}{V} = \frac{4.2172 \times 10^{-22}\ g}{0.04745 \times 10^{-21}\ cm^3} = 8.8876 \frac{g}{cm^3}$

Crystallographic Directions, and Planes

Now that we know how atoms arrange themselves to form crystals, *we need a way to identify directions and planes of atoms*.

Why?

✓ Deformation under loading (سلب.) occurs on certain crystalline planes and in certain crystallographic directions.

✓ Before we can predict how materials fail, we need to know what modes of failure are more likely to occur.

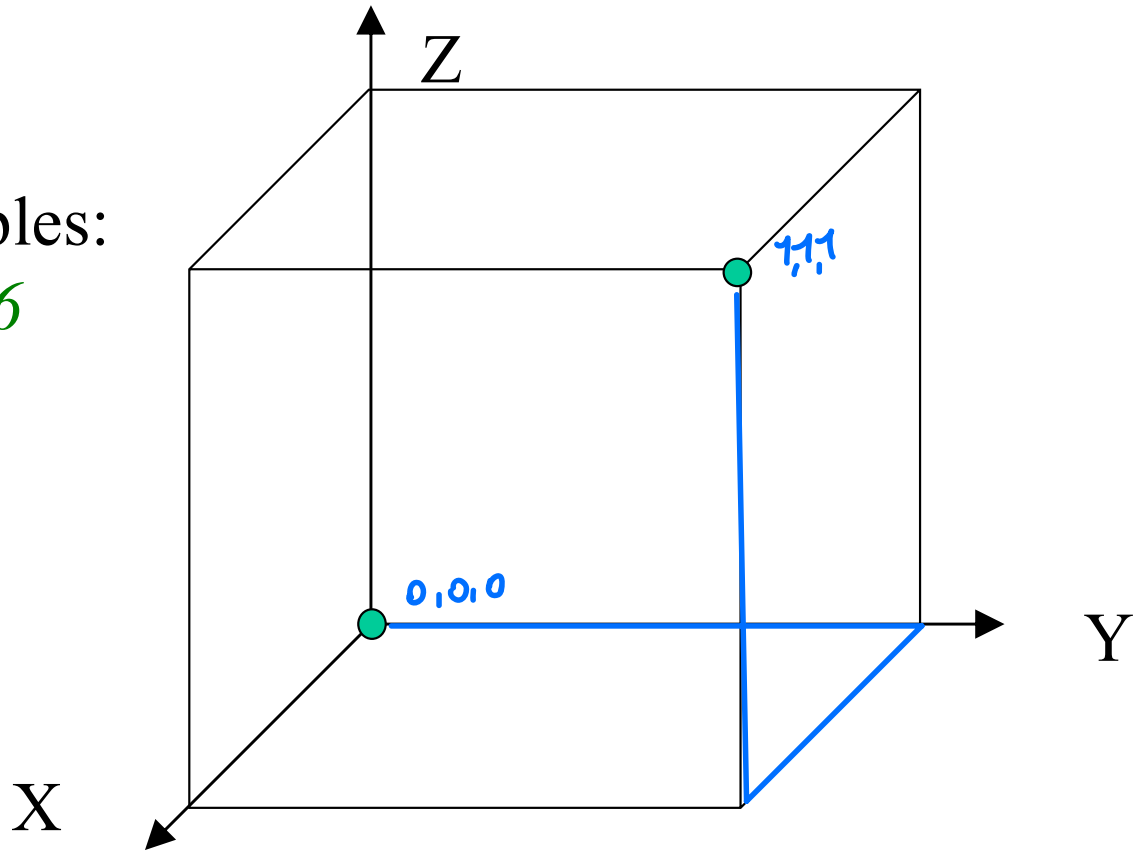
✓ Other properties of materials (*electrical conductivity, thermal conductivity, elastic modulus*) can vary in a crystal with orientation.

Directions and planes are described using three integers - Miller… Indices

Point Coordinates

Point position specified in terms of its coordinates as fractional multiples of the unit cell edge lengths

See examples:
3.5 and 3.6



General Rules for Lattice Directions, Planes & Miller Indices

- Miller indices used to express lattice *planes* and *directions*
- x, y, z are the axes (on arbitrarily positioned origin)
  - *in some crystal systems these are not mutually ⊥*
- a, b, c are lattice parameters (*length of unit cell along a, b, c sides*)
- h, k, l are the Miller indices for planes and directions - expressed as planes: (hkl) and directions: [hkl]

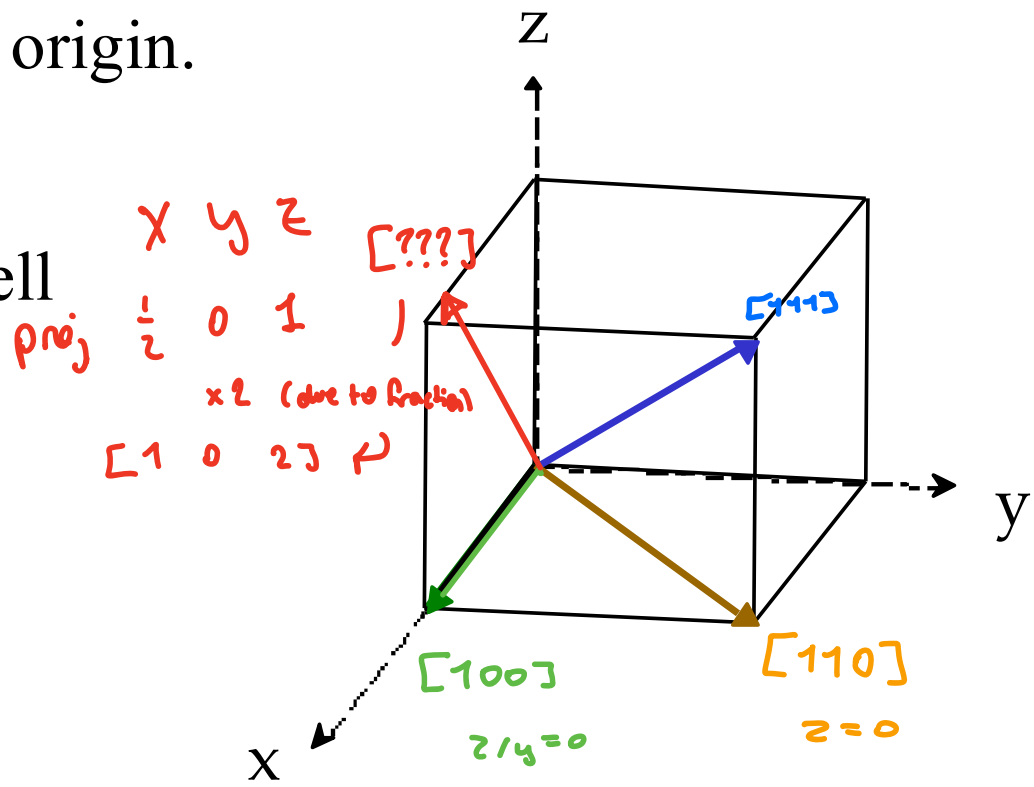
- Conventions for naming
  - There are NO COMMAS between numbers
  - Negative values are expressed with a bar over the number
    - *Example: -2 is expressed* $\bar{2}$

- Crystallographic direction:
  - [123]
  - [100]
  - … etc.

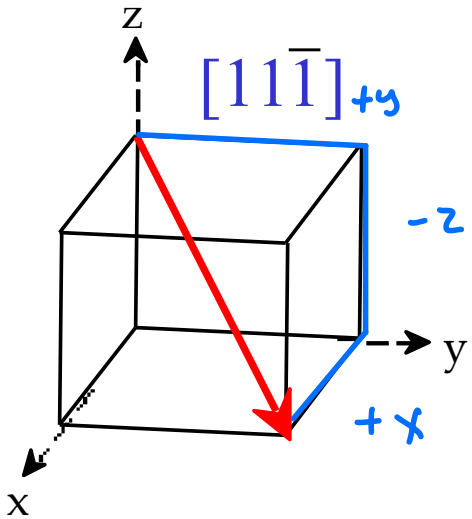
Miller Indices for Directions

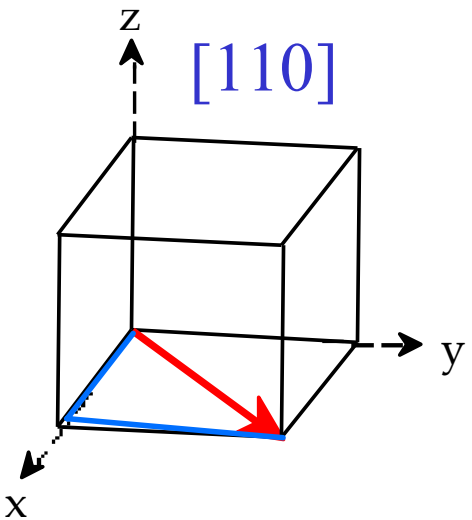
Method

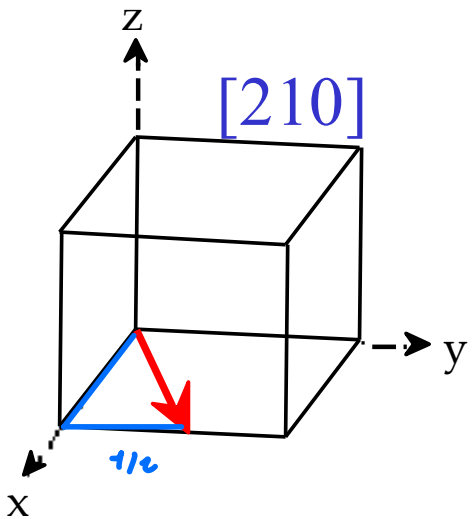
- Draw vector, define tail as origin.
- Determine length of the vector projection in unit cell dimensions, a, b, and c
- Remove fractions by multiplying by smallest possible factor
- Enclose in square brackets

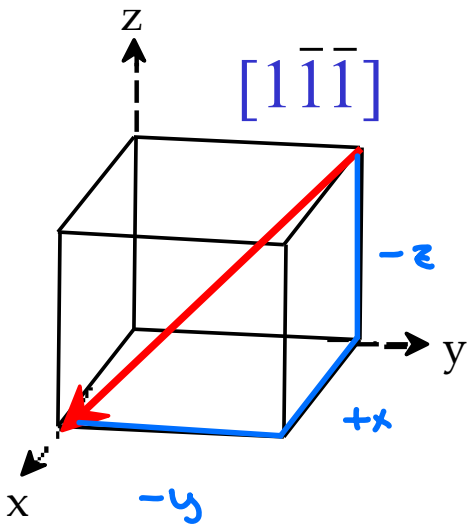


Example - Naming Directions





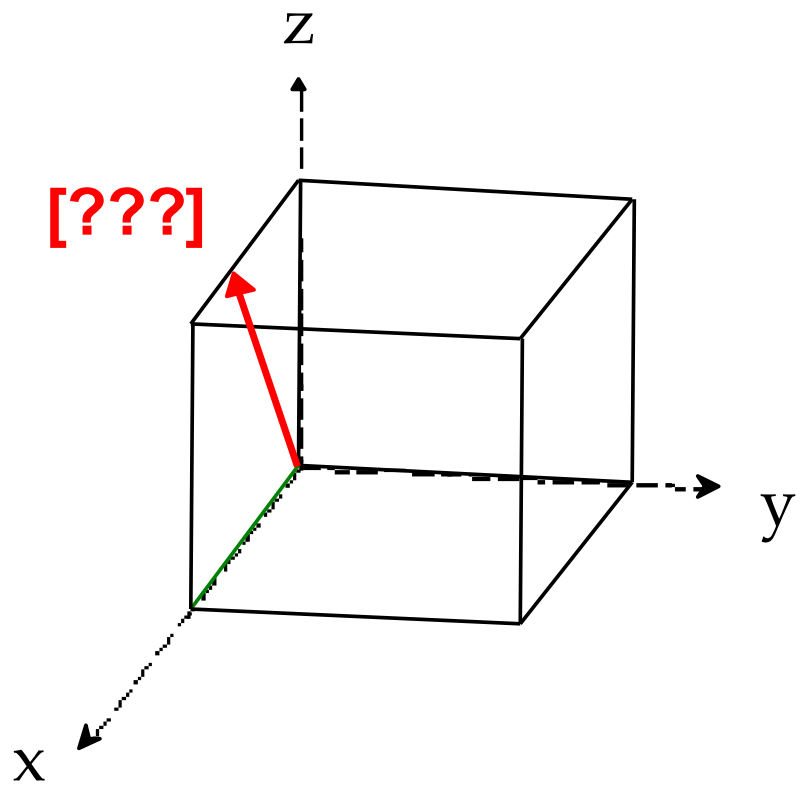




Miller Indices for Directions

Use the QR code shown in the classroom. This one won't work.

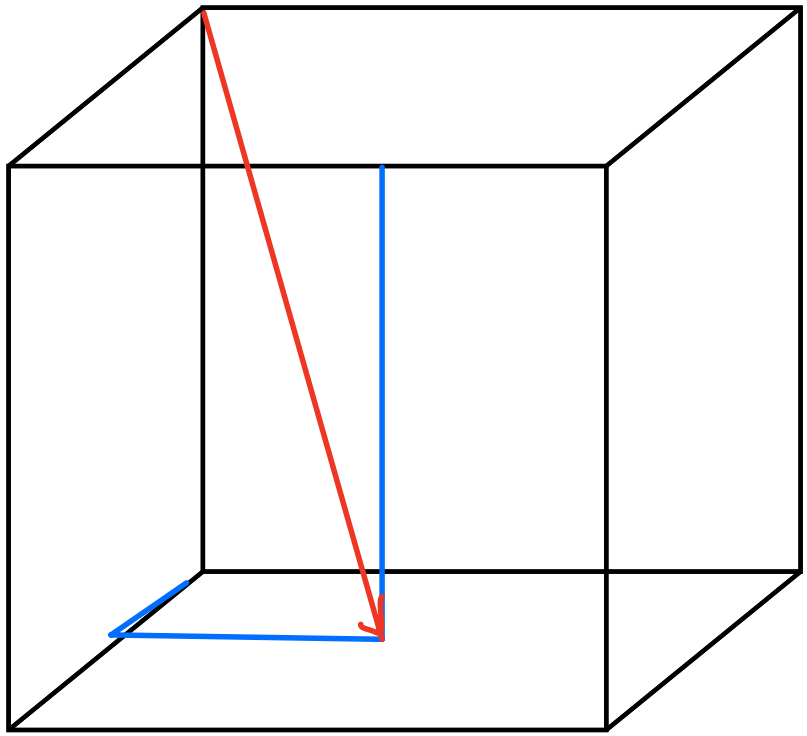
What is [? ? ?]



Example - Drawing Directions

- Draw $[11\bar{2}]$ and $[\bar{1}\bar{1}\bar{1}]$

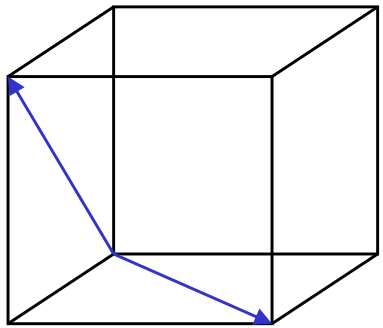
$\frac{1}{2}\ \frac{1}{2}\ -1$



The HW problems will give you more practice with directions

Families of Directions

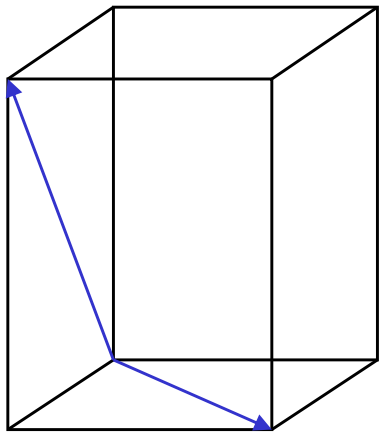
- Equivalence of directions



Only true for cubic

[101] = [110]

cubic



[101] ≠ [110]

tetragonal

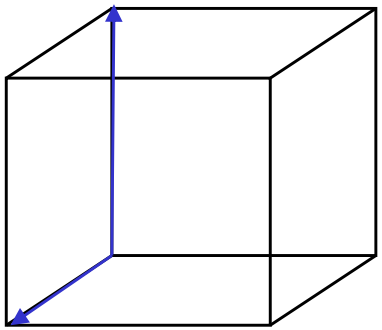
- <123> Family of directions

  For cubic, any coords containing 1, 2 and 3 are eq regardless of order/sign

  - [123], [213], [312], [132], [231], [321]
    - *only in a cubic crystal*

*In the cubic system directions having the same indices regardless of **order** or **sign** are equivalent.*

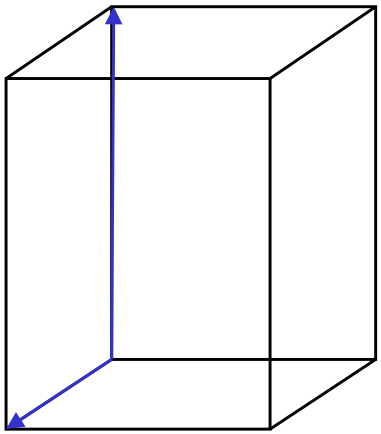
Isotropy vs. Anisotropy in Single Crystals



[100] = [001]

cubic

Isotropic



[100] ≠ [001]

tetragonal

Anisotropic

- Properties are independent of direction → isotropic material
- Directionality of properties → anisotropic material

Miller Indices for Planes

brackets or imp [] = directions

- (hkl) Crystallographic plane
- {hkl} Family of crystallographic planes
  - e.g. (hkl), (lhk), (hlk) … etc.

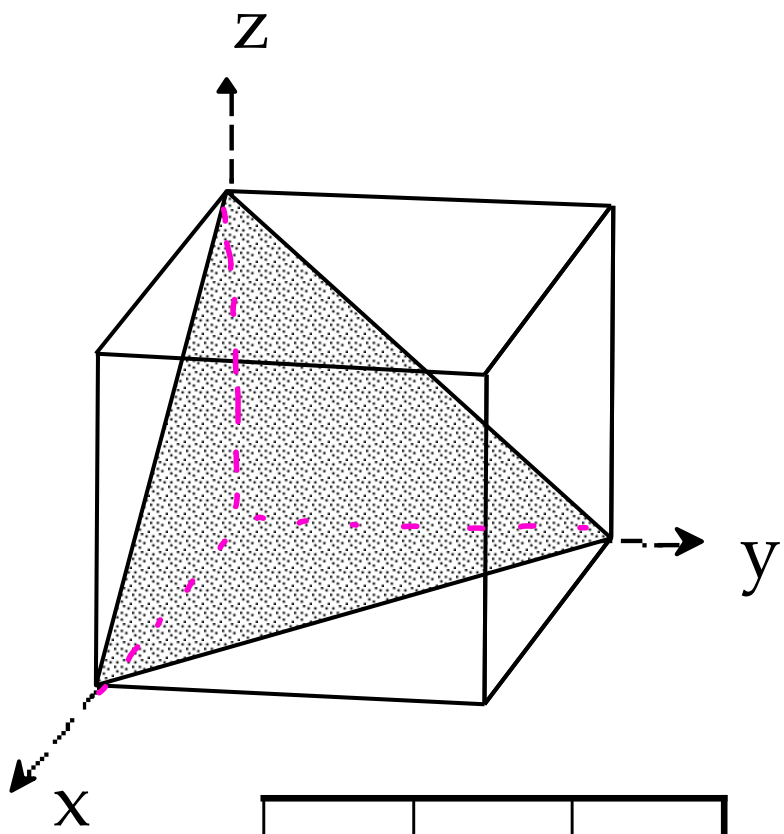
*In the cubic system planes having the same indices regardless of **order** or **sign** are equivalent*

- Hexagonal crystals can be expressed in a four index system (u v t w)
  - *Can be converted to a three index system using formulas*

Miller Indices for ***PLANES***

Method

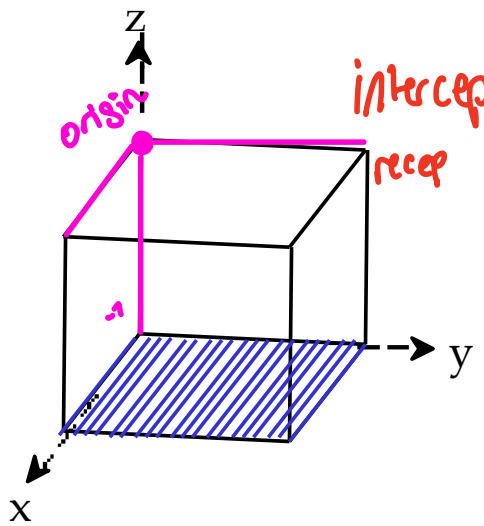
- If the plane passes through the origin, select an equivalent plane or move the origin
- Determine the intersection of the plane with the axes in terms of a, b, and c
- Take the reciprocal *(1/∞ = 0)*
- Convert to smallest integers *(optional)*
- Enclose by parentheses



	x	y	z
Intercepts	1	1	1
Reciprocals	1	1	1

see example 3.10

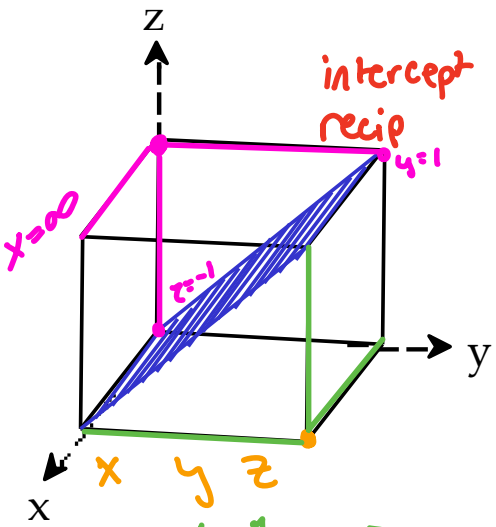
Crystallographic Planes



∅ intercept

	x	y	z
intercept	∞	∞	-1
recip	0	0	-1

$(0\,0\,\bar{1})$



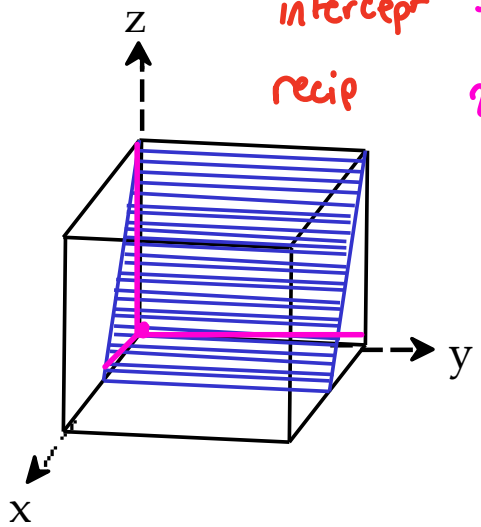
	x	y	z
intercept	∞	1	-1
recip	0	1	-1

$(0\,1\,\bar{1})$

equivalent

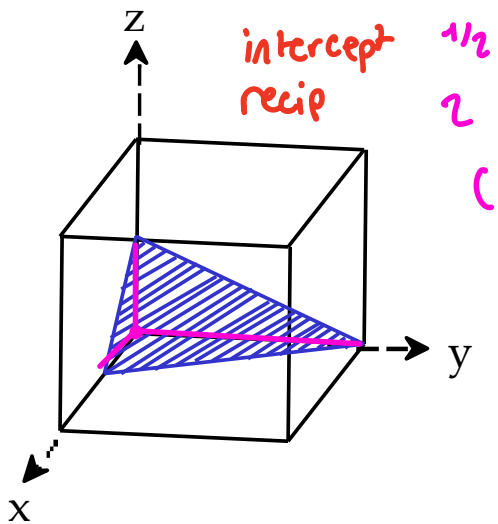
	x	y	z
intercept	∞	-1	1
recip	0	-1	1

$(0\bar{1}1)$



	x	y	z
intercept	1/2	∞	1
recip	2	0	1

(201)



	x	y	z
intercept	1/2	1	1/2
recip	2	1	2

(212)

Next time:

Linear and Planner Densities

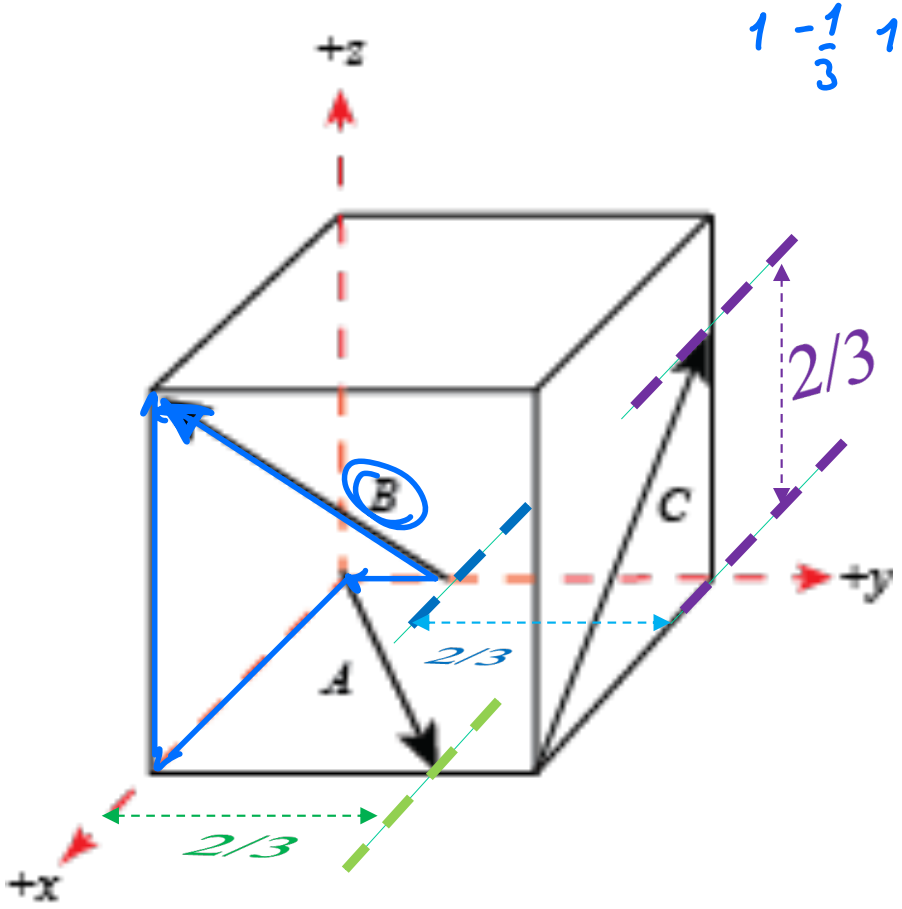
Outline:

- Atomic Densities
  - Linear Density
  - Planar Density
- Single- vs poly- crystalline materials
- X-ray Diffraction
- Example
- Polymorphism and Allotropy

Review - Directions

Use the QR code shown in the classroom. This one won't work.

Which direction is $[3\bar{1}3]$



Atomic Densities

- Linear Density
  - Number of atoms per length whose centers lie on the direction vector for a specific crystallographic direction.

→ atom nucleus (don't count if not through center)

Linear Density = $\frac{\text{Number of atoms centered on direction vector}}{\text{length of direction vector}}$

- Planar Density
  - Number of atoms per unit area that are centered on a particular crystallographic plane.

instead of line, this is AREA

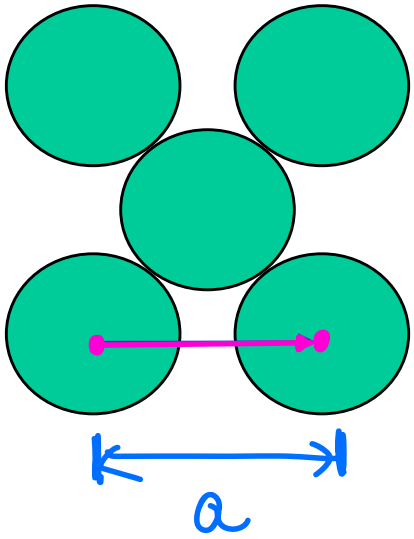
(must go through center)

Planar Density = $\frac{\text{Number of atoms centered on a plane}}{\text{area of the plane}}$

Linear Density

- Calculate the linear density of the [100] direction for the FCC crystal

atom center passed / length = linear ρ



$(L_D = n/L_L)$ linear density

$n = 1$ atoms

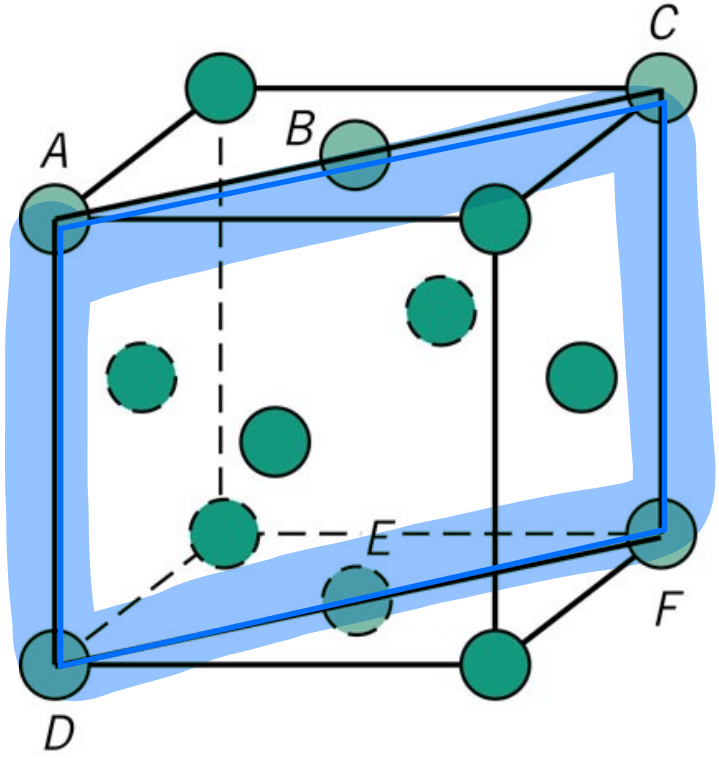
$L_L = a$ line length

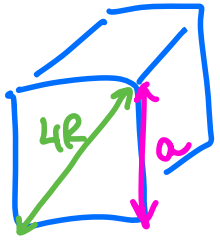
For FCC $a = 2R\sqrt{2}$

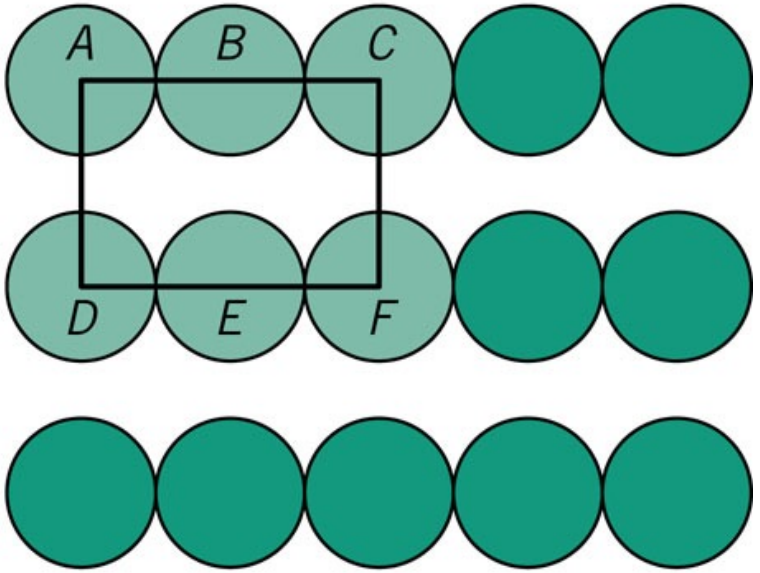
$L_D = 1/(2R\sqrt{2})$

Planar Density

- Calculate the planar density of the (110) plane for the FCC crystal



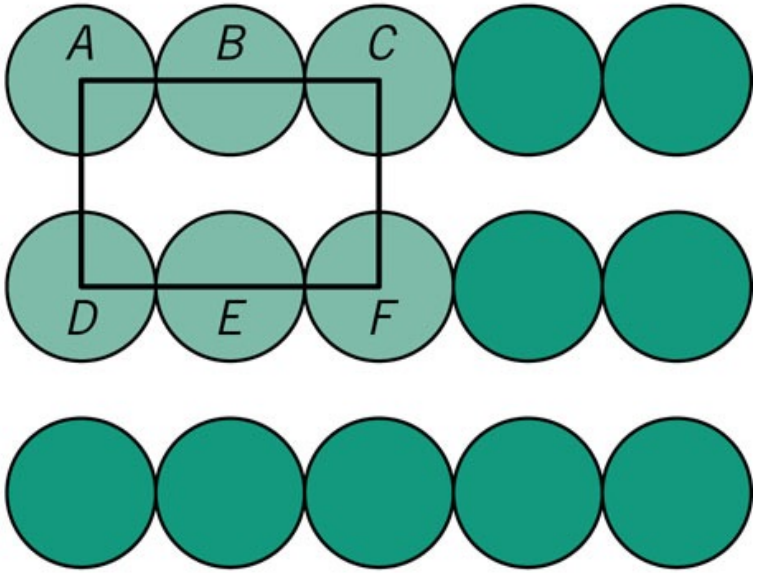




- Compute planar area
- Compute number of atoms

For an atom to be counted, it has to be centered on that plane.

Planar Density



Plane area $A_P = (AC)\times(AD)$

$AC = 4R$

$AD = 2R\sqrt{2}$

$A_P = (4R)(2R\sqrt{2}) = 8R^2\sqrt{2}$

Number of atoms = 2

$$PD = \frac{n}{A_P} = \frac{1}{4R^2\sqrt{2}}$$

Linear and Planar Density

- Why do we care?
- Properties, in general, depend on linear and planar density.
- Examples:
  - Speed of sound along directions
  - Slip (…… *in metals*) depends on linear and planar density
  - Slip occurs on planes that have the greatest density of atoms in direction with highest density

    we would say along closest packed directions on the closest packed planes

Crystals As Building Blocks

- *Some* engineering applications require single crystals:

Single crystal

(Courtesy P.M. Anderson)

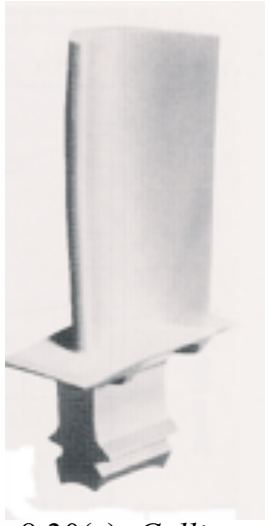


Fig. 8.30(c), *Callister 6e.*
(courtesy of Pratt and Whitney)

- Turbine blades
- Nickel alloy – single crystal
- to improve high temp. mechanical properties

(Courtesy GE Superabrasives)

diamond single crystals for abrasives

- *Most* engineering materials are polycrystals

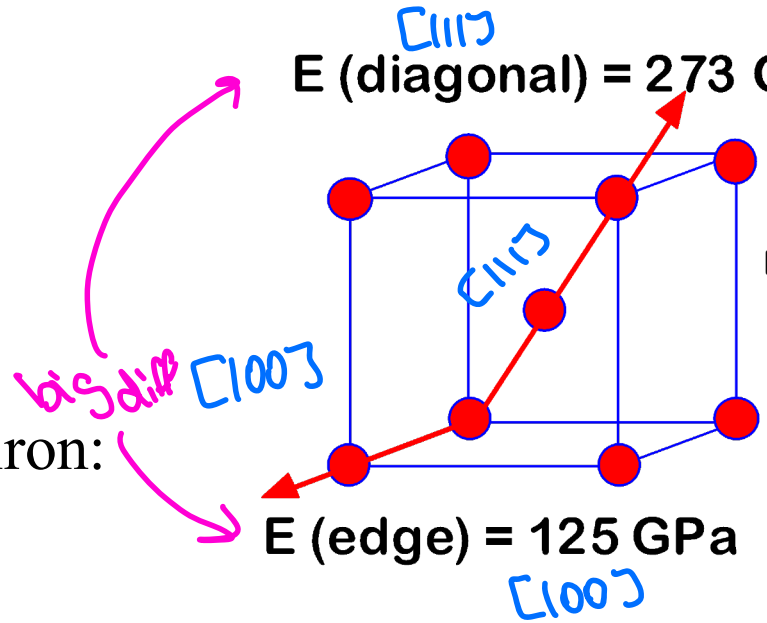
diff colours = diff orientation

- Each "grain" is a single crystal.
- If crystals are randomly oriented, overall component properties are not directional.
- Crystal sizes typically range from 1 nm to 2 cm (i.e., *from a few to millions of atomic layers*).

Nb-Hf-W plate with an electron beam weld

Single Vs Polycrystals

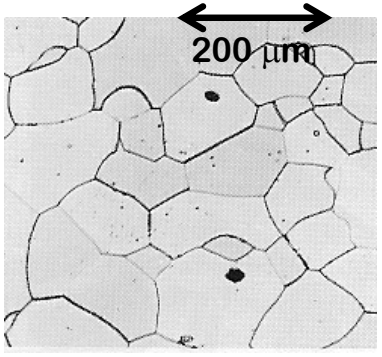
- Single Crystals
  - Properties vary with direction: anisotropic
  - Example: the modulus of elasticity (E) in BCC iron:



Data from Table 3.3, *Callister 6e.*

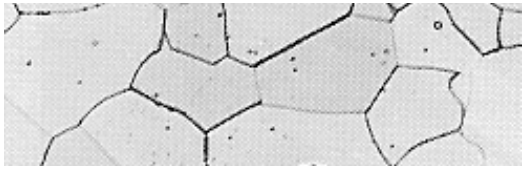
- Polycrystals
  - Properties may/may not vary with direction.
  - If grains are randomly oriented: isotropic
    ($E_{poly\ iron}$ = 210 GPa)
  - If grains are textured, anisotropic.

grain orientation



Adapted from Fig. 4.12(b), *Callister 6e.*

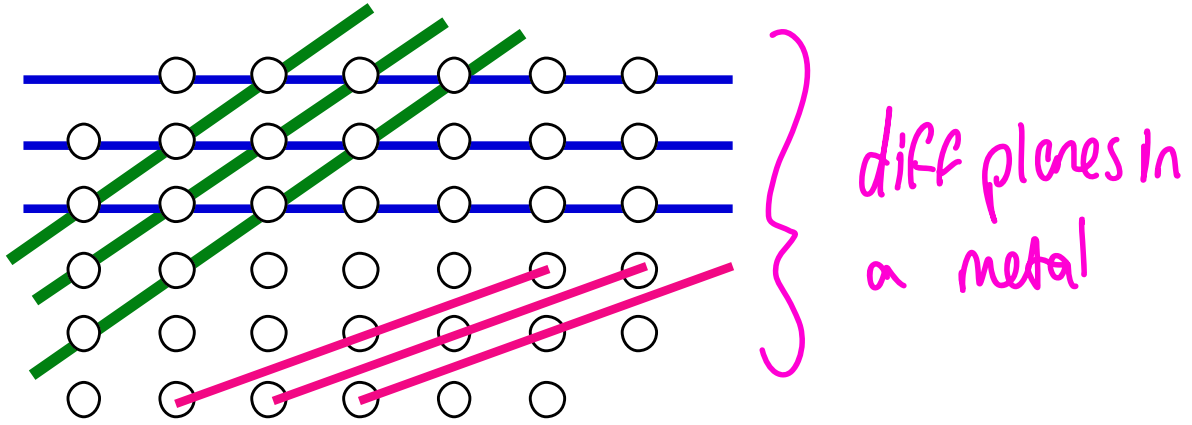
hammered down → grains change



Textured grains

Inter-Planar Spacing & X-Ray Diffraction

2D



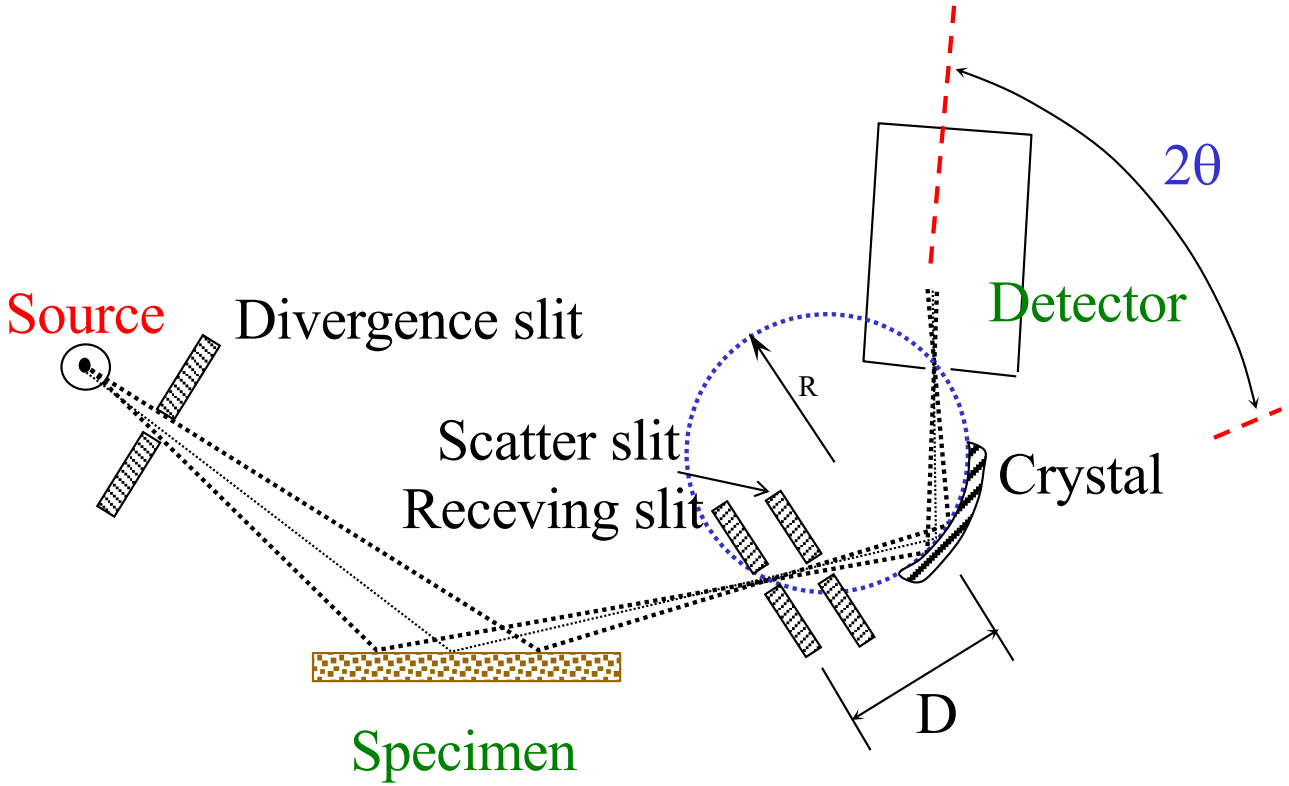
- Inter-planar spacing
  - The inter-planar spacing in a particular direction is the distance between equivalent planes of atoms
- The existence of planes and the distances between different sets of them in a material are characteristic for each material

  ➔Hence, inter-planar spacings are measured by x-ray diffraction to identify unknown materials!

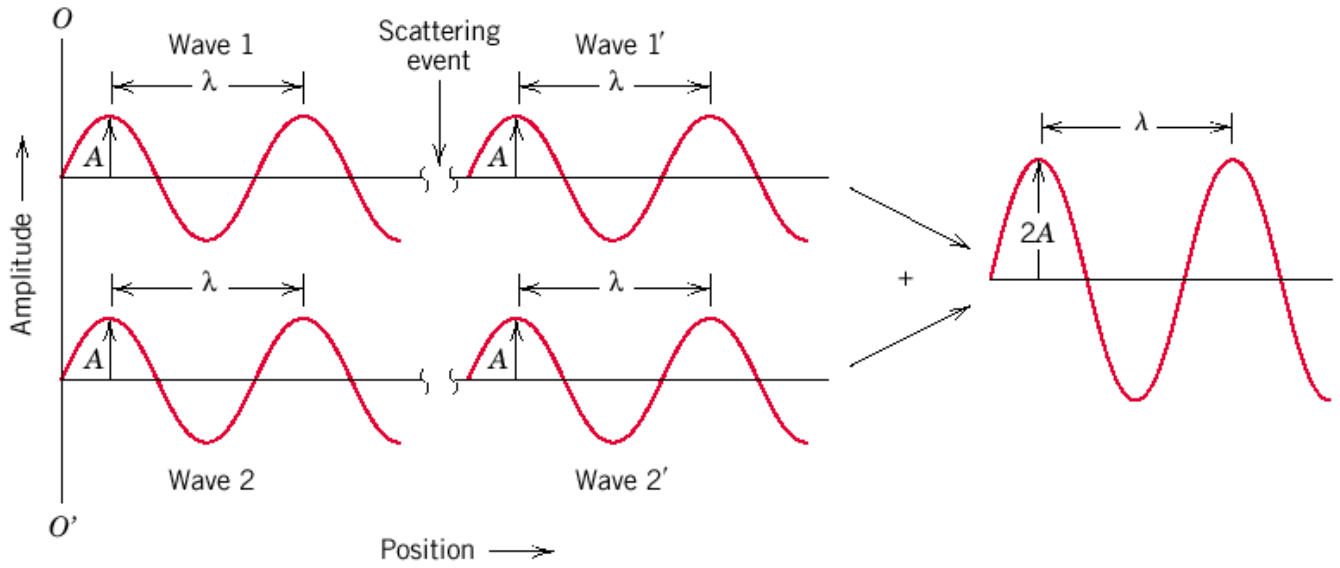
X-Ray Diffraction

- Can be used to determine crystal structure
  - identify unknown materials
  - measure lattice parameters
- X-rays are a form of electromagnetic radiation that have high energies and short wavelengths.
- Diffraction occurs whenever a wave encounters a series of regularly spaced obstacles that;
    - Can scatter the wave
    - Have a spacing comparable to the wavelength
- X-ray wavelength (λ) ~ inter-atomic spacing.
- Other techniques such as neutron or electron diffraction, also, can be used.

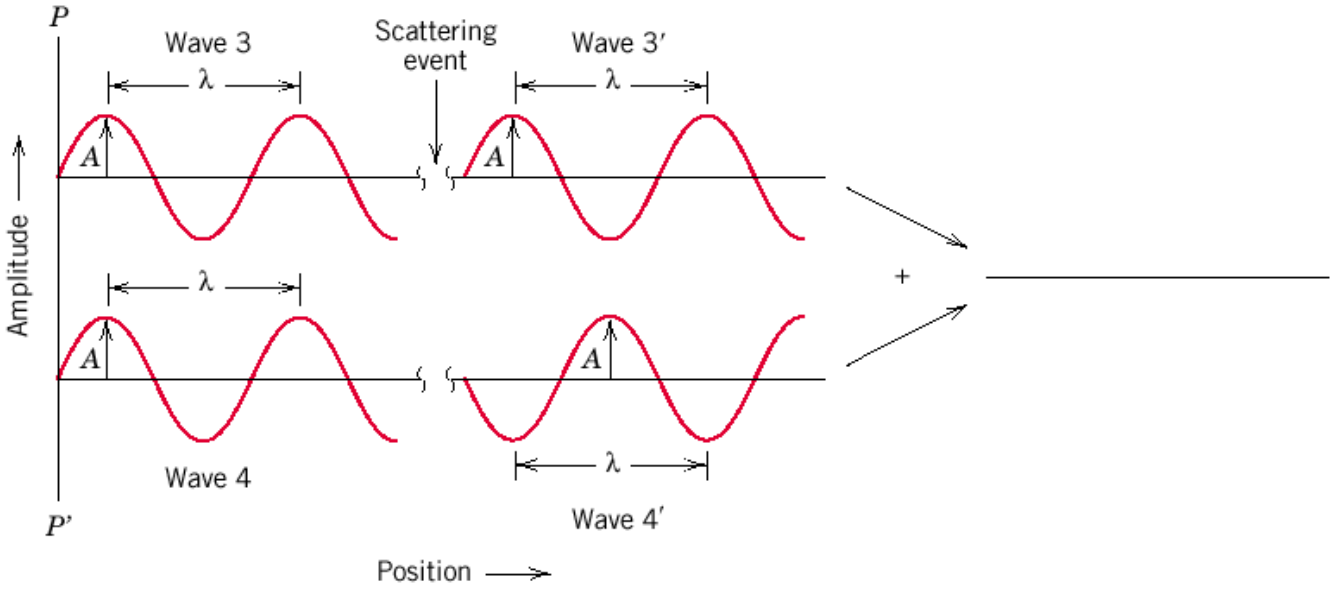
X-ray Diffractometer



Constructive & Destructive Interference

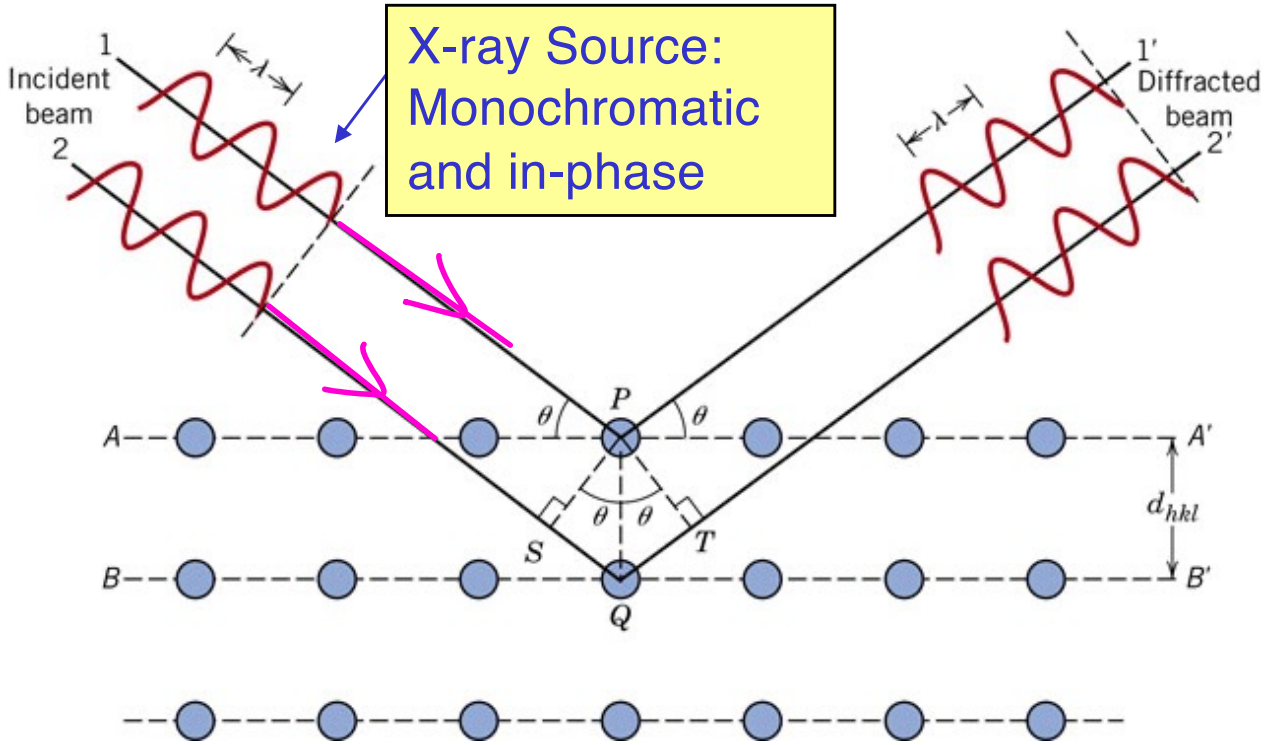


constructive



destructive

Bragg's Law



The law: For constructive interference, the additional path length SQ+QT must be an integer number of wavelength (λ).

$$n\lambda = SQ + QT = d_{hkl}\sin\theta + d_{hkl}\sin\theta = 2\, d_{hkl}\sin\theta = n\lambda$$

n = 1,2,3…order of reflection

Bragg's Law

$$n\lambda = d_{hkl}\sin\theta + d_{hkl}\sin\theta$$

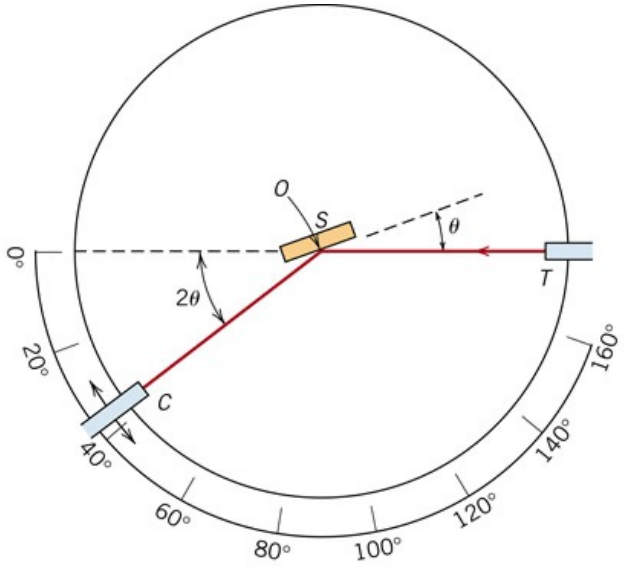
$$n\lambda = 2d_{hkl}\sin\theta$$

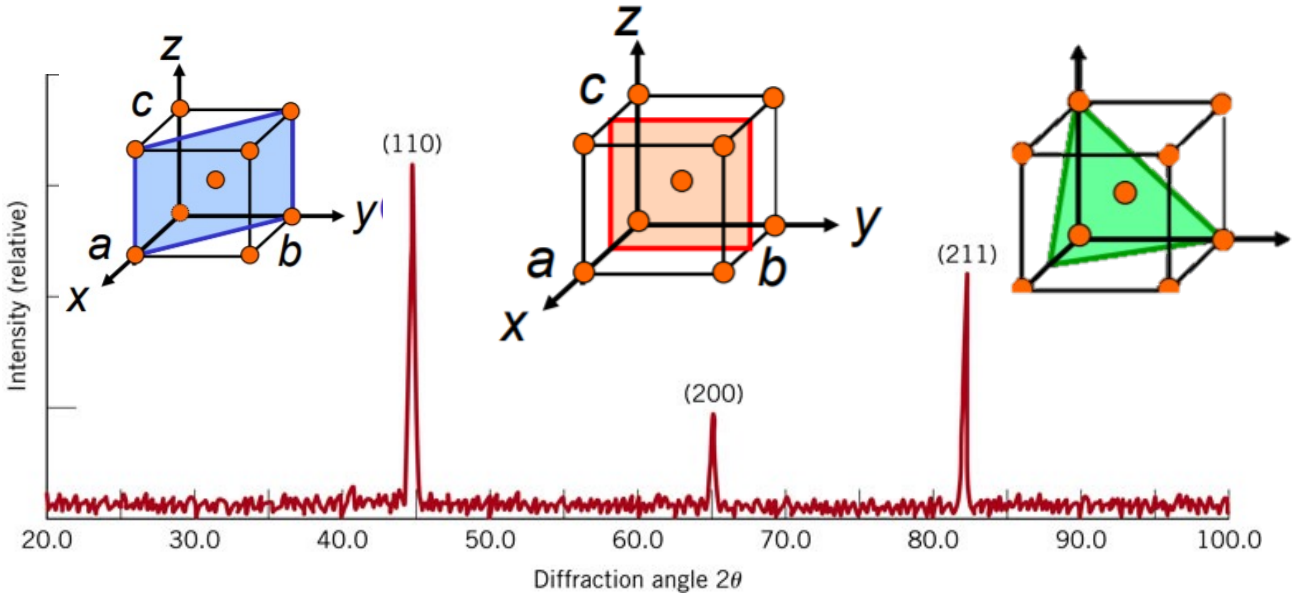
- We have a simple expression relating the x-ray wavelength and interatomic spacing to the angle of the diffracted beam.
- If **Bragg's law** is not satisfied, then the interference will be nonconstructive in nature so as to yield a very low-intensity diffracted beam.
- Magnitude of difference between two adjacent and parallel planes of atoms is function of Miller Indices and the lattice parameter. For cubic symmetry:

$$d_{hkl} = \frac{a}{\sqrt{h^2 + k^2 + l^2}}$$

Diffractometer Technique

- Use powder (or polycrystalline) sample to guarantee some particles will be oriented properly such that every possible set of crystallographic planes will be available for diffraction.





Each material has a unique set of planar distances and extinctions, making X-ray diffraction useful in analysis of an unknown material.

Example

For BCC iron (Fe), compute: (220)

(a) the interplanar spacing of the (220) planes, and

(b) the diffraction angle at which the (220) reflection occurs.

- Given: *lattice parameter of Fe, a = 0.2866 nm;*
- *X-ray wavelength, λ = 0.1790 nm;*
- *first-order reflection (n = 1).* ('if n isn't given then it's 1')

$$d_{hkl} = \frac{a}{\sqrt{h^2 + k^2 + l^2}} \quad \Rightarrow \quad d_{hkl} = \frac{0.2866\,nm}{\sqrt{2^2 + 2^2 + 0^2}} = 0.1013\,nm$$

(2 2 0)

$$\sin\theta = \frac{n\lambda}{2 d_{hkl}} \quad \Rightarrow \theta = \sin^{-1}\left(\frac{1(0.1790nm)}{2(0.1013nm)}\right) \Rightarrow \theta = 62.13^\circ \qquad 2\theta = 124.26^\circ$$

$\theta = 62.13^\circ$ $\qquad 2\theta = 124.26^\circ$

(usually answer written in 2θ unless told diff)

Polymorphism and Allotropy

- Some materials may have more than one crystal structure depending on temperature and pressure - called *POLYMORPHISM*
- Carbon
  - graphite
  - diamond
- Iron BCC and FCC

Next topic:

Imperfections in Solids

Imperfections in Solids

In this topic we will try to answer the following questions:

- What types of defects arise in solids?
- Are these defects undesirable?
- How do defects affect material properties?
- Can the number and type of defects be varied and controlled?

Imperfections in Solids

There is no such thing as a perfect crystal!

- Thermodynamically "impossible"
- "defects" lower the energy of a crystal & make it more stable
- always have vacancies and impurities, to some extent

Defect does not necessarily imply a bad thing, examples:

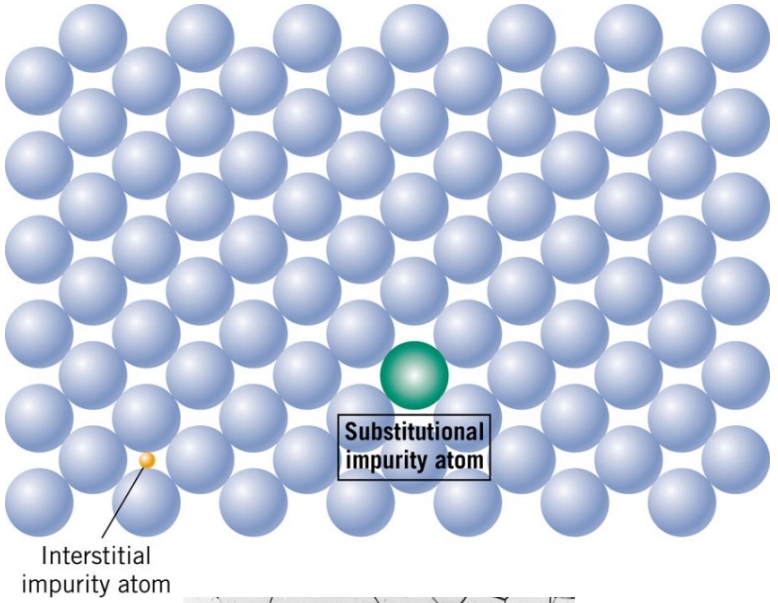
- addition of C to Fe to make steel
- addition of Cu to Ni to make thermocouple wires
- addition of Ge to Si to make thermoelectric materials
- addition of Cr to Fe for corrosion resistance
- introduction of *grain boundaries* to strengthen materials

*"**Defect**" (in this context) can be either desirable or undesirable. In general, a defect simply refers to a disruption in the crystalline order of an otherwise periodic material.*

Types of Imperfections

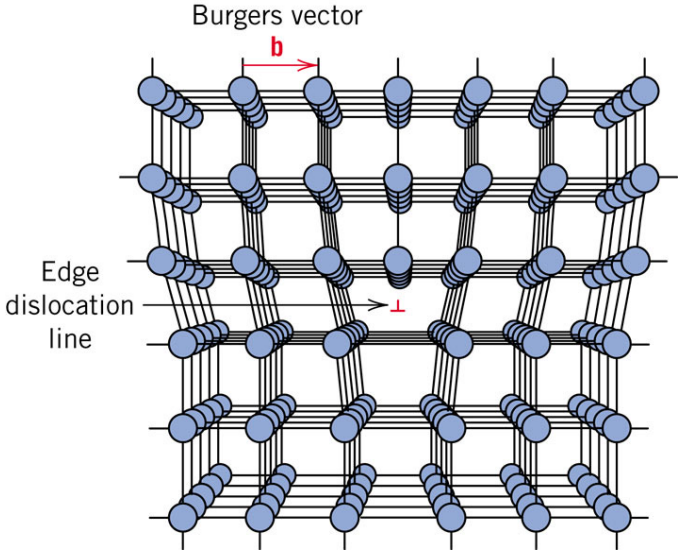
1- Point defects:

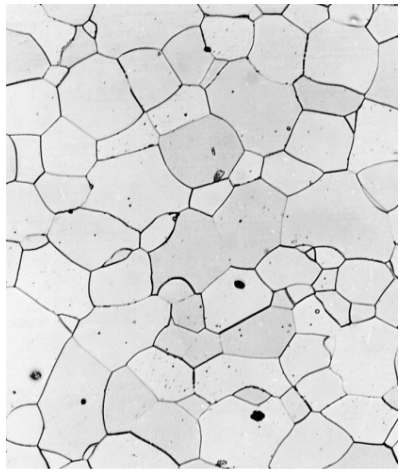
- *Vacancy atoms*
- *Interstitial atoms*
- *Substitutional atoms*



2- Line defects

- *Dislocations*

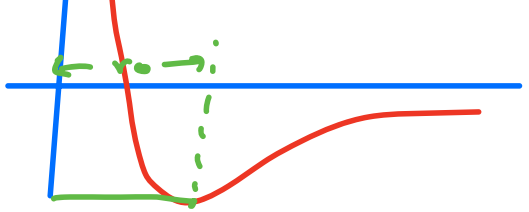




3- Area defects:

- *Grain Boundaries*

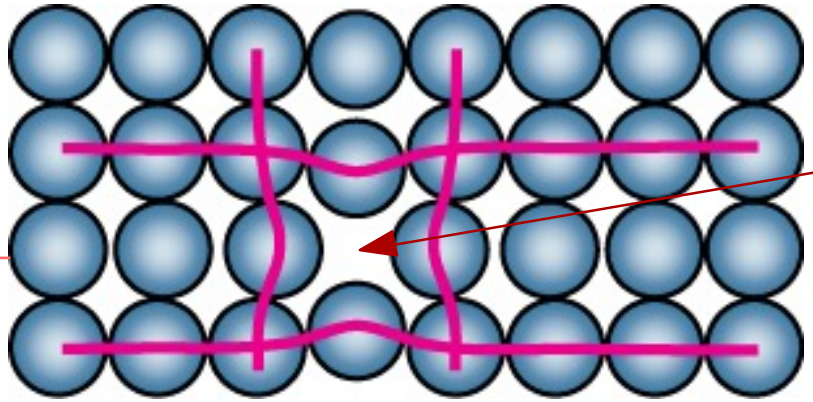
Point Defects



vacancy → a lattice site that is missing an atom:

All crystals contain some vacancies.

distortion of planes



missing atom

Vacancy

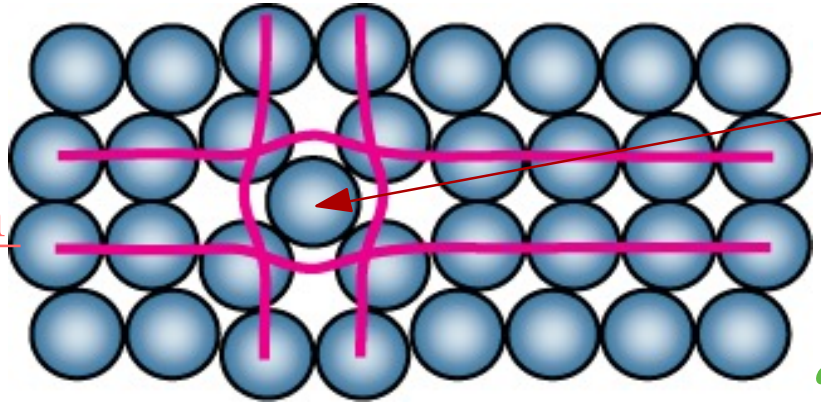
missing atom

self-interstitial → an atom from the crystal that crowds its way into an otherwise empty void between atoms

distorts the planes and almost destroys the order

↑E req.

distortion of planes



self-interstitial

atom squeezed inside

*Self-interstitials are far less common than vacancies because of the relatively large **energy** required to squeeze an atom into the small voids between existing sites.*

Equilibrium Concentration: *Point Defects*

- Equilibrium concentration varies with temperature!

$$\frac{N_D}{N} = \exp\left[\frac{-Q_D}{kT}\right]$$

No. of defects

N: No. of potential defect sites.

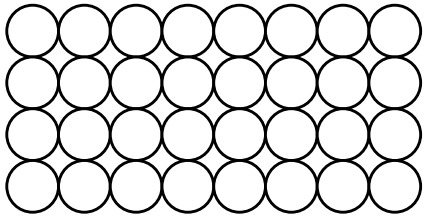
Q_D: Activation energy (Energy required for the activation of the defect)

k: Boltzmann's constant

(1.38 x 10^{-23} J/atom K)

(8.62 x 10^{-5} eV/atom K)

T: Temperature



Each lattice site is a potential vacancy site

This is called an *Arrhenius* equation, after Svante August Arrhenius, a Swedish chemist who won the *Nobel* prize in chemistry in 1903.

Observing Equilibrium Vacancy Concentration

- Increasing temp. causes surface island of atoms to grow.
- Why? ↑T = ↑Area

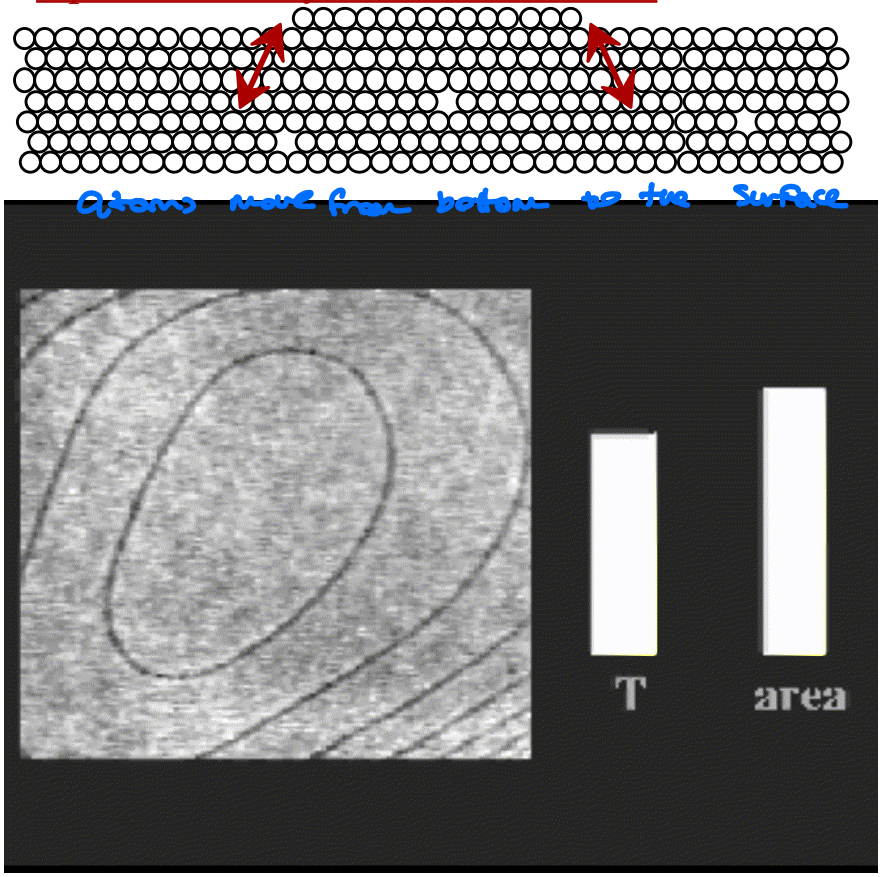
- *Because the equil. vacancy concentration increases via atom motion from the crystal to the surface, where they join the island.*

So, can we observe this?

Yes!

For e.g. the low energy electron microscope view of a (110) surface of NiAl shows how the surface islands grow with temp.

Island grows/shrinks to maintain equil. vancancy conc. in the bulk.

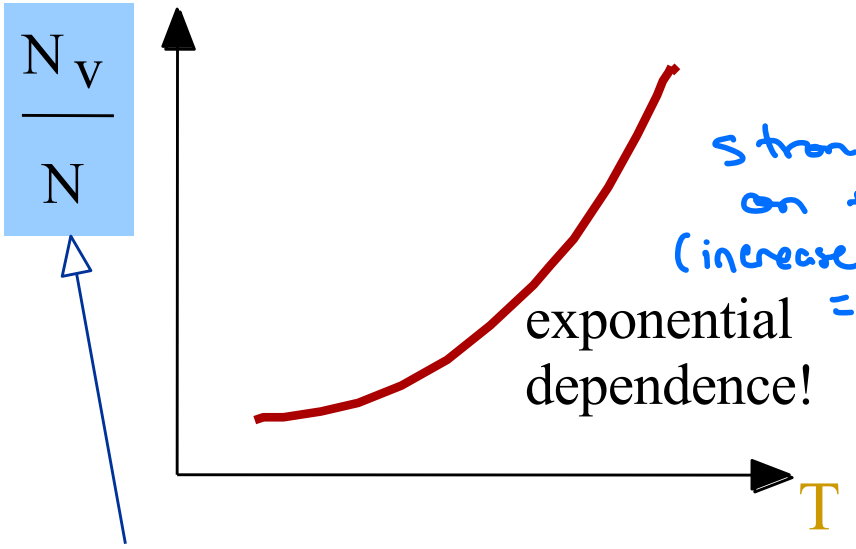


Reprinted with permission from (K.F. McCarty, J.A. Nobel, and N.C. Bartelt, "Vacancies in Solids and the Stability of Surface Morphology", Nature, Vol. 412, pp. 622-625 (2001). Copyright (2001) Macmillan Publishers, Ltd.

Measuring Activation Energy

$$N_v = N_o \exp\left(-\frac{Q}{kT}\right)$$

- We can get Q from an experiment.

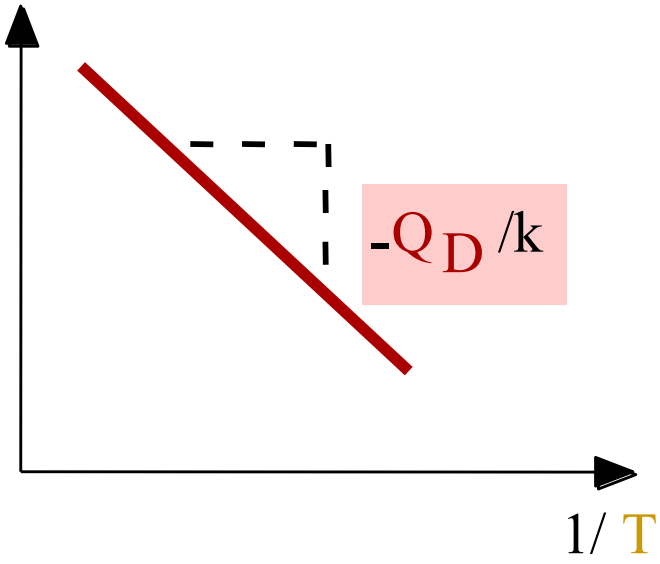


defect concentration

taking logarithms of both sides of the above equation:

$$\ln\left[\frac{N_v}{N_o}\right] = -\frac{Q}{k}\frac{1}{T}$$

plot $\ln\left[\frac{N_v}{N_o}\right]$ versus $\frac{1}{T}$:



Note: exp(-Q/kT) *is a ……. function of temperature*

Example:

Calculate the fraction of atom sites that are vacant for lead at its melting temperature of 327°C (600 K). Assume an activation energy of 0.55 eV/atom.

*To compute the fraction of atom sites that are vacant in lead at 600 K, we must employ **Arrhenius** Equation (eqn. 4.1).*

As stated in the problem, Qv = 0.55 eV/atom. Thus,

$$\frac{N_v}{N_0} = \exp\left(-\frac{Q_v}{kT}\right) = \exp\left[-\frac{0.55\ \text{eV/atom}}{(8.62\times10^{-5}\ \text{eV/atom·K})(600\text{K})}\right] = 2.41\times10^{-5}$$

Point Defects

impurity → addition of an atom of a different species than the "host" or matrix

- Alloys – other types of atoms are deliberately added to give the material certain properties
  - *May or may not result in the same crystal structure*
  - *May or may not result in secondary phases*

Example 1: add 1 at.% Sn to Pb

the tin can replace the lead if is a little is added so it keeps the same crystal structure

i.e., of every 100 Pb lattice sites, 1 is occupied by an Sn atom

Result: ..single..phase..alloy

*same **crystal structure** as pure Pb*

Example 2: add 25 at.% Sn to Pb

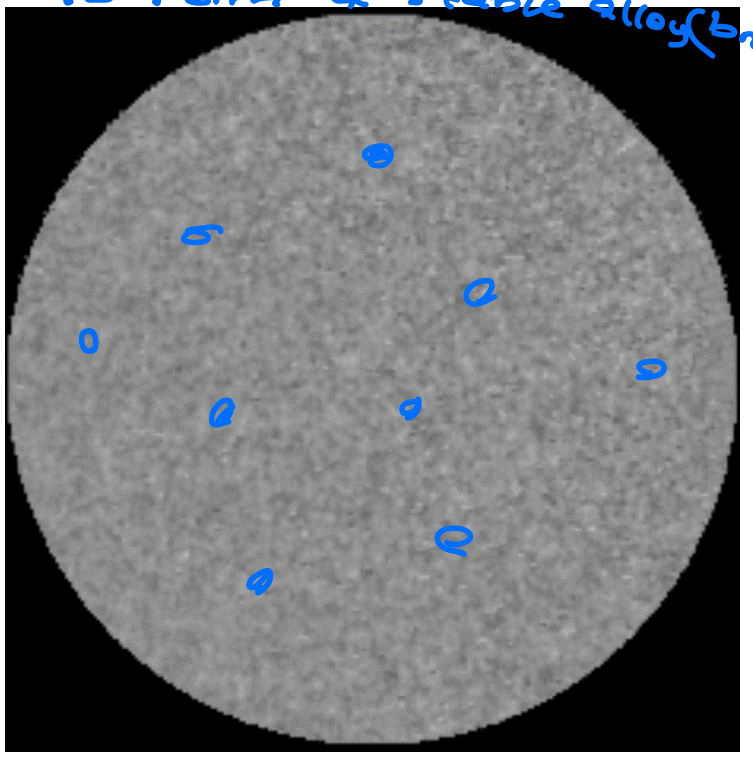
if too much is added it enters second-phase and changes

Result: an alloy with ...two-..phase. microstructure (distinct regions of Sn)

*"**solubility**" of Sn (in the solid state) is exceeded*

Alloying a Surface

- Low energy electron microscope view of a (111) surface of Cu.
- Sn islands move along the surface and alloy the Cu with Sn atoms, to make bronze.
- The islands continually move into unalloyed regions and leave tiny bronze particles in their wake.
- Eventually, the islands disappear.



Reprinted with permission from: A.K. Schmid, N.C. Bartelt, and R.Q. Hwang, "Alloying at Surfaces by the Migration of Reactive Two-Dimensional Islands", Science, Vol. 290, No. 5496, pp. 1561-64 (2000). Field of view is 1.5 μm and the temperature is 290K.

Solid Solution

- A homogeneous distribution of two or more elements.
- "solute" atoms are added without altering the crystal structure or resulting in formation of a new phase.
- Solid solution is a particular type of alloy
- Two types: **substitutional** and **interstitial**

replaces some ↙ Cu in Sn C in Fe ↘ squeezes into pre existing space

"solvent" – the host material, usually the element or compound present in the greatest amount.

"solute" – the minor phase, added to the solvent. Usually the element or compound present in the lower concentration.

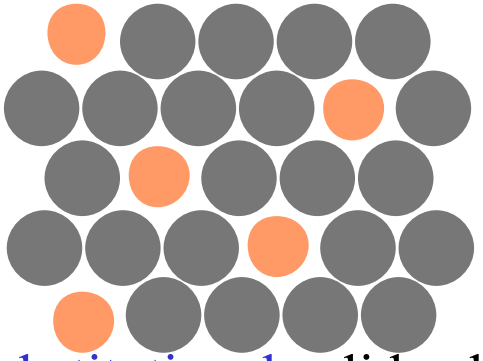
"phase" is a region of uniform composition or crystal structure

What would a solid solution look like?

Point Defects in Alloys

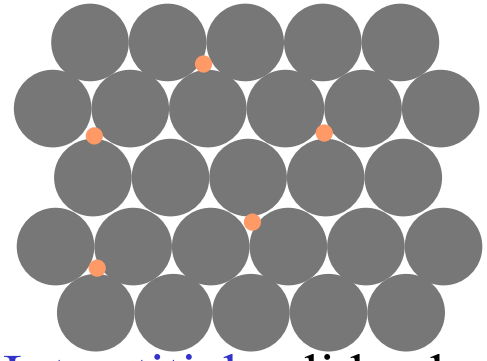
Two outcomes if impurity (B) added to host (A):

- Solid solution of B in A (i.e., random dist. of point defects)



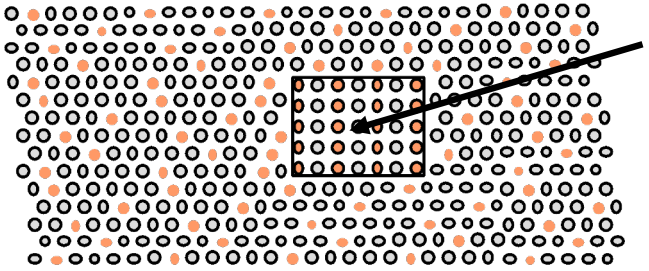
Substitutional solid soln.
(e.g., Cu in Ni)

OR



Interstitial solid soln.
(e.g., C in Fe)

- Solid solution of B in A plus particles of a new phase (usually for a larger amount of B)



Second phase particle
- *different* composition
- *often different* ..structure...

Solid Solution

Not all elements form solid solutions. There are very specific rules that govern the extent to which solid solutions can form.

For ***appreciable*** **substitutional** solubility to occur the following factors must hold:

- Atomic size factor

  the difference in atomic radii between the two atoms must be $\leq \pm 15\%$ (within 15%.)

- Electronegativity

  the difference in electronegativity between the two atoms must be $\leq \pm 0.4$ *eV* (9x)

  (i.e. large differences $\rightarrow$ *compound formation (intermetallics))*

- Crystal structure

  The crystal structure of each element must be the same

- Valence

  For a given solvent, a solute with a higher valency is more likely to be soluble than one of lower valency, all other factors being equal.

Composition and Conversion wt% ↔ at.%

Mixtures or alloys of two (or more) elements, are quantitatively described in terms of composition:

Consider a mixture of B in A:

$$\text{weight percent B} = \frac{\text{mass of B}}{\text{total mass of A + B}} \times 100\% = C_B$$

$$\text{atomic percent B} = \frac{\text{\# moles of B}}{\text{total \# moles of A + B}} \times 100\% = C'_B$$

- Conversion between wt % and at% in an A-B alloy:

$$C_B = \frac{C'_B A_B}{C'_A A_A + C'_B A_B} \times 100 \qquad C'_B = \frac{C_B/A_B}{C_A/A_A + C_B/A_B}$$

- Basis for conversion:

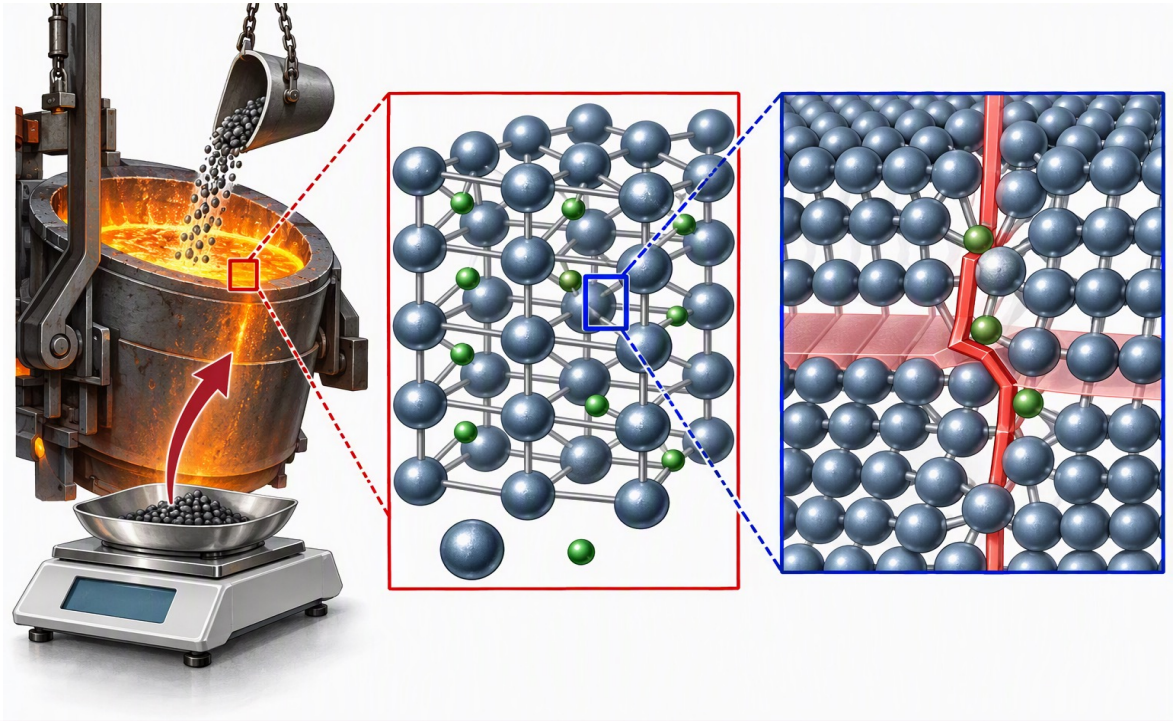
mass of B = moles of B x A_B

mass of A = moles of A x A_A

- *In industry usually use weight % for alloys: easier to add (x) kgs of Mg to (Y) tonnes of Al.*
- *For chemical/scientific measures often need **atomic %***

Why convert wt% to at.%?

- We've understood where solute atoms go.
- We now know how to convert a production recipe into that atomic percentage.



Example
0.80 wt% C steel

Take a 100 g basis:

C: 0.80 g ÷ 12.01 = 0.0666 mol
Fe: 99.20 g ÷ 55.85 = 1.776 mol

at.% C = 0.0666/(0.0666+1.776) × 100

≈ 3.6 at.% C

Meaning: about 3.6 of every 100 atoms are C, mostly in interstitial sites.

- For PROCESSING → weigh alloy additions (wt%)
- To predict properties → often convert to at.% → identify defects / phases (mostly R&D)

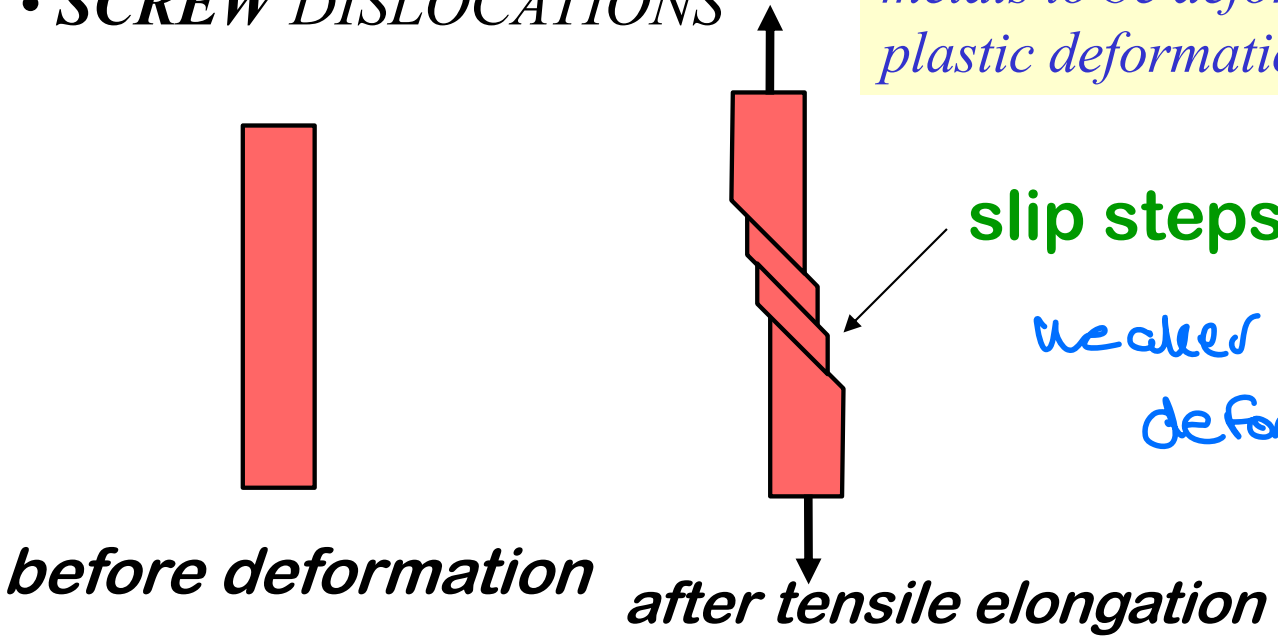
Example: interstitial C distorts the Fe lattice and impedes **dislocation** motion, linking composition to strengthening.

Linear Defects

Dislocations

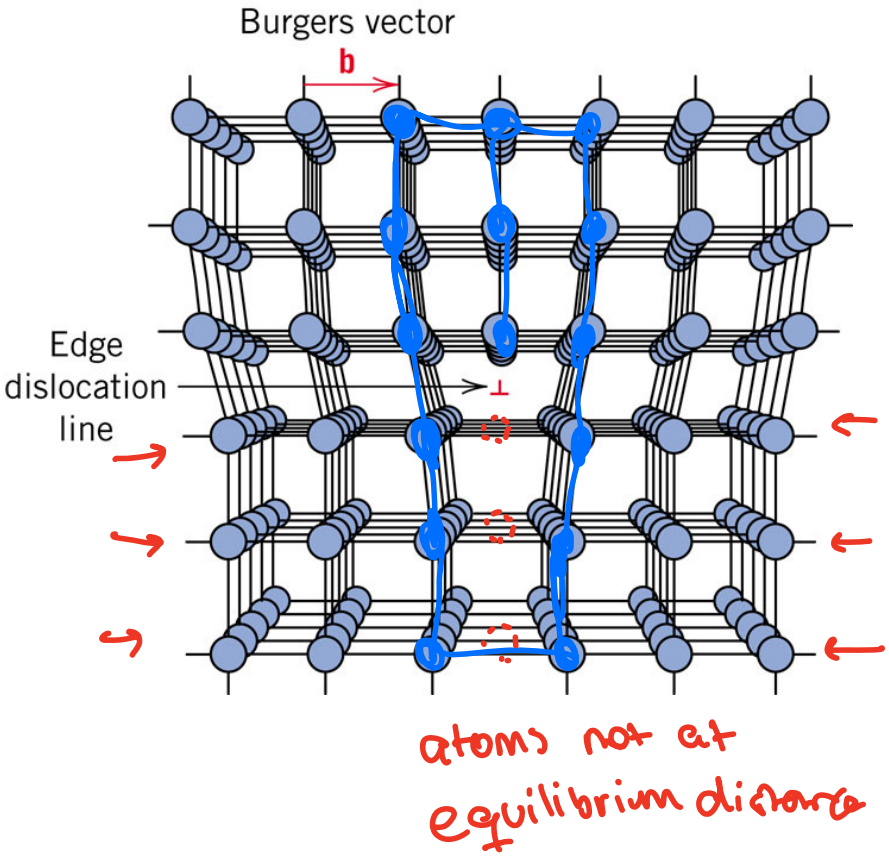
- Dislocations result from solidification from the melt, from mechanical work (*e.g., rolling, drawing, compressive impact, tensile or shear stress*), or from thermal stresses
- It is very difficult to prepare a **dislocation-free** crystal!!!
- 2 Types:
  - ***EDGE*** *DISLOCATIONS*
  - ***SCREW*** *DISLOCATIONS*

Dislocations *make metals weaker than they should be, BUT also allow metals to be deformed (ie. allow plastic deformation). (Chp. 6)*



Edge Dislocation

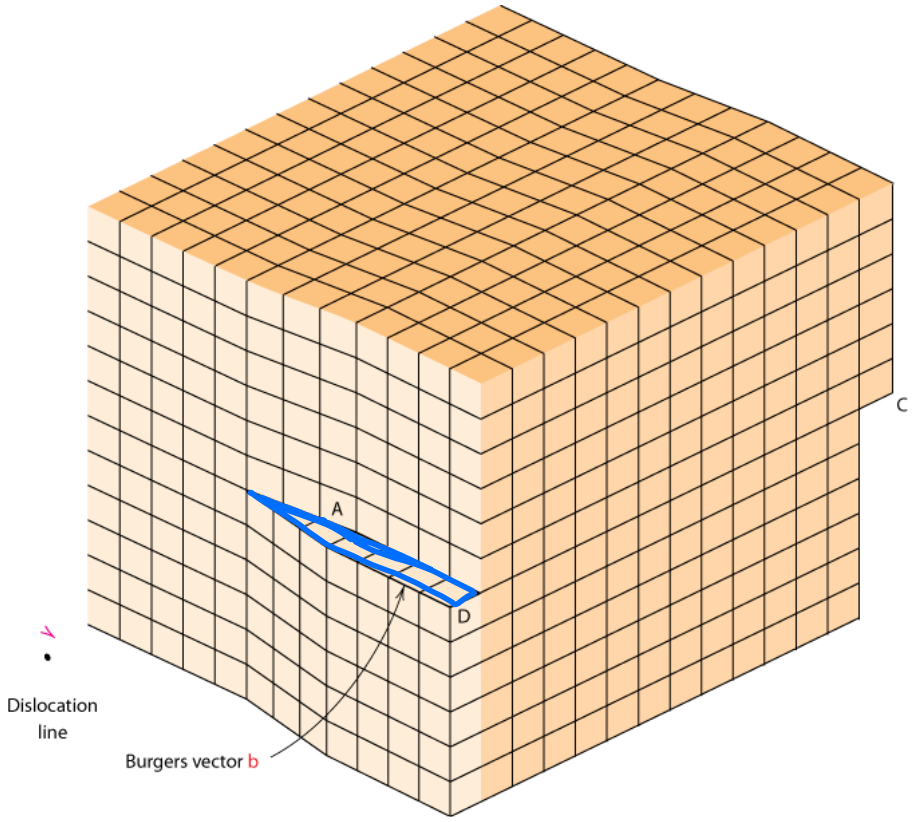
- *Think of edge dislocation as an extra half-plane of atoms inserted in a crystal.*
- Misalignment of atomic planes due to the extra half plane.

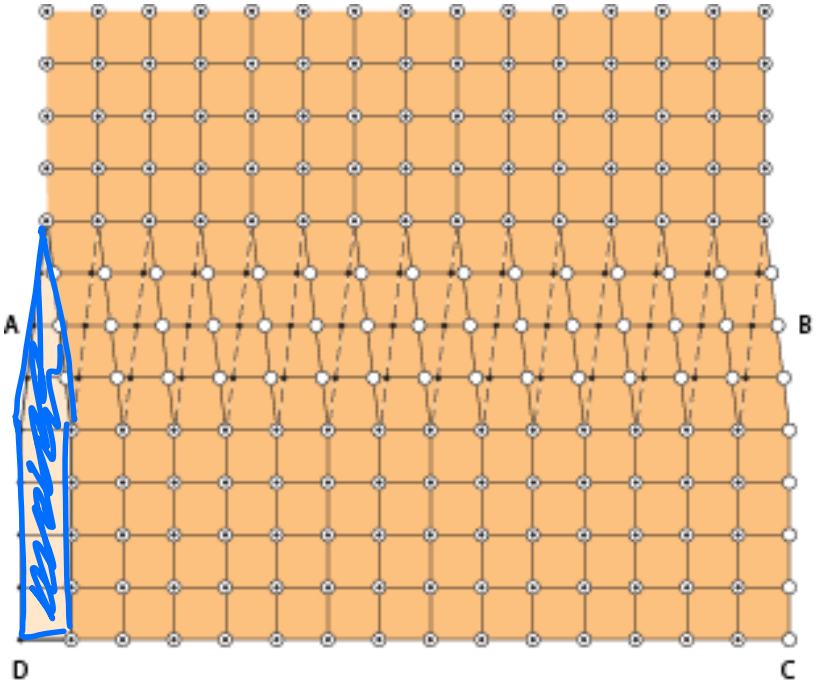


Burger's vector (b) = magnitude + direction of lattice distortion.

Screw Dislocation

can't exist w/out edge

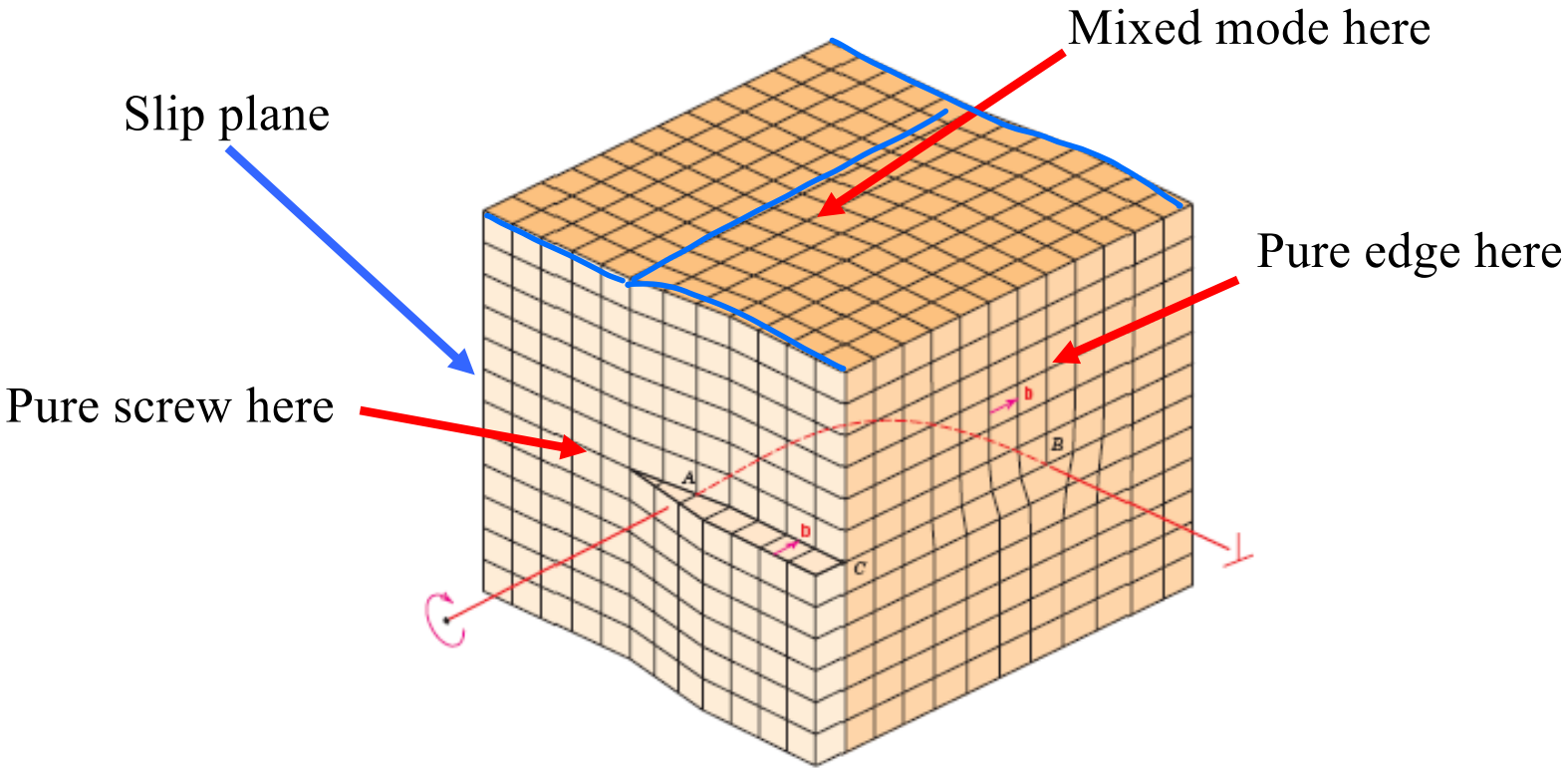




- *Crystal* is "cut halfway through and then slide sideways“
  helical path through structure hence “screw”.
- The motion of a screw dislocation can be thought of in terms of tearing a sheet of paper.

Dislocations

Usually, dislocations have both an edge and a screw character; i.e., they are ..mixed.... dislocations:



Summary

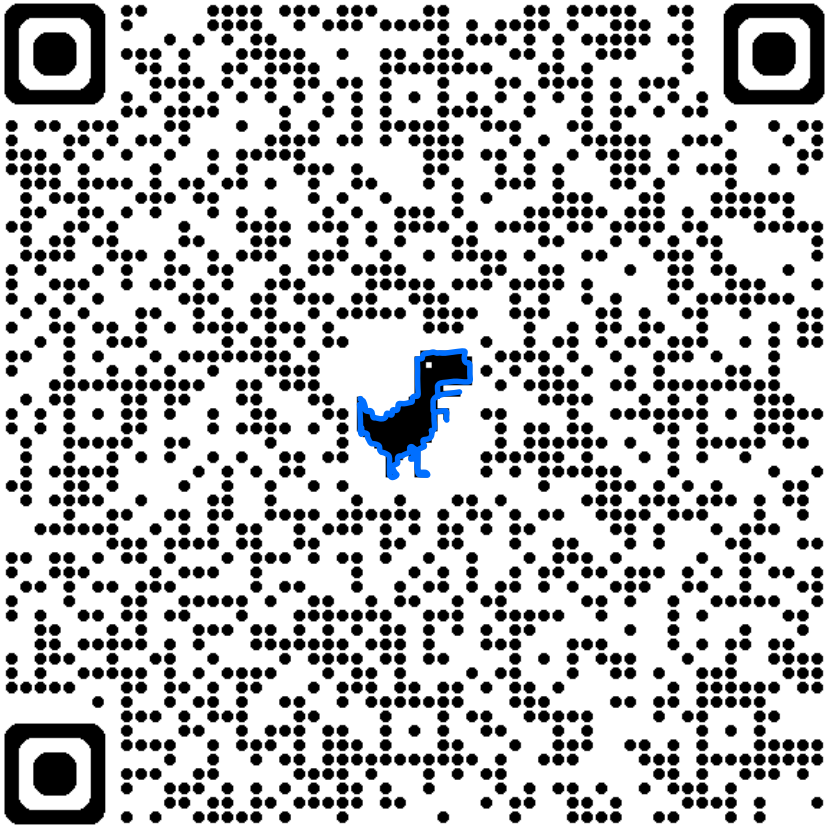
Which statement is MOST correct regarding defects in solids?

A. Defects may be either desirable or undesirable, and may be either intentionally introduced or naturally occurring.

B. All defects are undesirable because they disrupt the crystal structure.

C. Desirable defects are always intentionally introduced.

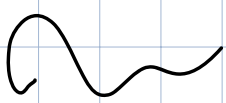
D. Unintentional defects are always harmful.



Use the QR code shown in the classroom. This one won't work.

Next time:

Grain Boundaries and Microstructure

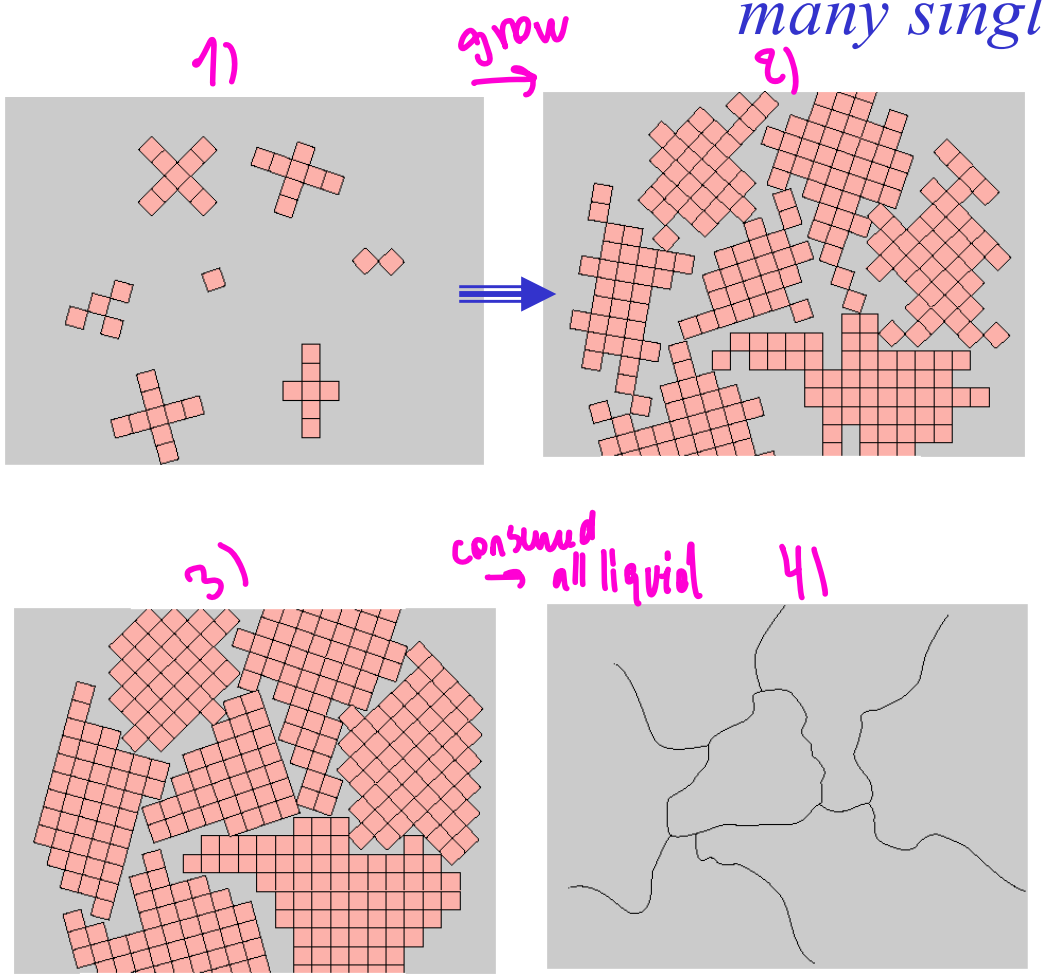


Outline:

- Crystallization
- Grain boundaries
- Grain size determination
- Stacking faults
- Types of microscopes
  - *LOM*
  - *SEM*
  - *TEM*
  - *SPM*

Polycrystalline Materials

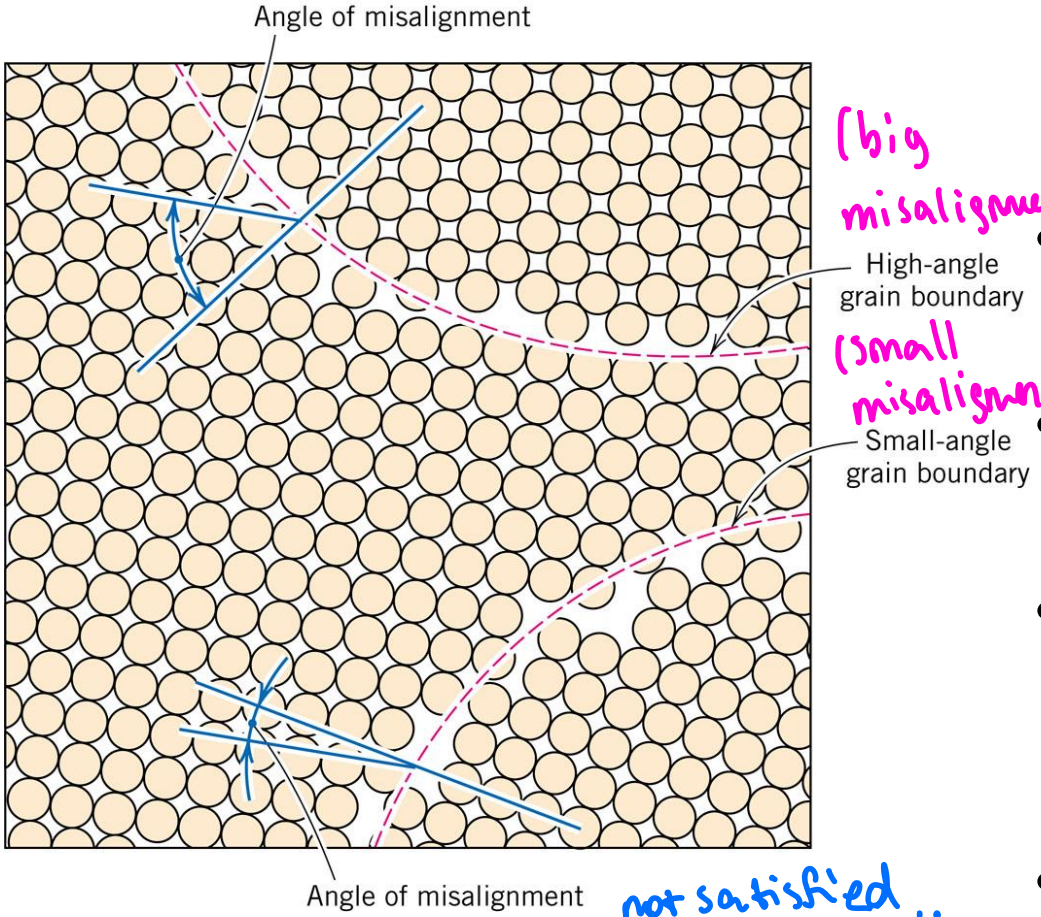
Most materials are polycrystalline and are made of many single crystals



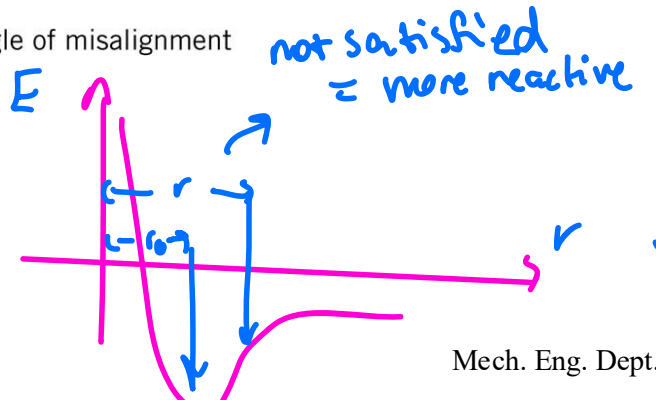
- during *solidification* the crystals nucleate and grow from the liquid in a *random orientation*
- the *grains* impinge on each other when the solidification is complete
- junction of grains are grain boundaries

you can increase grain boundaries by making the grain smaller

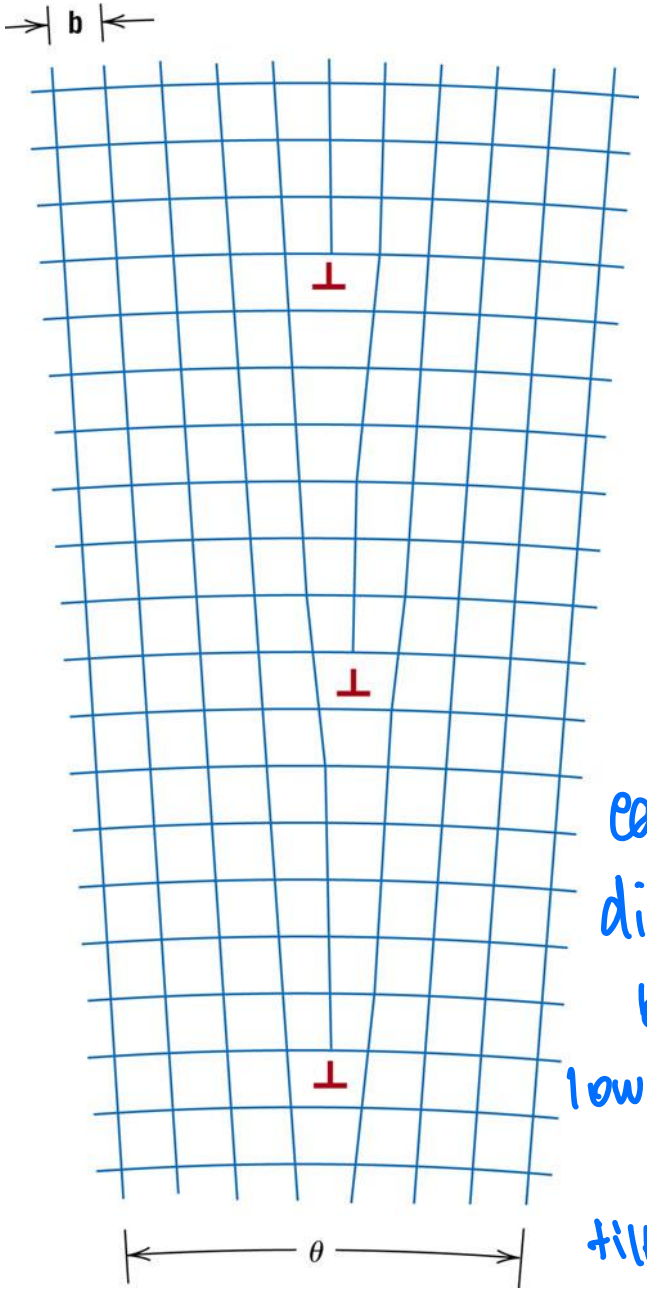
Grain Boundaries



- Occurs due to the crystallographic mismatch when two grains meet
- when mis-orientation is large → ***high angle*** *grain boundary*
- when mis-orientation is small, → ***low angle*** *grain boundary*
- atoms are *less bonded* and the atomic packing is lower than in the grain (*lower coordination*)
- the result is an energy difference → *interfacial surface energy or grain boundary energy*



Grain Boundaries



edge dislocation but w/ low angle grain ⇓ tilt boundary (edge)

- grain boundaries are more chemically reactive
- segregation of impurities due to higher energy
- total grain boundary area smaller in *coarse grained* than *fine grained* material
- low angle grain boundary is described as an array of dislocations
  - *tilt boundary* (edge)
  - *twist boundary* (screw)

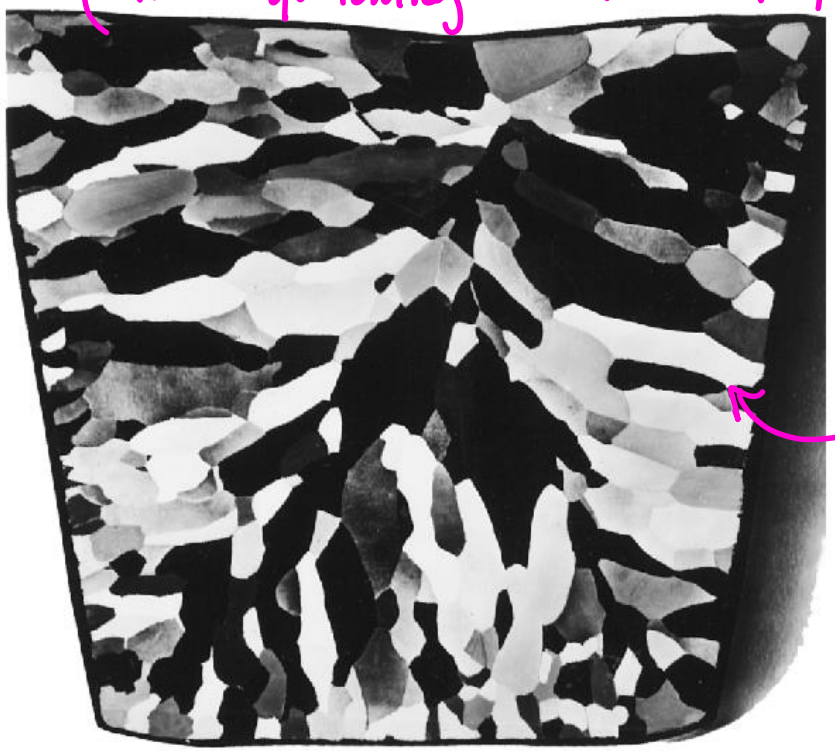
Smaller the grains ⇒ bigger the surface area

like grains of sand in jar

slow = big

Observation of Grain Structure

cooling fast = small grain
(like quenching sword in water)

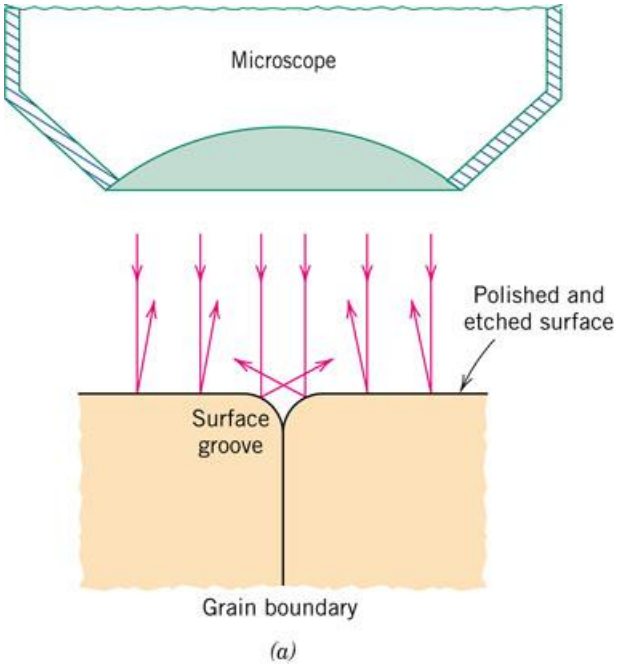


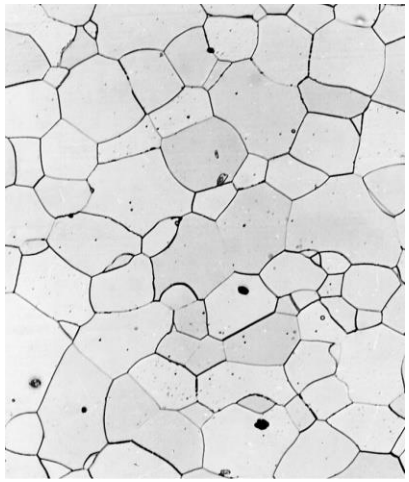
ex of slow cooling

FIGURE 4.10 *High-purity polycrystalline lead ingot in which the individual grains may be discerned.*

- *Macrostructure* can be observed with naked eye
- coarse grains can be revealed this way (e.g. Al streetlight posts, and zinc galvanized garbage cans)
- microstructure ... is when the grains can only be observed with a microscope
- imaged using a camera for archiving
  → *photomicrograph*

Sample Preparation for Microscopy





otherwise you only see scratches, dirt, fingerprints, ect

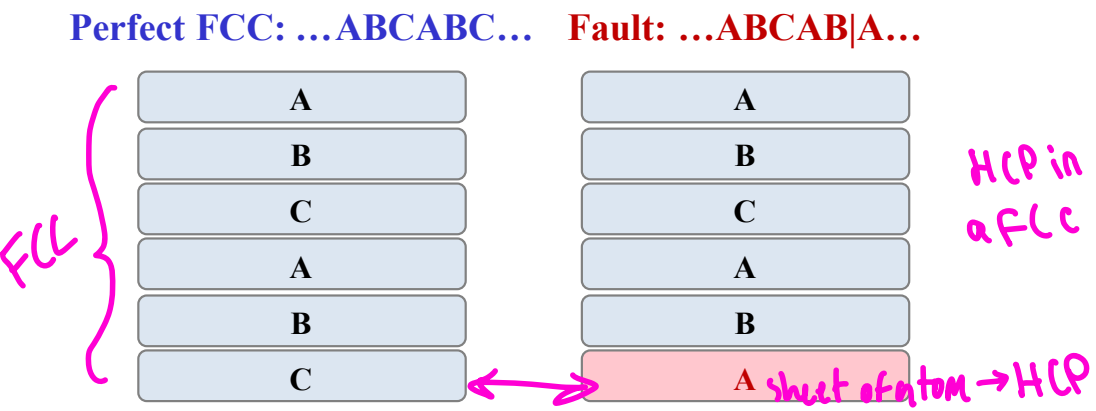
- Preparation requires meticulous *grinding* and *polishing* of the surface
- the microstructure is revealed by attack using …… (chemical reagents)
  - preferential attack of grain boundaries
- effect is that these features scatter the incident light and create **optical contrast**

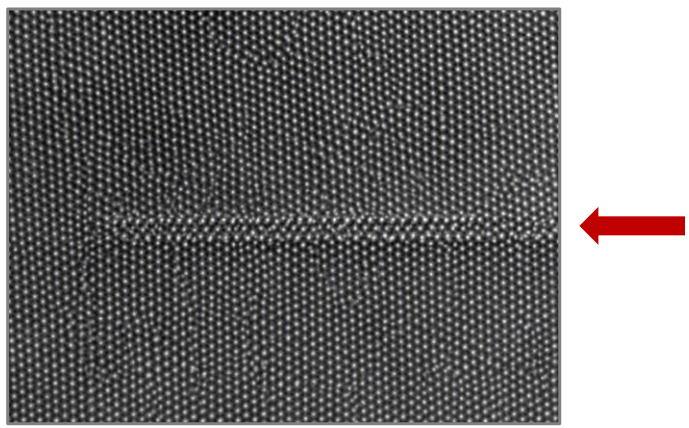
Stacking Faults

- An **area (planar) defect** – a mistake in the stacking sequence of atomic planes.
- Recall: FCC stacks as **ABCABCABC…** and HCP stacks as **ABABAB…**
- A stacking fault is when the order is broken, e.g. **ABCAB**ABC… – one plane (C) is missing.
- The atoms are still in proper positions; only the **order of the planes** is wrong, so a thin HCP-like layer appears inside the FCC crystal.

*Because the mistake is small, its energy is low – stacking faults are common. They form during **solidification and plastic deformation** and affect how easily a metal deforms.*

Compare the stacking sequences:

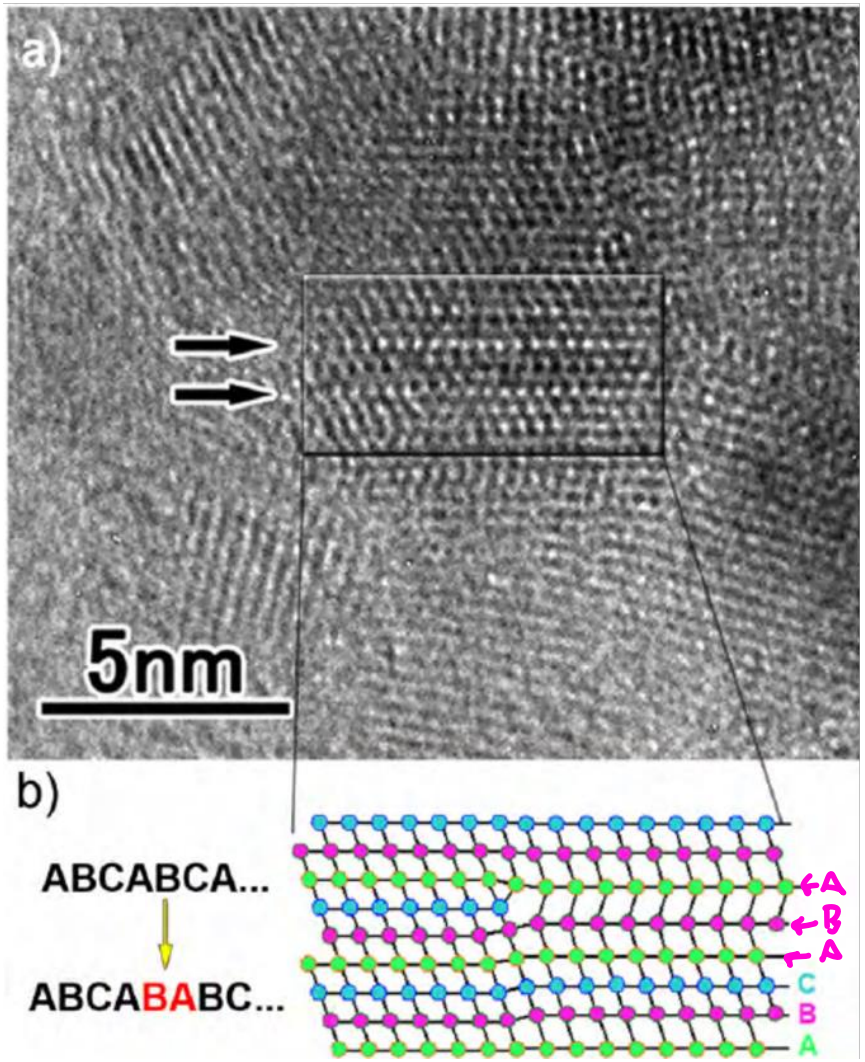




High-resolution TEM: the faulted plane appears as a break in the regular stacking

Stacking Faults

Wurtzite-type ZnS nanoparticles by pulsed electric discharge

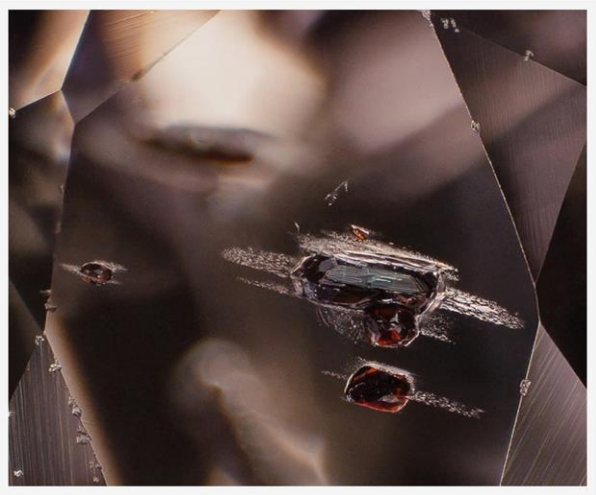


Omurzak et al. (2011) Nanotechnology.

HRTEM image of a stacking fault (a), denoted by the arrows, and (b) schematic illustration of the stacking fault in the crystal shown in (a).

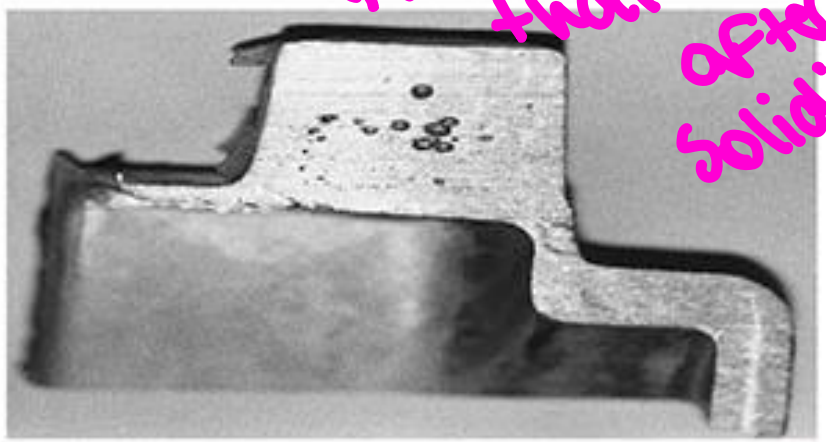
Three-dimensional (*volumetric*) defects

- Inclusions
- precipitates
- Porosities or voids
- Cracks

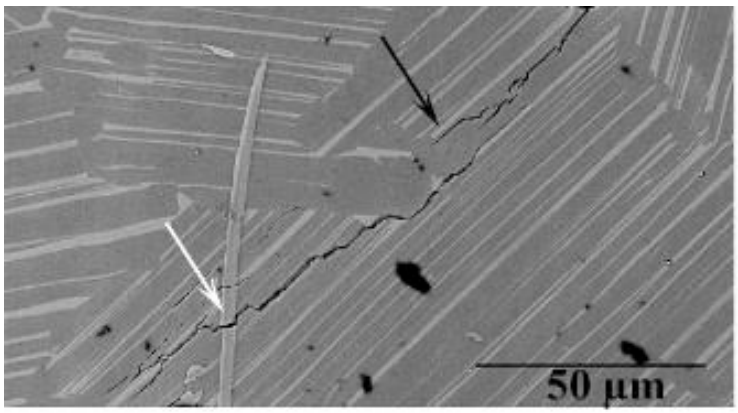


Sapphire crystal containing Rutile inclusions

trapped gas that stayed after solidification



Gas porosity in metallic component

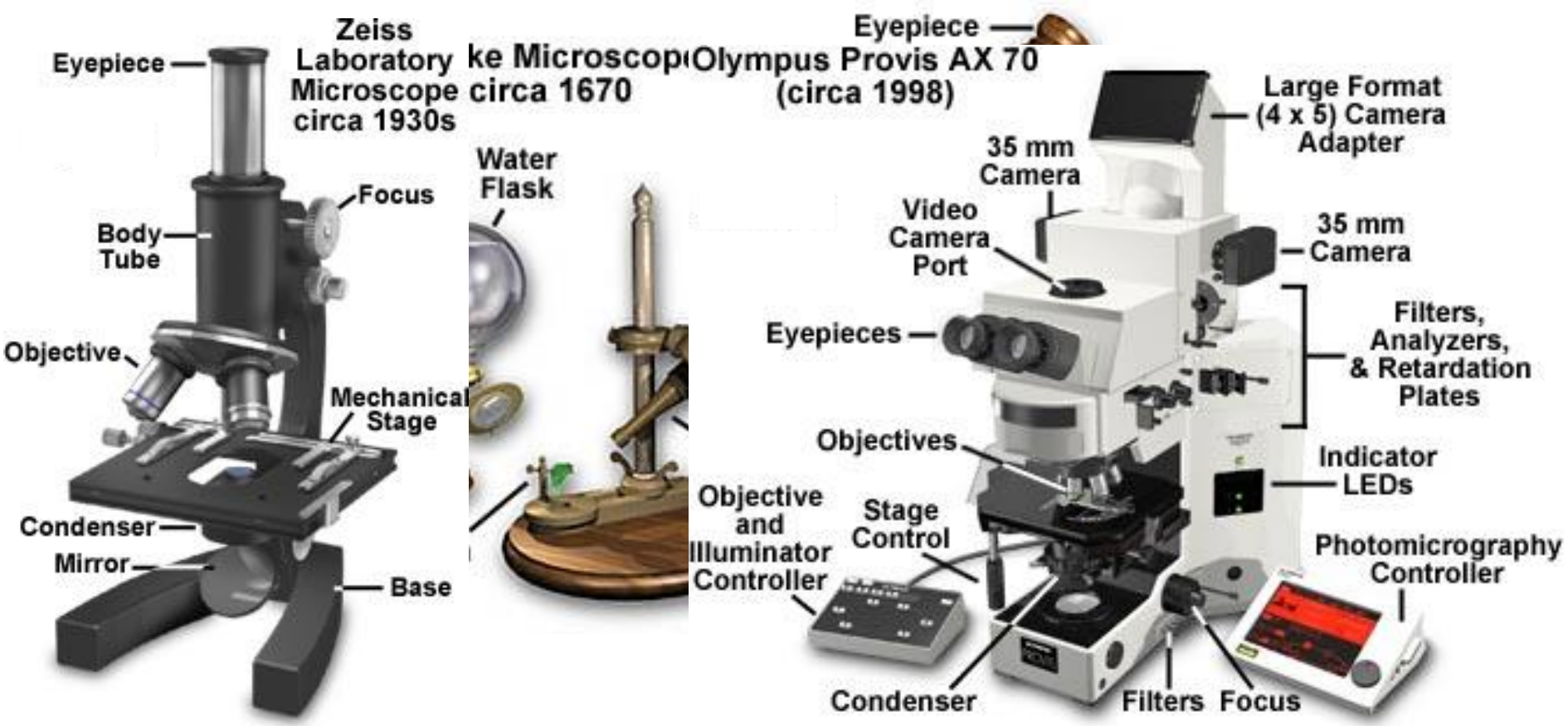


Cracks in titanium aluminide

Types of Microscopy

- *Optical microscopy is limited to x2000 magnification*
- *electron microscopy* uses electron waves (very small $\lambda \sim 0.003$nm) rather than light ($\lambda \sim 500$nm).
- can reveal microstructural features down to atomic scale (x1,000,000)
- scanning electron microscope (…….)
  - sample preparation similar to optical microscopy
  - can be used to observe fracture surface (*fractography*)
- transmission electron microscope (……..)
  - samples are very small
  - requires very thin (*electron transparent*) samples

Optical Microscopy



Review Optical Microscopy

A polished metal sample is placed under an optical microscope, but no grains can be seen — the surface just looks like a uniform bright mirror. After the sample is etched, the grains and their boundaries become clearly visible. What is the best explanation? etching attacks grain boundaries

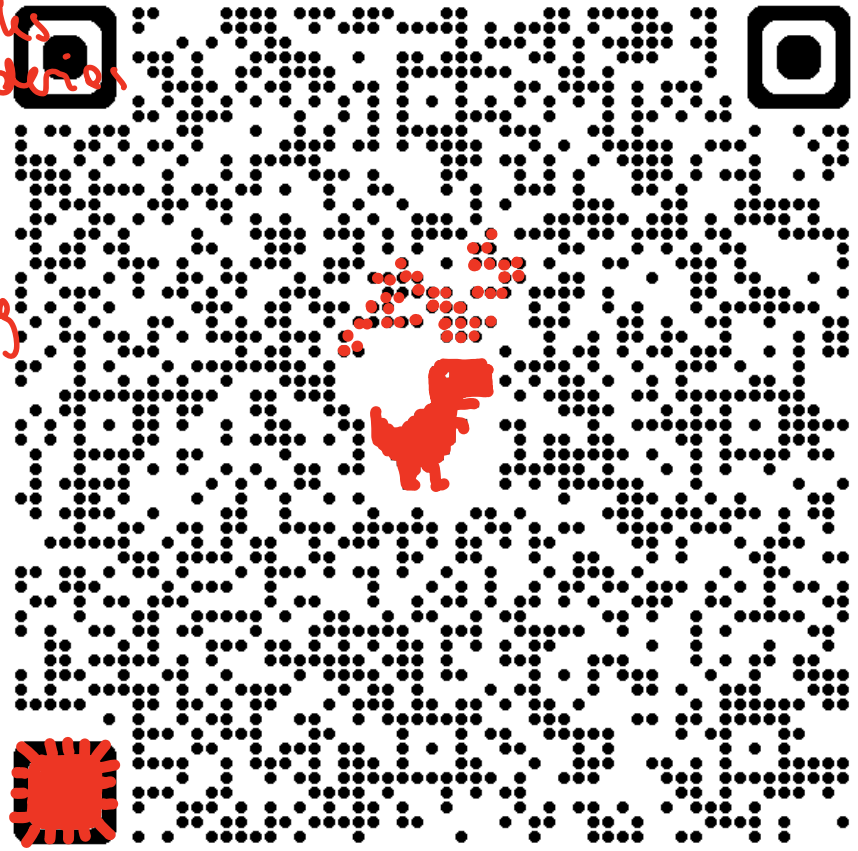
A. Etching increases the magnification so that the grains become large enough to see.

↳ colour doesn't come from etching

B. Etching colours each grain with a different chemical dye, and the optical microscope detects the colours.

C. Etching attacks grain boundaries preferentially, so the surface is no longer uniform and these features scatter the incident light differently, creating contrast.

D. Before etching the sample was a single crystal.



Use the QR code shown in the classroom. This one won't work.

Scanning Electron Microscope (SEM)

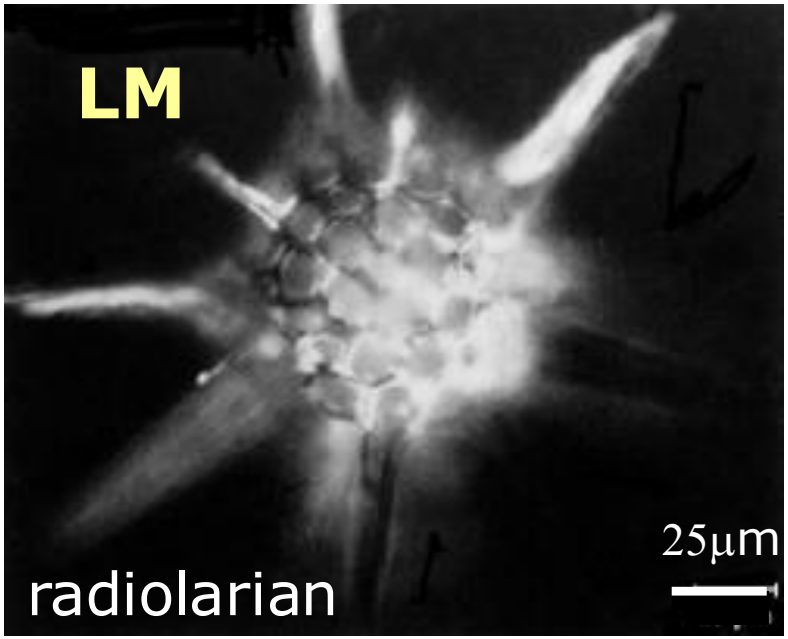
The SEM is designed for direct studying of the surfaces of solid objects

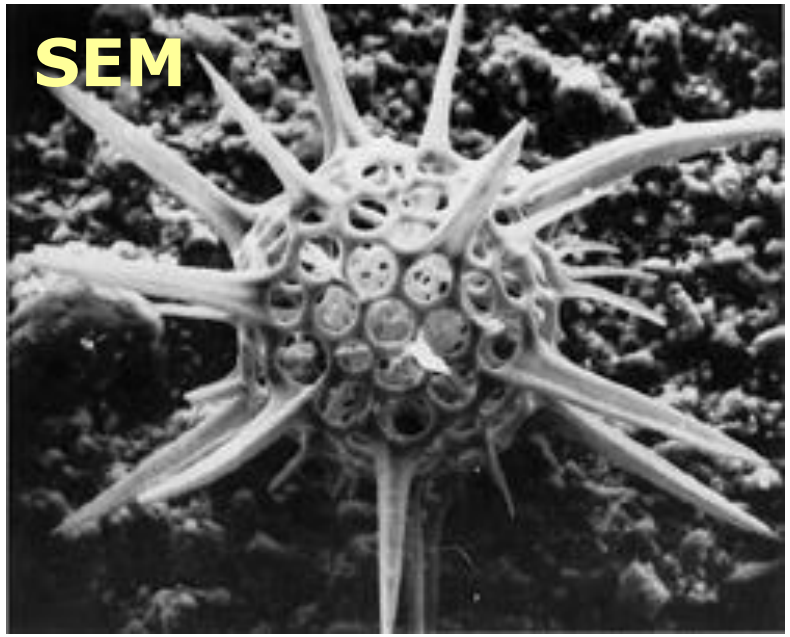
Scanning electron microscope (SEM) is a microscope that uses electrons rather than light to form an image. There are many advantages to using the SEM instead of an OM.

Advantages of Using SEM over LM

- **The SEM has a large depth of field,**

  This allows a large amount of the sample to be in focus at one time and produces an image that is a good representation of the three-dimensional sample.



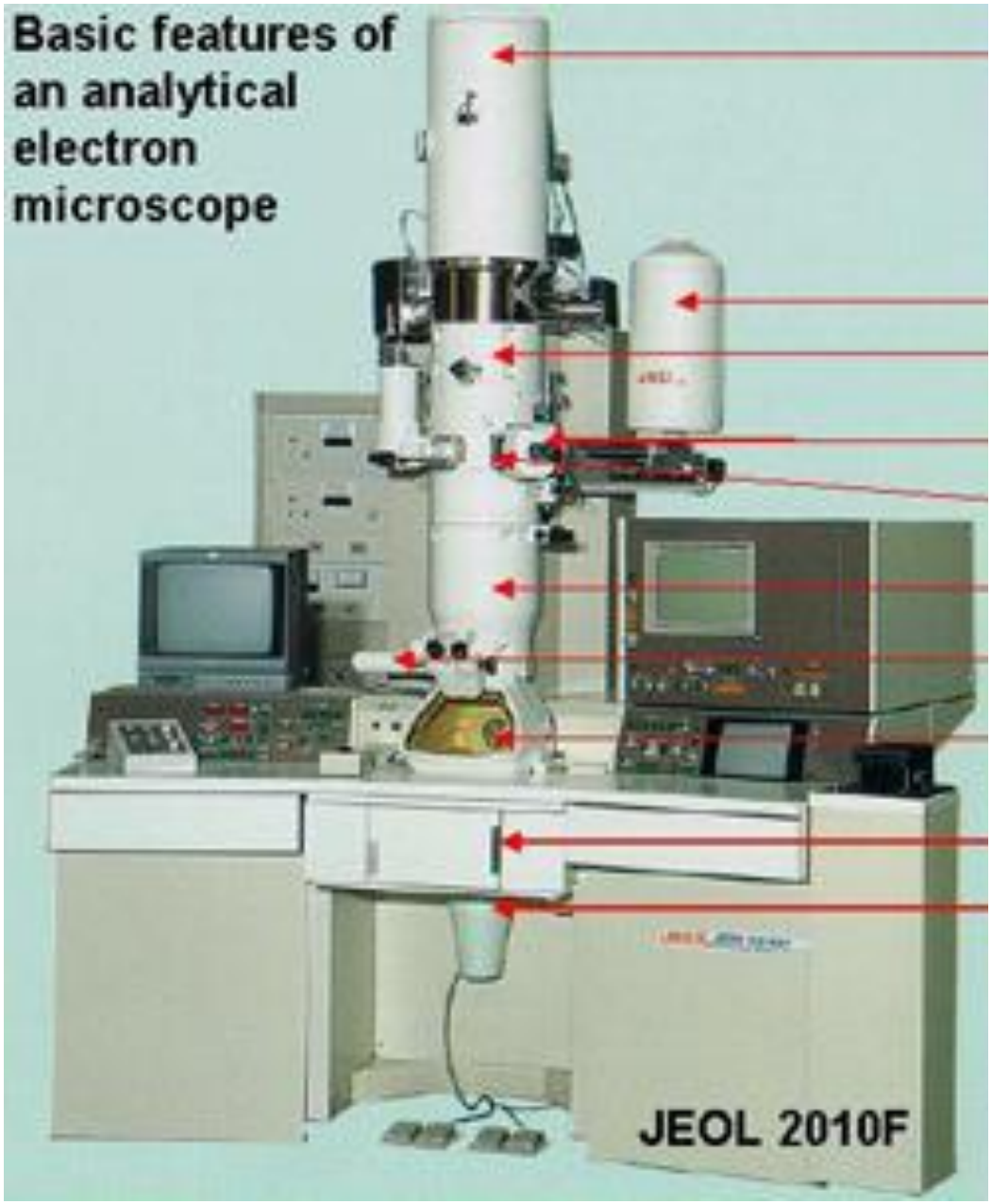


↑FOV
=
↑resolution
topogr
aphy

- **The SEM also produces images of high resolution;** ***fine features can be closely examined at a high magnification.***

- ***The combination of higher magnification, larger depth of field, greater resolution makes the SEM one of the most heavily used instruments in research areas and industries, especially in semiconductor industry.***

Transition Electron Microscope (TEM)



Electron Gun

EDS Detector
Condenser Lens

Specimen Holder
Objective Lens

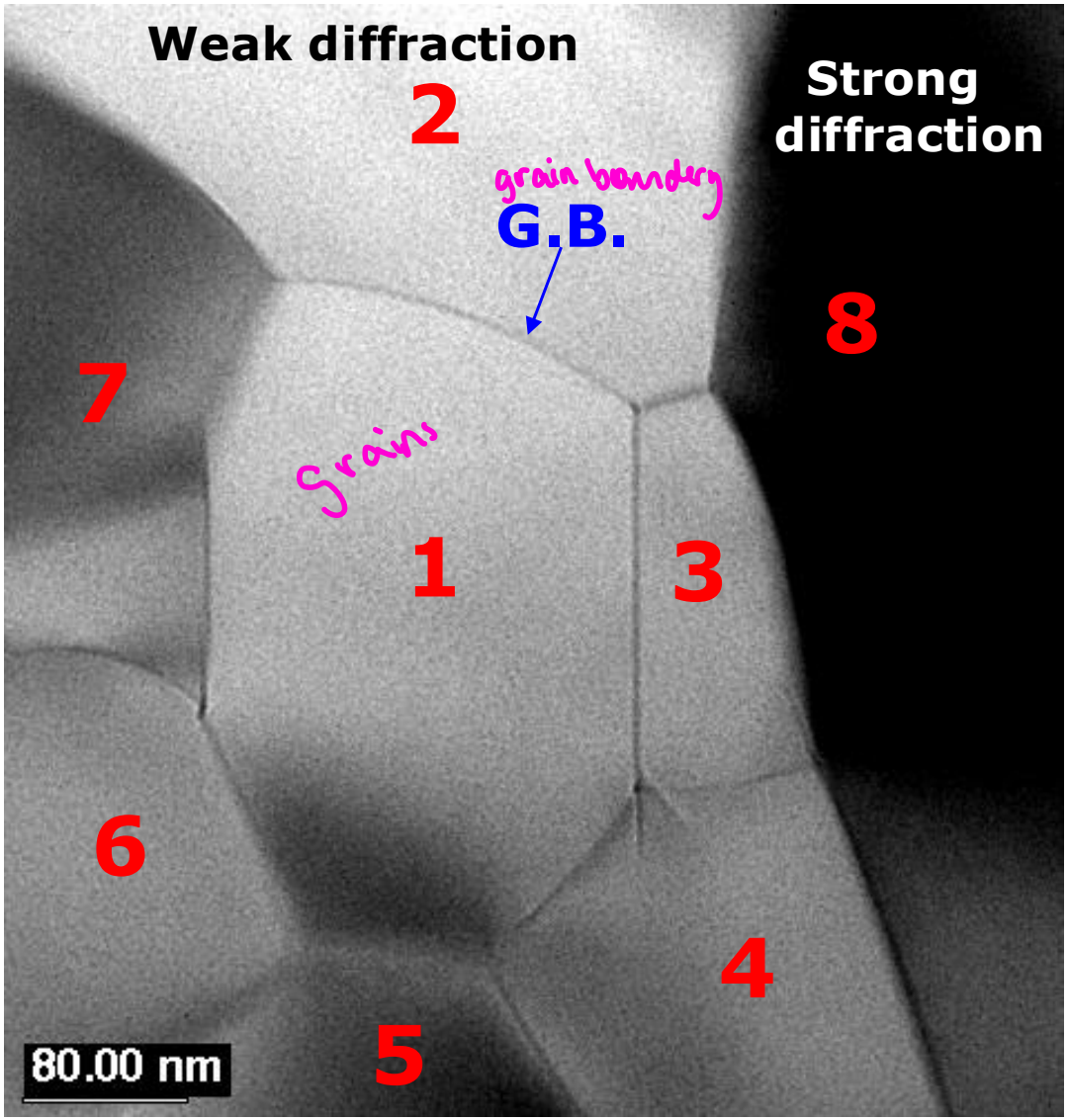
Magnifying Lenses

HAADF Detector
Viewing Chamber

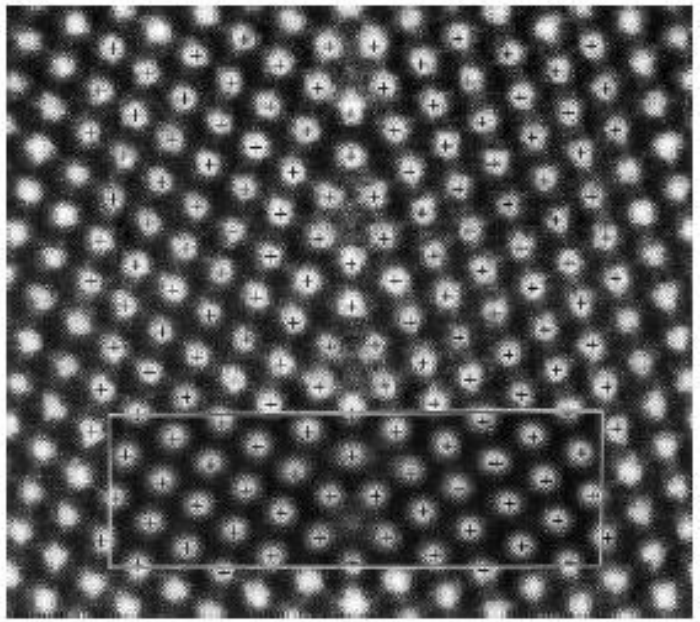
Camera Chamber

STEM Detector or EELS

Transition Electron Microscope (TEM)



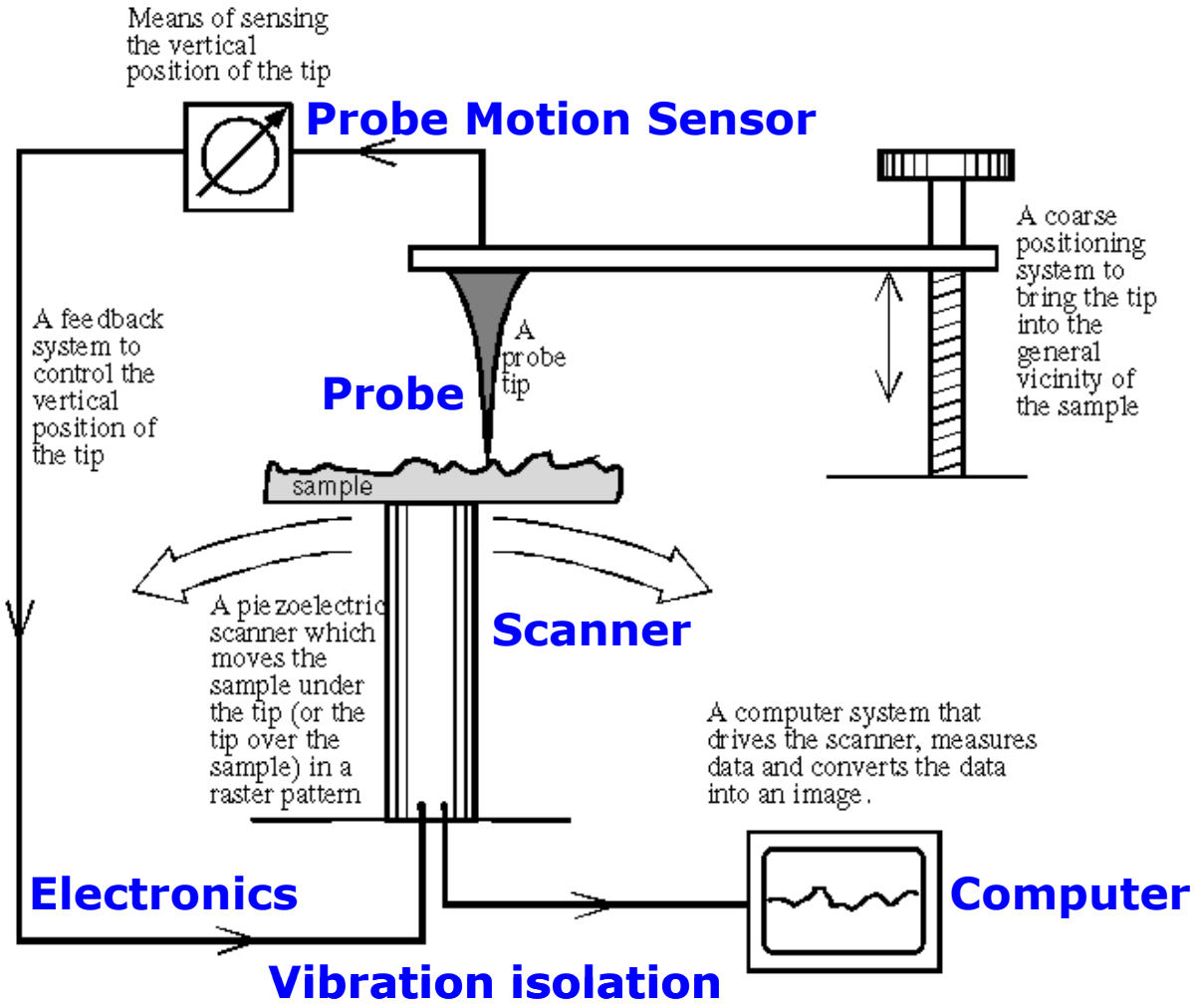
8 grains are in different orientations



High-resolution TEM image of a tilt grain boundary in Al, Sandia National Lab.

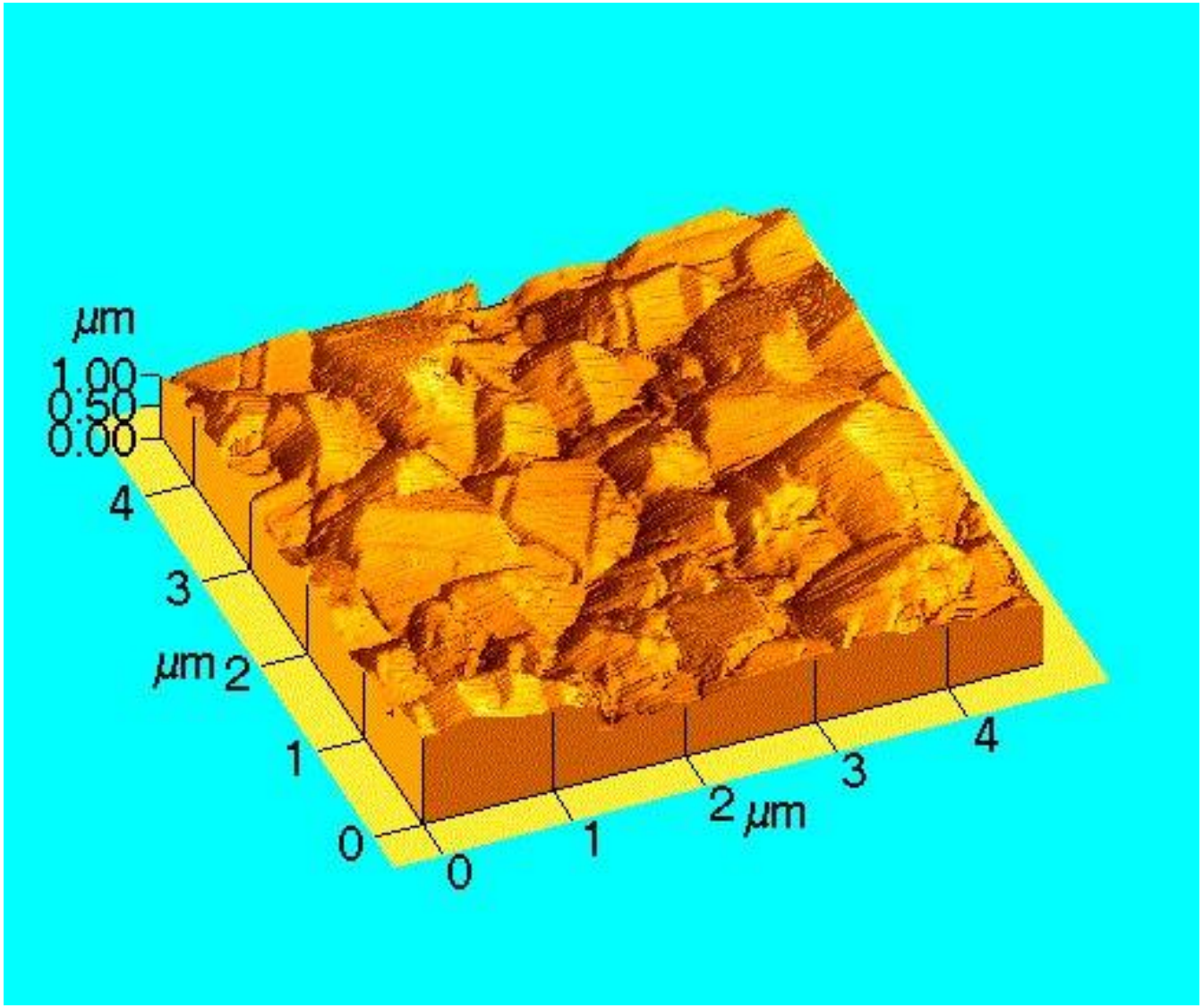
Scanning Probe Microscope (SPM)

SPMs are used for studying surface topography and properties of materials from the *atomic* to the *micron* level.



↓ topography

SPM



Microscopy

Question:

→ they don't depend on pcs ←

Why were commercial TEMs developed from about 1938 and SEMs from about 1965, whereas SPMs were not around before 1980's?

↳ depends on pc

Answer:

TEMs and SEMs were not limited by computers technology initially as images could be easily collected on photographic film which has been around for a long time, with high resolution. On the other hand, SPMs ***depend*** *on computers to construct the 3D images.*

General Resolution of Microscopes

Type of Microscope	Approx. Resolution
Human Eye	2'500'000 Å
Optical Light (OLM)	3000 Å
Scanning Electron (SEM)	10-50 Å
Transmission Electron (TEM)	2-5 Å, near atomic
Scanning Probe (SPM)	0.1 Å, atomic

Review Electron Microscopy

A fracture surface is very rough, with peaks and valleys. Why does the SEM give a clear image of it while the optical microscope gives an image where only part is in focus?

in slide 14

A. The SEM has a much larger depth of field.

B. The SEM has higher magnification.

C. The SEM uses colour.

D. The fracture surface must be etched first.

Use the QR code shown in the classroom. This one won't work.

Next topic:

Diffusion

Diffusion

Atoms movements in materials

Movement of atoms in **solids**, liquids and gases is very important

Examples: Hardening steel, chrome-plating, gas reactions, Si wafers .. etc.

Today we will discuss:

- Atomic mechanisms of diffusion
- Mathematics of diffusion
- Factors affecting diffusion
- Examples
- Steady state diffusion
- Non-steady state diffusion
- Summary

Diffusion

temp is the vibration of atoms

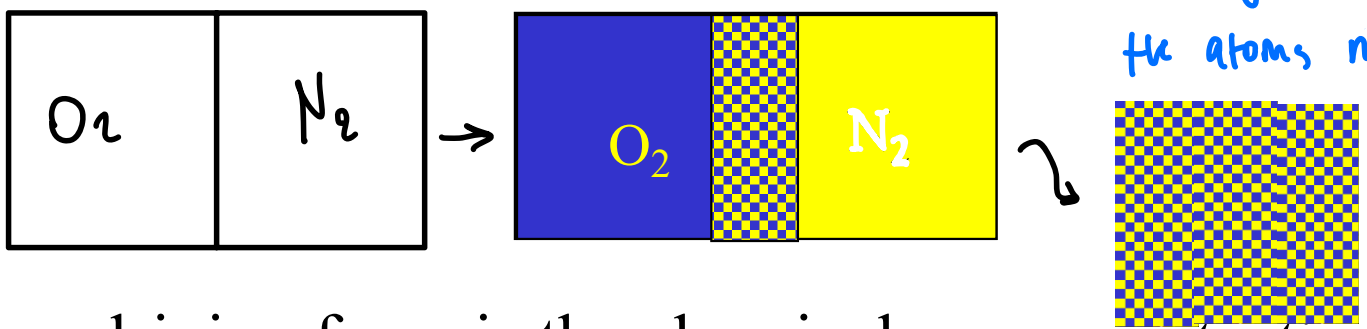
↑ T = ↑ vibration

- Diffusion is the mass transport through atomic motion at high temperature
- can have *self-diffusion* or *interdiffusion* between two materials
  - Self-diffusion occurs in pure elements

ex Fe and Fe

O_2 and N_2

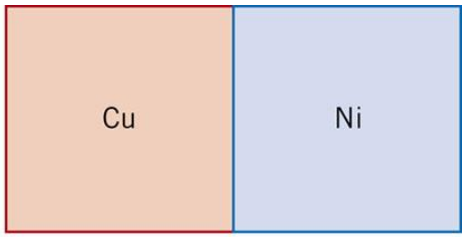
Example: Two chambers, each containing a different gas, separated by a removable barrier; when the barrier is pulled away, *interdiffusion* occurs



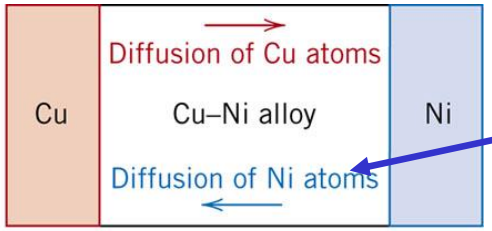
the high temperature makes the atoms move and stick

driving force is the chemical or *concentration gradient* through materials or *diffusion couple*

Diffusion in Solid Materials

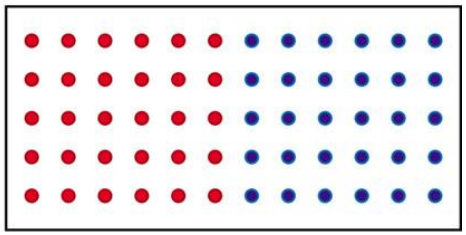


(a)



(a)

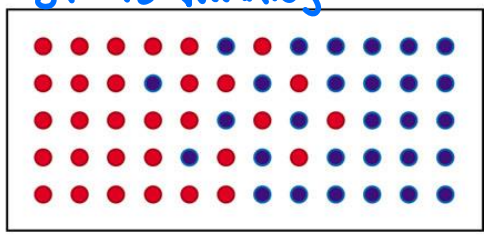
Diffusion Couple



(b)

Heat

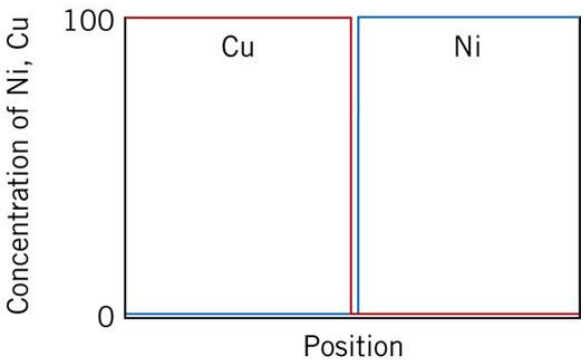
Starts mixing



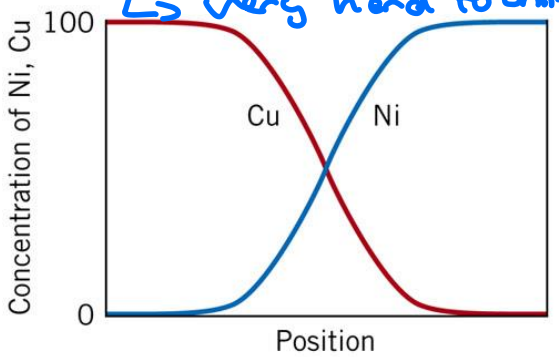
(b)

temperature should be high enough to overcome energy barriers to atomic motion.

permanent mixing
↳ very hard to unmix



(c)



(c)

Heat causes atoms to vibrate

- Vibration amplitude increases with temperature
- If heating continues *melting occurs when vibrations are sufficient to rupture bonds*. Faster diffusion takes place in the liquid state.

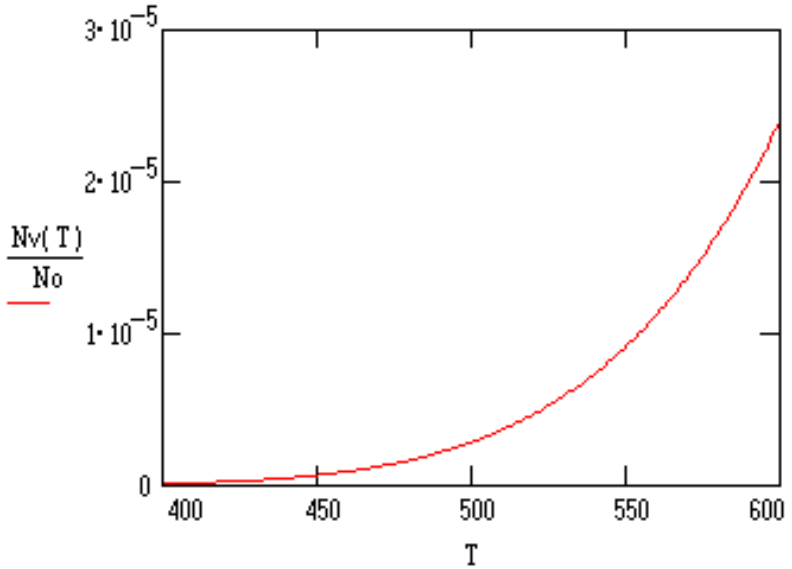
The process of substitutional diffusion requires the presence of ..vacancies..
(Vacancies give the atoms a place to move)

Recall Arhenius equation

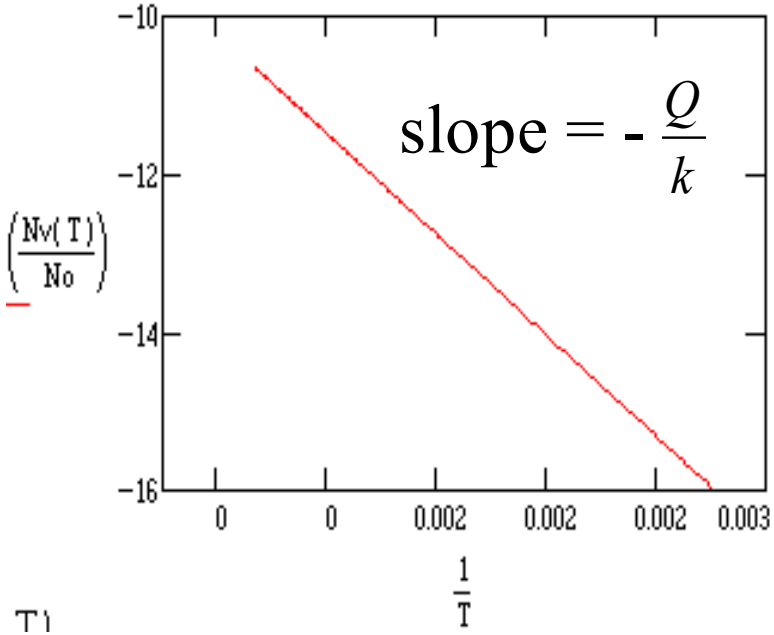
$$N_v = N_o \exp\left(-\frac{Q}{kT}\right)$$

where
N_v = equilibrium # of vacancies
N_o = total number of lattice sites
k = Boltzman's constant
Q = activation energy for vacancy formation

Example: lead (Pb)



Example: lead (Pb)

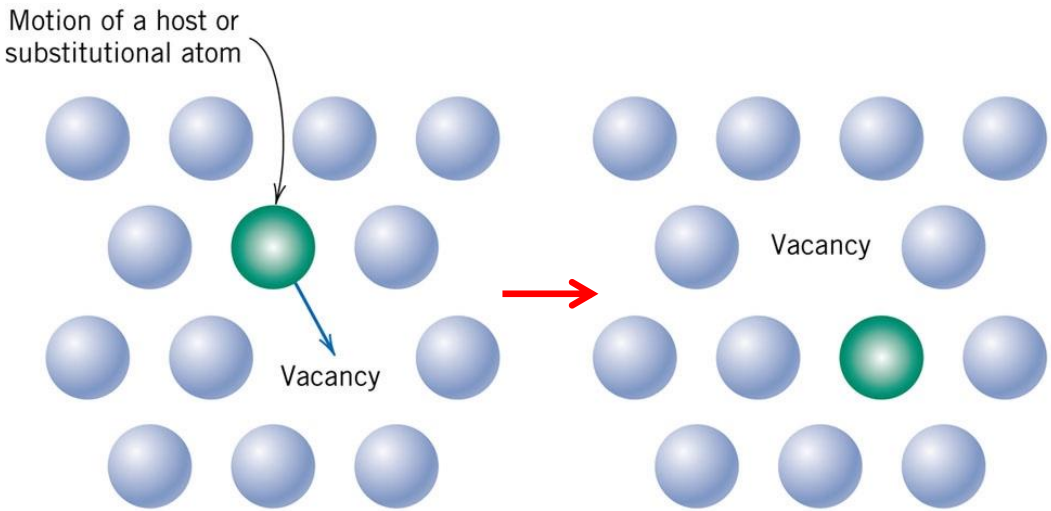


As T ↑, number of vacancies ↑, and energy ↑, so diffusion is faster

↑T → faster diffusion

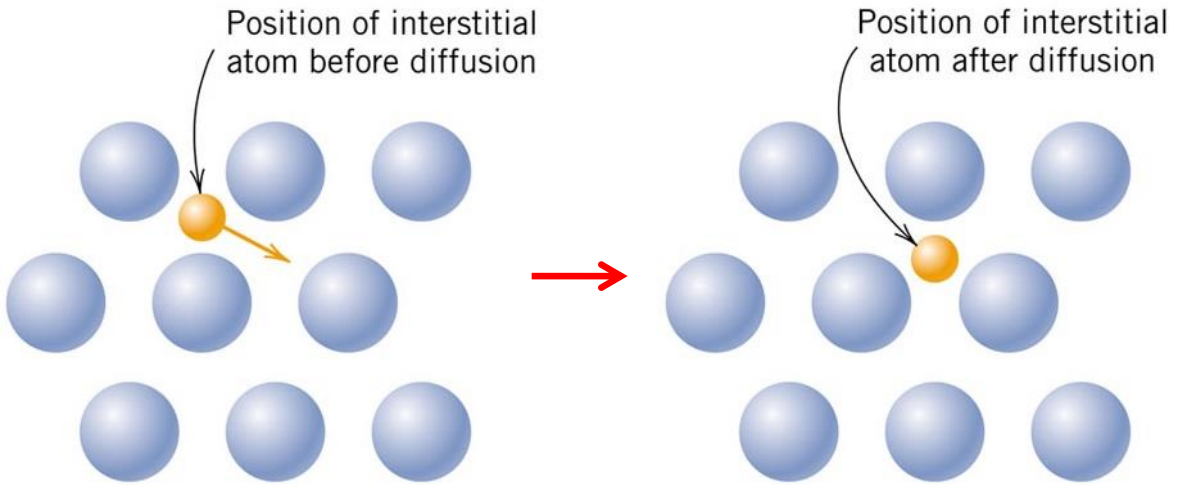
Diffusion Mechanisms

- Atoms are constantly in motion and vibrating
- change of atomic position requires:
  - vacant site
  - energy to break atomic bonds
- Two types of diffusion mechanism:
  - **vacancy** diffusion
  - **interstitial** diffusion
- movement of vacancies in one direction is equivalent to atomic movement in the opposite direction

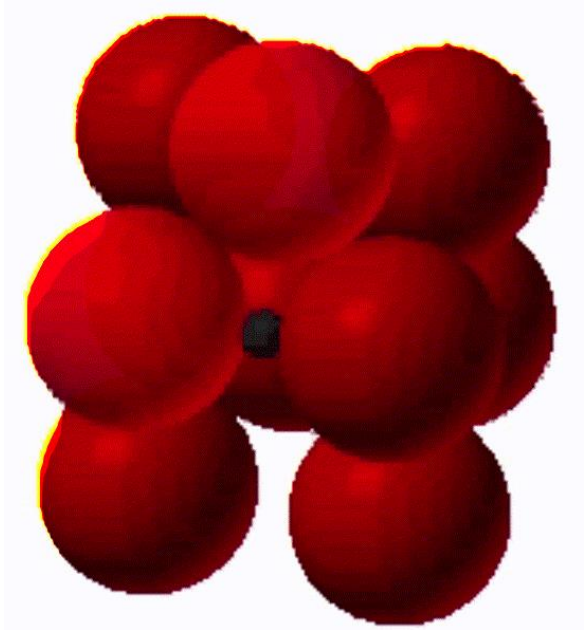


Vacancy diffusion

INTERSTITIAL DIFFUSION



Interstitial diffusion



- Interstitial atom moves from one interstitial site to another (*empty*)
- Energy needed, again to squeeze past atoms.
- Example: ……
- Usually much faster because many more empty interstitial sites and no vacancies are required

Diffusion

- Diffusion is a …time dependant.. process

➔ rate of diffusion is important

- diffusion *flux* (J) is defined as the mass, M, diffusing through unit area, A, per unit time, t

A : Area of filter

t = time

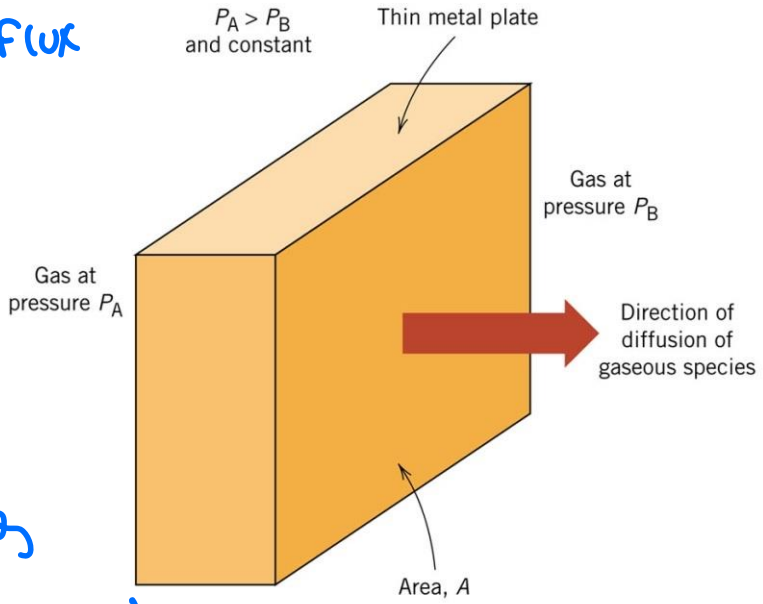
$$J = \frac{M}{At} \quad \text{or} \quad J = \frac{1}{A}\frac{dM}{dt}$$

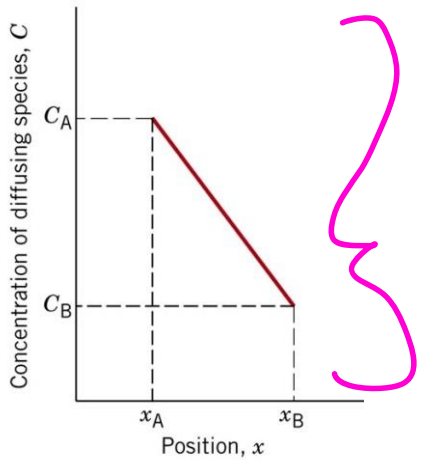
flow inside a "filter"

If flux does not change with time: ..steady state.....

J = diff flux

↳ constent flux





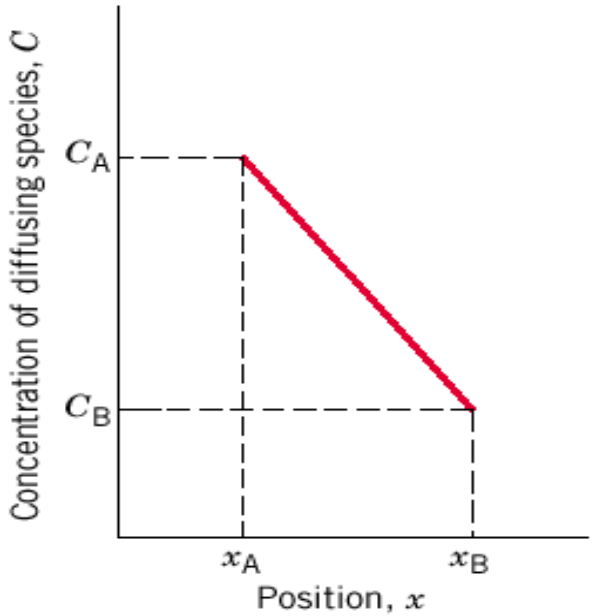
hydrogen diffusing through metal

Steady-State Diffusion

- Concentration gradient = $\frac{dC}{dx}$
- Fick's 1st law:

$$J = -D\frac{dC}{dx}$$

D = -ve



Where D is the ***diffusion coefficient*** (diffusivity) or speed of diffusion *(m^2/s)*.

↳ speed of diffusion

-ve *because atoms diffuse down concentration gradient*

• Example of *steady-state diffusion* is gas diffusing through a metal plate (*gas pressure constant*).

Example

The purification of H_2 (gas) by diffusion through a Pd sheet. Compute the number of kilograms of hydrogen that pass per hour through a 5 mm thick sheet of Pd having an area of 0.20 m^2 at 500°C. Assume a diffusion coefficient of 1.0×10^{-8} m^2/s, that the concentrations at the high and low pressure sides of the plate are 2.4 and 0.6 kg of H_2 per m^3 of Pd, and that steady state conditions have been attained.

Recall, flux is mass per unit time per unit area. Thus, multiplying J by area and time will give total mass.

$$M = JAt = -DAt\frac{\Delta C}{\Delta x}$$

$$= -(1.0\times10^{-8}\ m^2/s)(0.2\ m^2)(3600\ s/h)\left[\frac{0.6-2.4\ kg/m^3}{5\times10^{-3}\ m}\right]$$

$$= 2.6\times10^{-3}\ kg/h$$

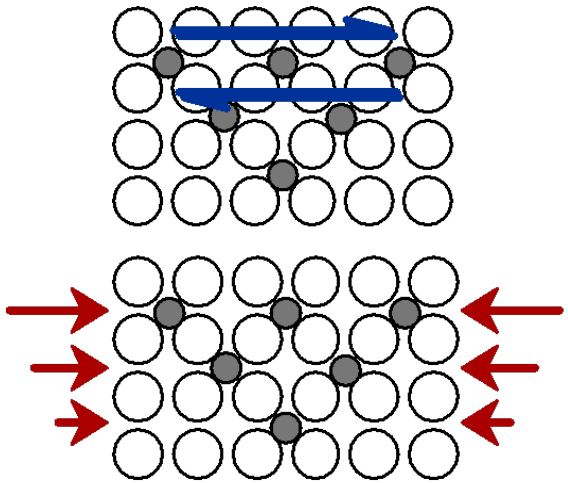
interstitial diffusion Case Hardening

Chapter 5 Callister 8e. (courtesy of Surface Division, Midland-Ross.)

- Example of interstitial diffusion is a case hardened gear.

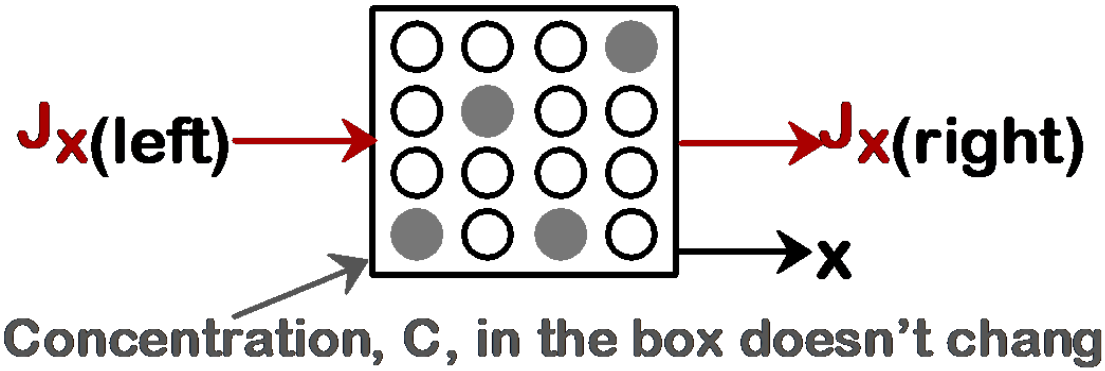
- Diffuse carbon atoms into the host iron atoms at the surface.

- Result: The "Case" is
  - hard to deform: C atoms "lock" planes from shearing.
  - hard to crack: C atoms put the surface in compression.



Steady State Diffusion: *Summary*

- Steady State: the concentration profile doesn't change with time.



Steady State:

$J_x(\text{left}) = J_x(\text{right})$

providing H to left = providing H to right

- **Apply Fick's First Law:** $J_x = -D\frac{dC}{dx}$
- **If $J_{x_left} = J_{x_right}$, then** $\left(\frac{dC}{dx}\right)_{left} = \left(\frac{dC}{dx}\right)_{right}$
- **Result: the slope, dC/dx, must be constant (i.e., slope doesn't vary with position)!**

Non-steady State Diffusion

- In most real situations diffusion is not ……steady state……
- Flux and concentration gradient vary with time

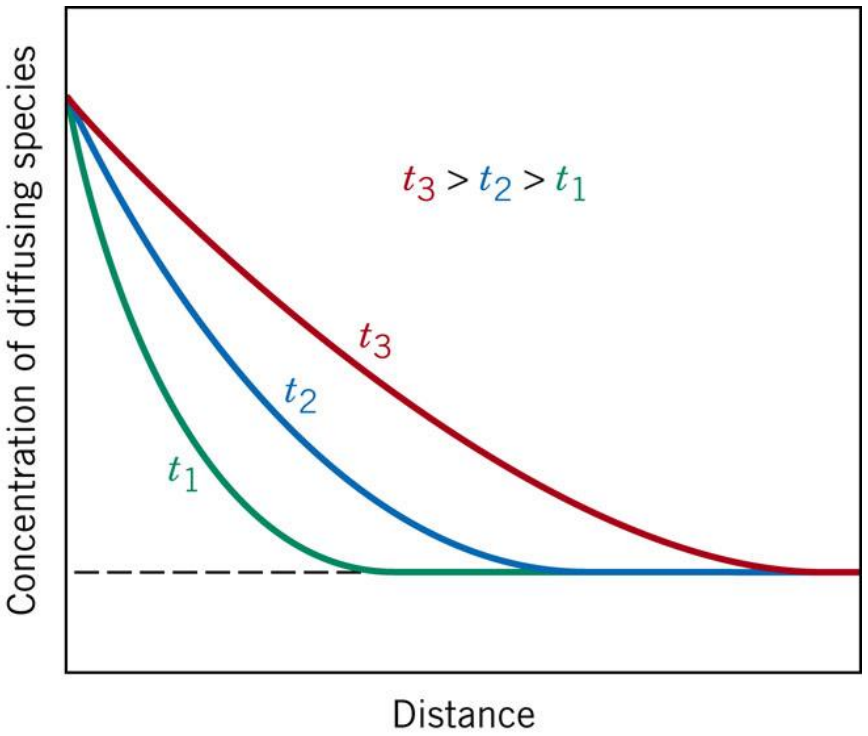


FIGURE 5. 6 Concentration profiles for nonsteady-state diffusion taken at three different times, t_1, t_2, and t_3.

• The changes of the concentration profile is given in this case by a differential equation, ……Fick's…… *second law*:

Fick's 2nd law

$$\frac{\partial C}{\partial t} = \frac{\partial}{\partial x}\left(D\frac{\partial C}{\partial x}\right)$$

• If diffusion coefficient is independent of *composition* then:

simplified

$$\frac{\partial C}{\partial t} = D\frac{\partial^2 C}{\partial x^2}$$

Solution of this equation is concentration profile as function of time, ……C(x,t)……

Concentration as a function of time relative to position

Fick's Second Law

$$\frac{\partial C}{\partial t} = D\frac{\partial^2 C}{\partial x^2}$$

- Solution requires boundary conditions.

  • *A useful solution is for a semi-infinite solid when the surface concentration remains constant.*

- What is semi-infinite solid?

  ✓ *semi-infinite bar:* none of the diffusing atoms reaches the bar end during the time over which the diffusion takes place.

  ✓ $l > 10\sqrt{(Dt)}$, *where* l*: bar length,* D*: diffusion coefficient and* t*: time*

Fick's Second Law - *Application*

- Simple boundary condition is where the surface concentration is held constant,
  - *e.g. gas phase with constant partial pressure at the surface*
- Conditions are:
  - before diffusion, solute atom have a homogeneous concentration of C_o
  - x is zero at the surface and increases with distance into the solid
  - *time is zero just before diffusion begins*

at beg. of membrane

- Mathematically, for $t = 0$, $C = C_o$ at $0 \le x \le \infty$

  $t > 0$, $C = C_s$ at $x = 0$ and $C = C_o$ at $x = \infty$

  C_s = constant surface concentration

- applying these boundary conditions gives:

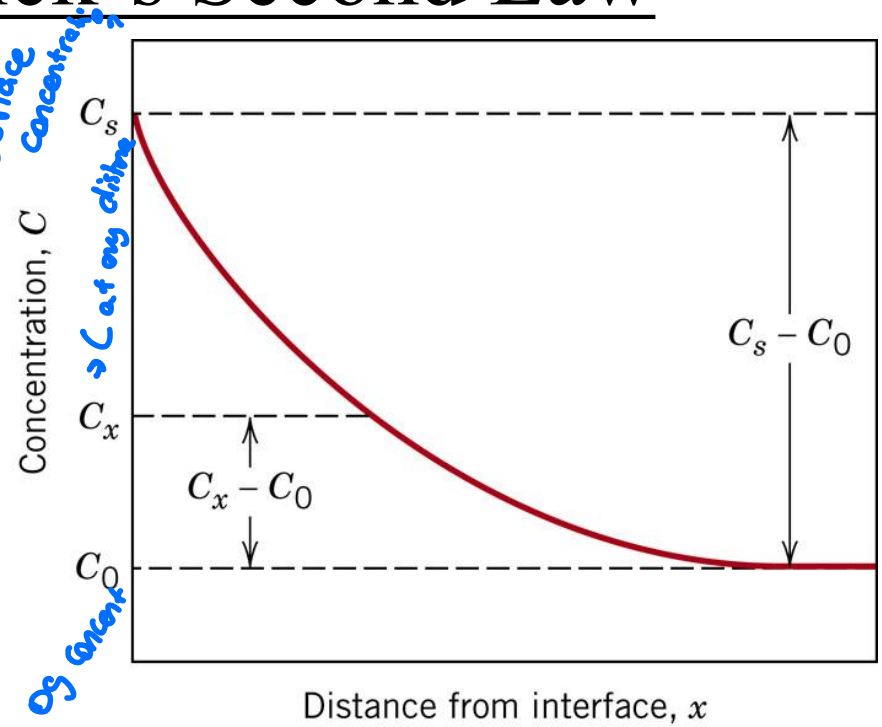
$$\frac{C_x - C_0}{C_s - C_0} = 1 - erf\left(\frac{x}{2\sqrt{Dt}}\right)$$

no need to memorize / do ts eq, just apply the variables

error function

Application of Fick's Second Law

$$\frac{C_x - C_0}{C_s - C_0} = 1 - erf\left(\frac{x}{2\sqrt{Dt}}\right)$$



C_s = Surface concentration which remains constant

C_0 = Initial concentration in solid

C_x = Concentration at distance x into sample after time t.

D = Diffusivity of solute in solvent, m^2s^{-1}

t = Time, seconds

erf = Gaussian error function, based on integration of the "bell shaped" curve

$$\mathbf{erf}\,(z) = (2/\sqrt{\pi})\int_0^z \exp(-y^2)\,dz \qquad \left(\frac{x}{2\sqrt{Dt}}\right) = z$$

Tabulation of Error Function Values

Table 5.1 Tabulation of Error Function Values

z	$erf(z)$	z	$erf(z)$	z	$erf(z)$
0	0	0.55	0.5633	1.3	0.9340
0.025	0.0282	0.60	0.6039	1.4	0.9523
0.05	0.0564	0.65	0.6420	1.5	0.9661
0.10	0.1125	0.70	0.6778	1.6	0.9763
0.15	0.1680	0.75	0.7112	1.7	0.9838
0.20	0.2227	0.80	0.7421	1.8	0.9891
0.25	0.2763	0.85	0.7707	1.9	0.9928
0.30	0.3286	0.90	0.7970	2.0	0.9953
0.35	0.3794	0.95	0.8209	2.2	0.9981
0.40	0.4284	1.0	0.8427	2.4	0.9993
0.45	0.4755	1.1	0.8802	2.6	0.9998
0.50	0.5205	1.2	0.9103	2.8	0.9999

$$\frac{C_x - C_0}{C_s - C_0} = 1 - erf(z)$$

$$\left(\frac{x}{2\sqrt{Dt}}\right) = z$$

say $z = 1.1$

then $erf(z) = 0.8802$

in the eq.

Special Case

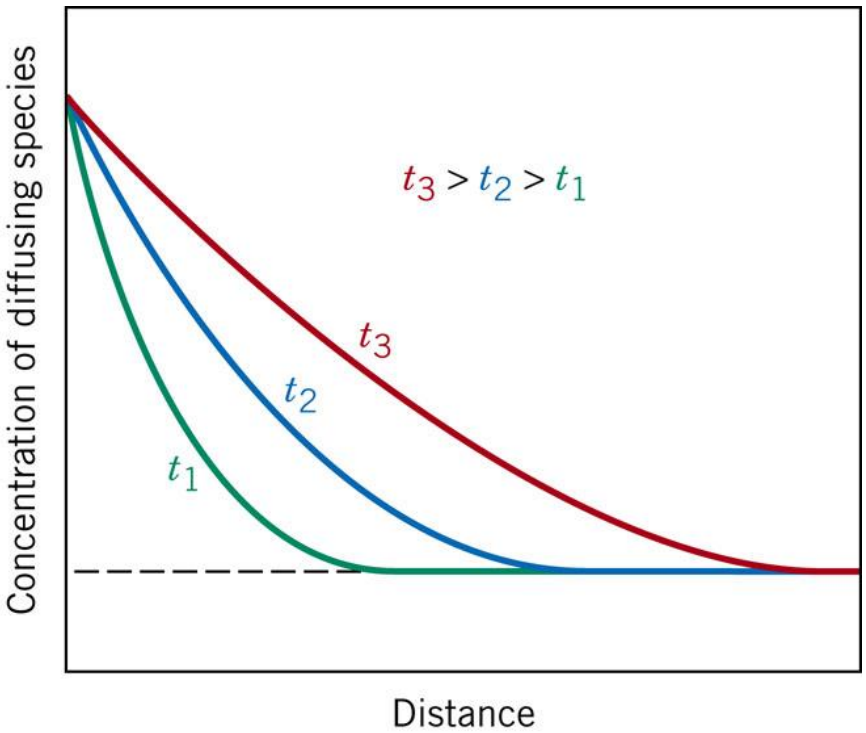


FIGURE 5. 6 Concentration profiles for nonsteady-state diffusion taken at three different times, t_1, t_2, and t_3.

- If it is desired to achieve a particular concentration of solute atoms in an alloy then:

$$\frac{C_1 - C_o}{C_s - C_o} = \text{constant}$$

and

$$\frac{x}{2\sqrt{Dt}} = \text{constant}$$

$$\frac{x^2}{Dt} = \text{constant}$$

- These equations facilitate the diffusion computation for this specific case (*see example 5.2*).

Factors Affecting Solid-State Diffusion

- Diffusing species and host material are important
  - smaller **atoms** can "squeeze" in between host atoms more easily
  - in lower **packing density** host material ⇒ easier for atoms to migrate with fewer bonds to expand
    - eg faster in more open lattice (BCC faster than FCC) *(APF....)*
  - in lower **melting point** host material ⇒ weaker bonds (easier to push apart)

Smaller atom diffuse easier

↑MP = harder diffusion

- *Temperature* has a strong effect on diffusion rates:

$$D = D_o \exp\left(-\frac{Q_d}{RT}\right)$$

Where: D_o determines if its self diffusion or not

D_o = temperature dependant pre-exponential constant

Q_d = *activation energy* for diffusion

R = gas constant (8.31 J/mol K)

T = absolute temperature

Activation energy is the barrier to diffusion and controls the diffusion coefficient, D

Typical Diffusion Data

Host crystal	Atom	D_o (cm^2/s)	q_d (eV/atom)
Cu	Cu	0.20	2.04
Cu	Zn	0.34	1.98
Ag	Ag	0.40	1.91
Ag	Au	0.26	1.98
Ag	Cu	1.2	2.00
Ag	Pb	0.22	1.65
U	U	0.002	1.20
Si	Al	8.0	3.47
Si	Ga	3.6	3.51
Si	In	16.0	3.90

Typical values for preexponential (D_o) and activation energy

(after Kittel, "Solid State Physics" 5th ed.)

Experimental Determination of Activation Energy

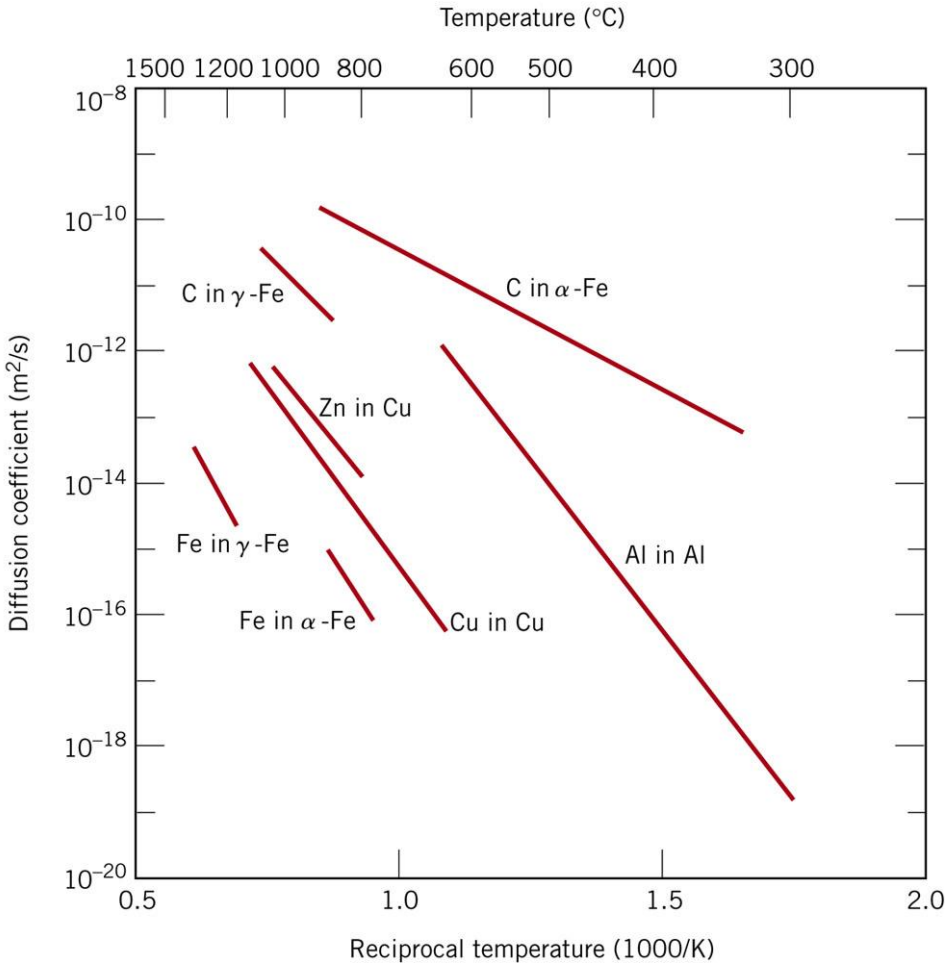


FIGURE 5.6 Plot of the logarithm of the diffusion coefficient versus the reciprocal of absolute temperature for several metals.

$$D = D_o \exp\left(-\frac{Q_d}{RT}\right)$$

- Taking logs:

$$\log D = \log Do - \frac{Qd}{2.3R}\left[\frac{1}{T}\right]$$

- plot log D vs. 1/T
- slope = ……-Qd/2.3R……
- intercept = ……Do……

Example

0,35% → 0% C decarburization C – 0

An FCC Fe-C alloy initially containing 0.35 wt. % C is exposed to an oxygen-rich (and carbon-free) atmosphere at 1400 K (1127°C). Under these conditions the carbon in the alloy diffuses toward the surface and reacts with the oxygen in the atmosphere; that is, the carbon concentration at the surface is maintained essentially at 0 wt. % C. (This process of carbon depletion is termed decarburization). At what position will the carbon concentration be 0.15 wt. % after a 10-hour treatment. The value of D at 1400 K is 6.9×10^{-11} m^2 /s.

$$\frac{C_x - C_0}{C_s - C_0} = \frac{0,15 - 0,35}{0 - 0,35} = 0,5714 = 1 - \operatorname{erf}\left(\frac{x}{2\sqrt{Dt}}\right)$$

$$\operatorname{erf}\left(\frac{x}{2\sqrt{Dt}}\right) = 0,4284 \qquad z = 0,4$$

$$x = 2(0,4)\sqrt{Dt} = (0,8004)\sqrt{(6,9 \times 10^{-11}\ m^2/s)(3,6 \times 10^4\ s)}$$

$$1,3 \times 10^{-3}\ m = 1,3\ mm$$

SUMMARY

Is diffusion faster, slower or same in the following scenarios?

1- Close-packed (slower ✓) vs open structure (faster)

2- Lower-melting (faster ✓) vs higher-melting-point materials (slower)

3- Covalent bond (slower ✓) vs secondary interactions (faster)

4- Smaller (faster ✓) vs larger atoms (slower)

Use the QR code shown in the classroom. This one won't work.

Next topic:

Mechanical properties of materials

Mechanical Properties I

↳ last topic on the midterm

Outline

- Introduction
- Mechanical testing
- Tensile test
- Elastic deformation

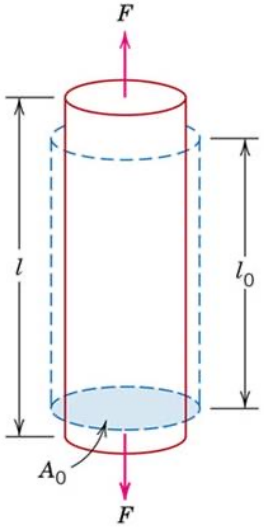
Mechanical Properties

- From an applications standpoint, one of the most important topics within Materials Sc. & Eng. is the study of how materials respond to external loading or deformation.
- Most components, even if used primarily for other property (e.g. electronic substrate) have to fulfil certain *mechanical functions* as well.
- Important mechanical properties are strength, hardness, stiffness, ductility and toughness.
- Laboratory testing to measure mechanical properties attempts to replicate the service conditions.
- Consistency is accomplished by using standardised test, so people are measuring same thing in the same way.
- **A**merican **S**ociety for **T**esting **M**aterials (ASTM) maintains and updates standards for mechanical properties.
- Several other standards organizations exist, e.g. SAE, ANSI, DIN….

Mechanical Testing of Metals

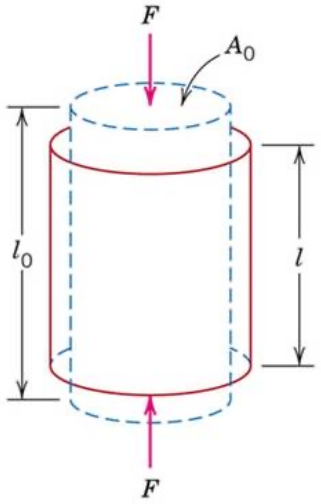
Loading can take any of the following forms:

tension



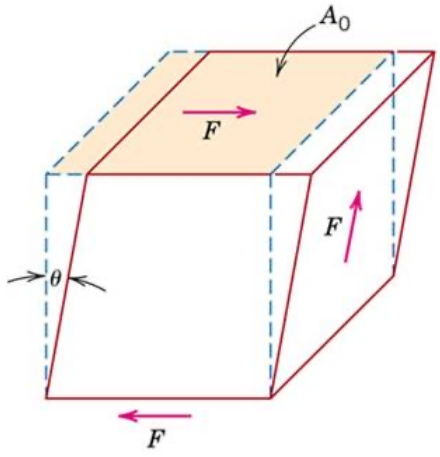
Pulling/ Stretching

compression



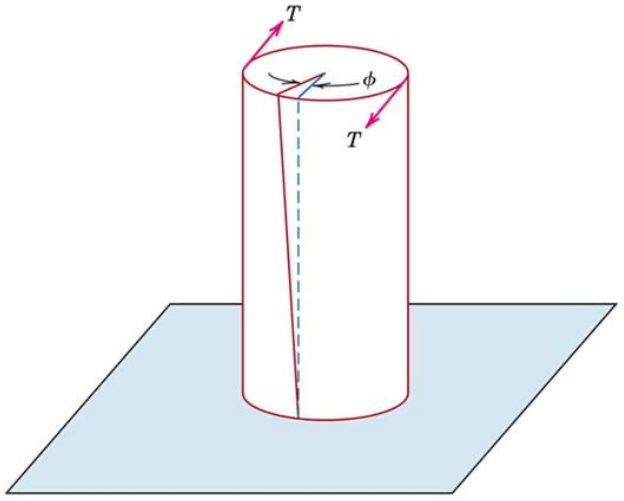
squeezing/ squashing

shear



sliding

torsion

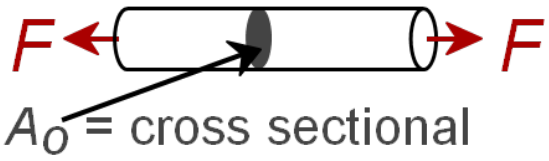


twisting

Different tests measure different types of loading conditions

Common States of Stress

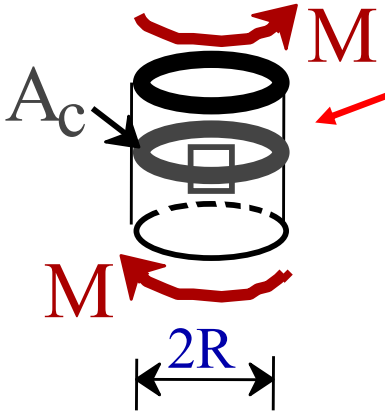
Simple tension: cable



$$\sigma = \frac{F}{A_o}$$

Ski lift *(photo courtesy P.M. Anderson)*

Torsion: drive shaft

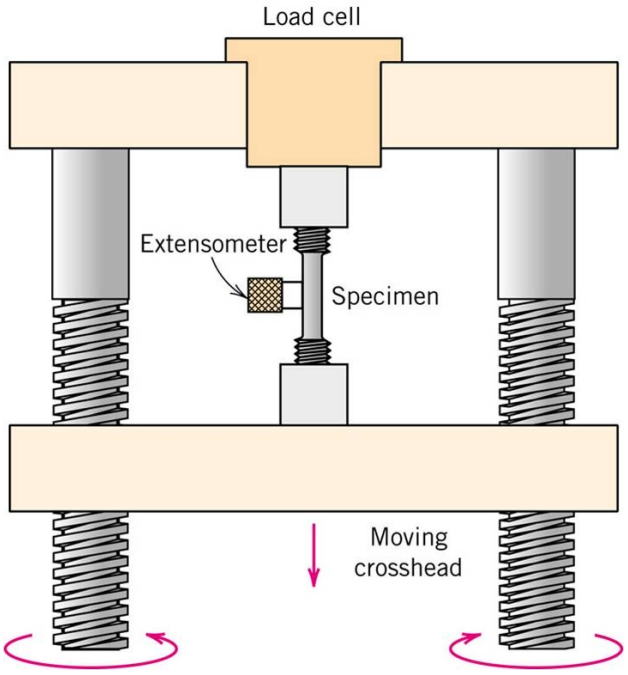


Simple compression:

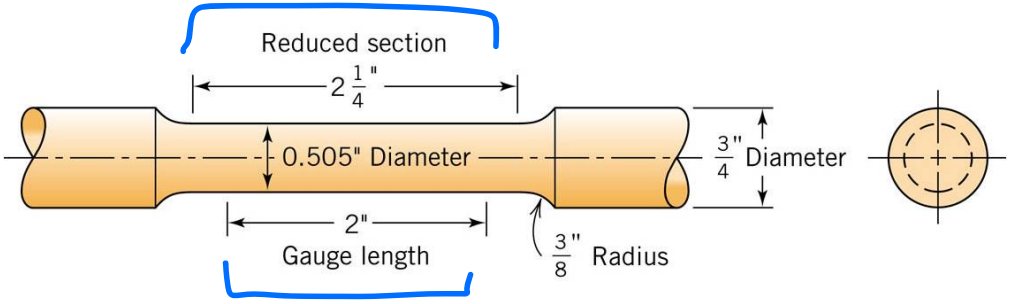
Canyon Bridge, Los Alamos, NM
(photo courtesy P.M. Anderson)

tries to stretch the sample and measures the force req to do so

Tensile Test



A typical tensile test machine



A typical standard tensile test specimen

- The material's response to the applied tensile or compressive load is a change in length.
- We can monitor the change in length very precisely with an instrument called an *extensometer*.

$$\varepsilon = \frac{l_i - l_o}{l_o} = \frac{\Delta l}{l_o}$$

We call this quantity strain

ε = change in length / original length

Where:

l = instantaneous length

l_o = initial length (length at various points during the test)

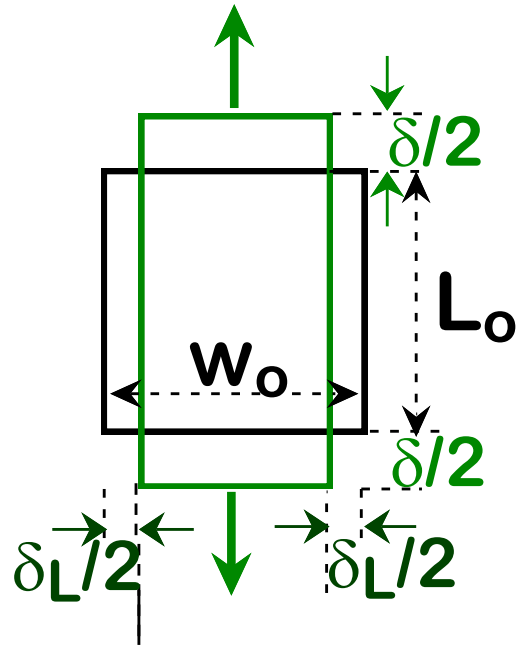
Strain is a dimensionless quantity (or, can be reported as m/m or in./in.)

Engineering Strain

- Tensile strain: $\varepsilon = \frac{\delta}{L_o}$

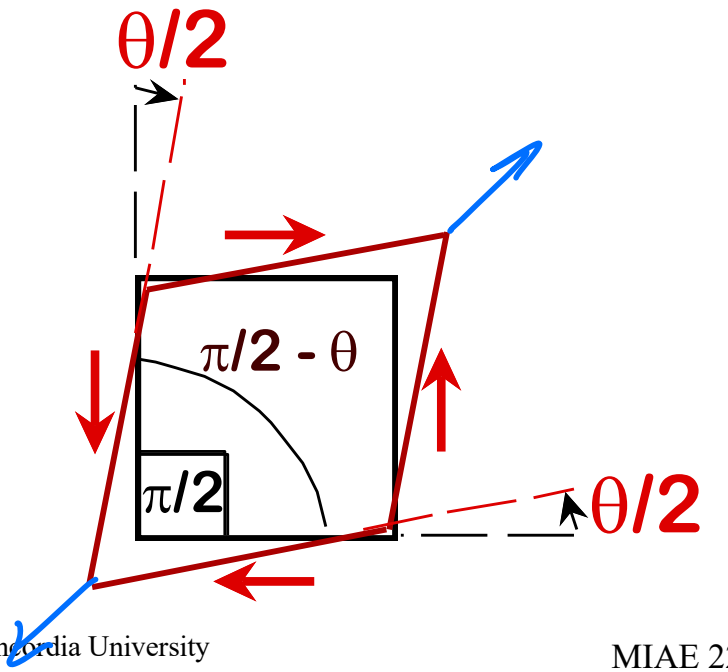
strain

- Lateral strain: $\varepsilon_L = \frac{-\delta_L}{w_o}$



- Shear strain:

$$\gamma = \tan\theta$$



Strain is always dimensionless.

Tensile Test

Typically, loading is normalized to cross sectional area:

$$\sigma = \frac{F}{A_o}$$

We refer to this ratio as the applied **stress**

when normalized to initial area, this is **engineering stress**.

when normalized to actual area, this is **true stress**.

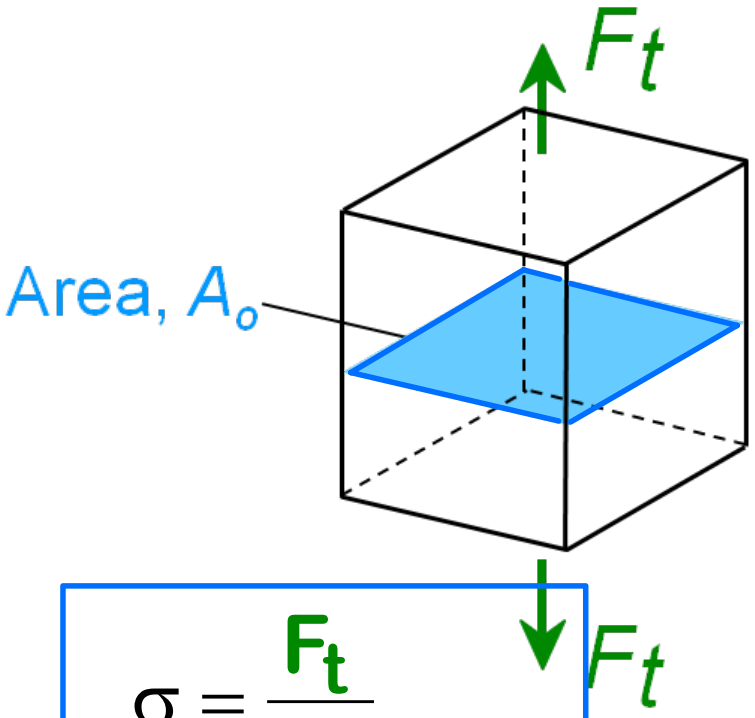
eng stress

Tensile test

- Load - elongation testing
- But a thin wire breaks under a lower tensile load than a thicker wire of the same material
- Engineering stress (σ) versus engineering strain (ε)

Engineering Stress

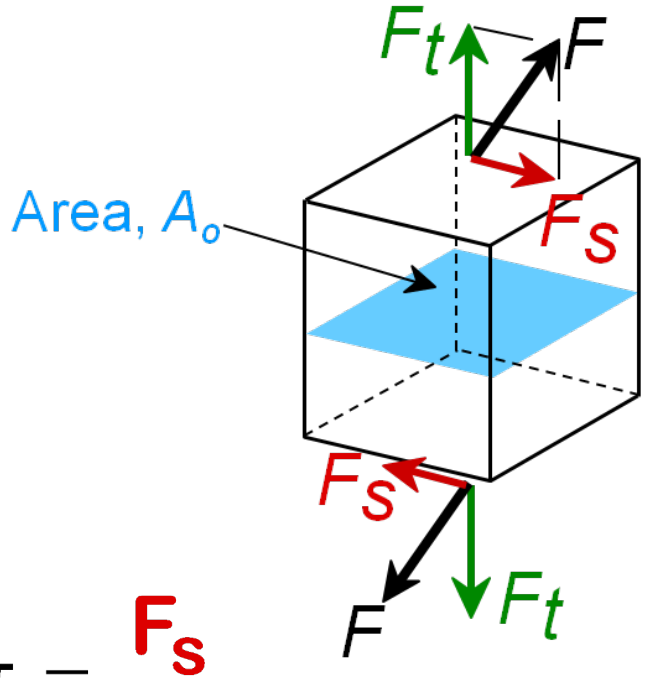
- Tensile stress, σ:



$$\sigma = \frac{F_t}{A_o}$$

A_o: original area before loading

- Shear stress, τ:

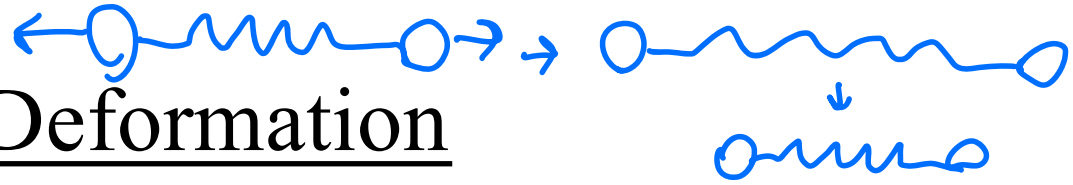


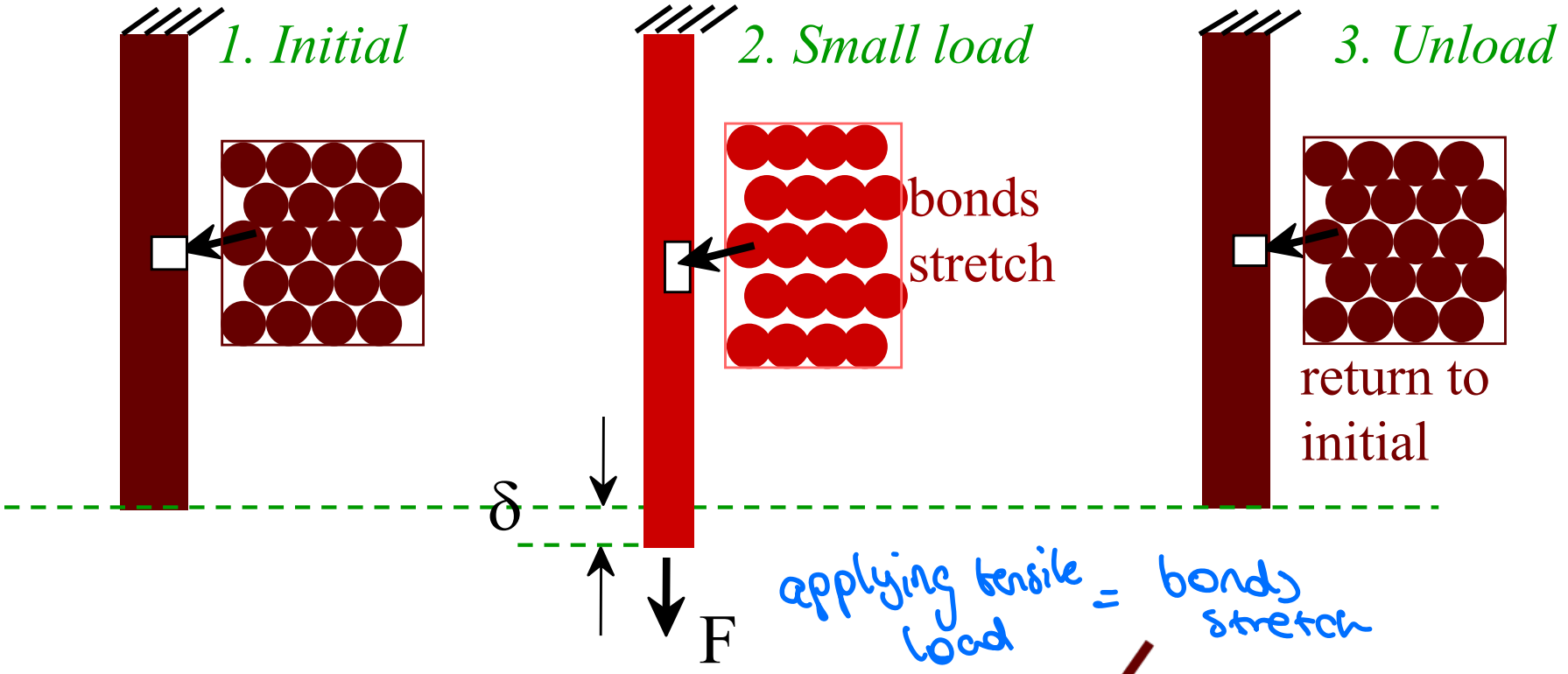
$$\tau = \frac{F_s}{A_o}$$

Shear stress

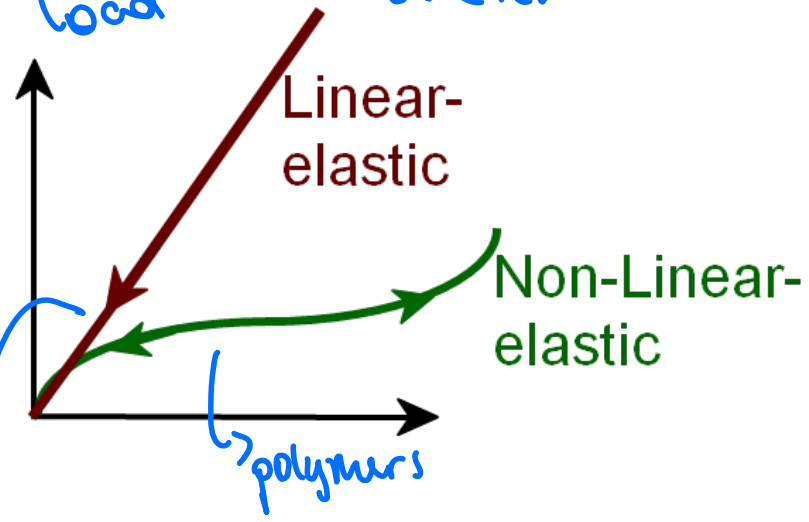
Stress has units: N/m^2 or lb/in^2

Elastic Deformation





applying tensile load = bonds stretch



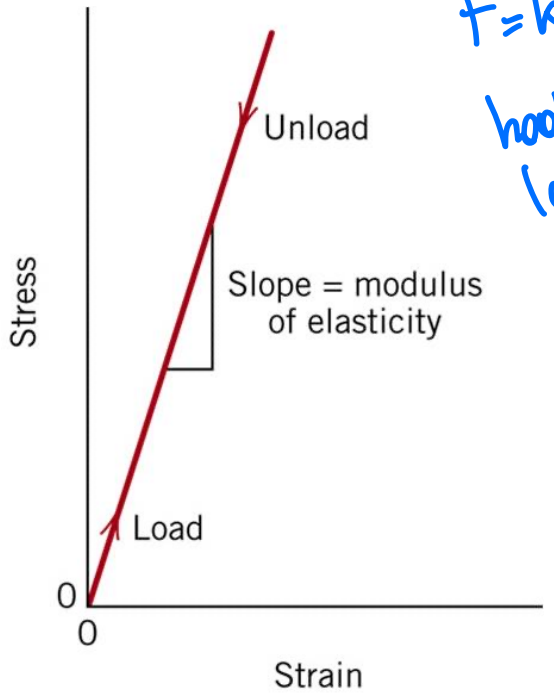
Elastic means reversible!

recoverable

Stress-Strain Behaviour - *Elastic Response*

Initially, stress and strain are directly proportional to each other

- *Rationale: atoms can be thought of as masses connected to each other through a network of* ***springs***.



$F = kx$

hooke's law

According to Hooke's law, the extension of a spring, x, and the applied force, F, are related by the spring constant, k:

$$F = - kx$$

$\sigma = E\varepsilon$ ↳ in terms of stress ↳ modulus of elasticity

The constant of proportionality, *Young's modulus or modulus of elasticity*, is a measure of the material's stiffness.

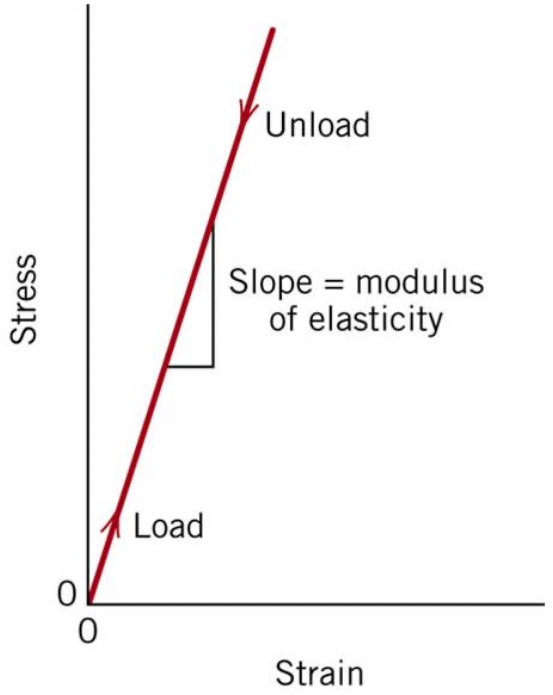
Materials possessing high stiffness: W, Ta, Mo → large... slope

Materials possessing low stiffness: Al, Cu, Ag → ..small... slope

Stress-Strain Behaviour - *Elastic Response*

In the elastic region a material returns to its original dimensions when load released, and we can write

$$\sigma = E\,\varepsilon \quad \text{or} \quad \frac{F}{A_o} = E\frac{\Delta l}{l_o}$$



Example:

A steel wire with a cross sectional area of 0.55 mm^2 and length of 10 m is extended *elastically* 1.68 mm by a force of 17.24 N. What is the modulus of elasticity for this steel specimen?

$A_o = 0.55\,mm^2$ $\quad l_o = 10\,m$ $\quad \Delta l = 1.68\,mm$

$F = 17.24\,N$ $\quad E = ?$

$\rightarrow \sigma = \frac{F}{A_o} = \frac{17.24\,N}{0.55\,mm^2} = 31.34\,N/mm^2 = 31.34\times10^6\,\frac{N}{m^2}$

$\rightarrow \varepsilon = \frac{\Delta l}{l_o} = (1.68\times10^{-3}\,m)/(10\,m) = 0.000168$

$\rightarrow E = \sigma/\varepsilon = (31.34\times10^6\,N/m^2)/$

Young's Modulus or Modulus of Elasticity

So E tells us how much something will stretch elastically when loaded, i.e. the **STIFFNESS** of that material

High E value - very stiff

Medium E value

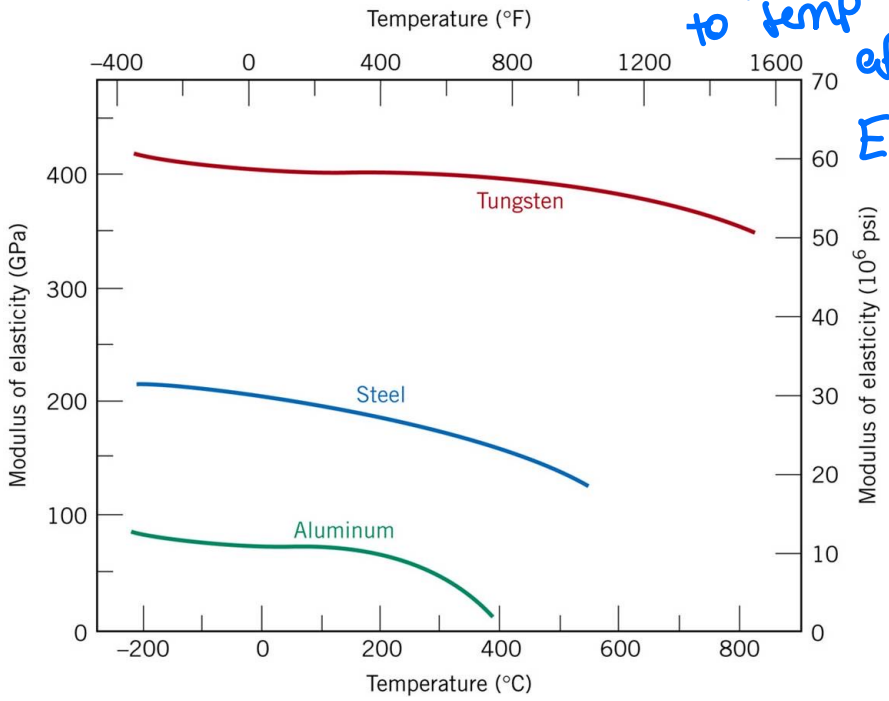
Low E value: - (not stiff)

Ceramics	300 GPa
Steel	207 GPa
Copper	110 GPa
Plastics	3 GPa

In general, the material's modulus (or stiffness) decreases with increasing temperature.

Can you think of why this happens?

because bonds become weaker



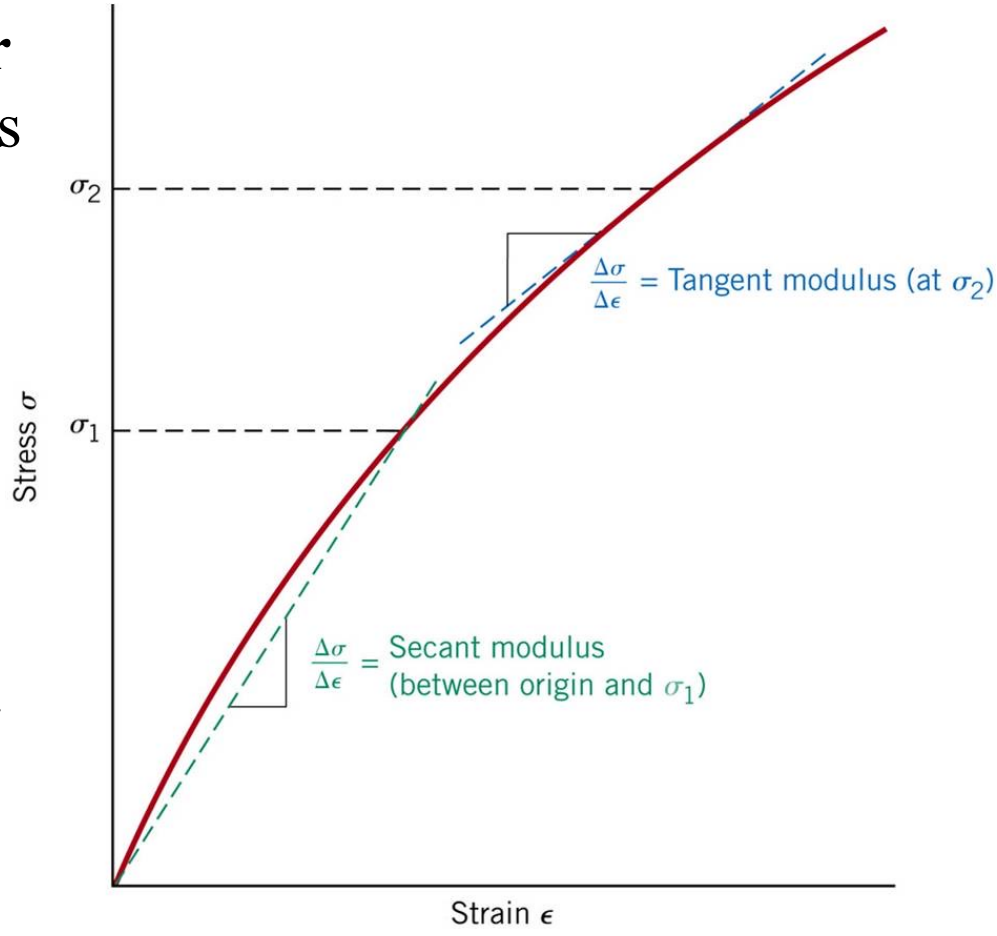
Temperature dependence of elastic moduli

Tangent and Secant Modulus

- Some materials do not show **linear** elastic region; their elastic region is non-linear. Cast iron, concrete, some polymers.

In this case E is harder to define:

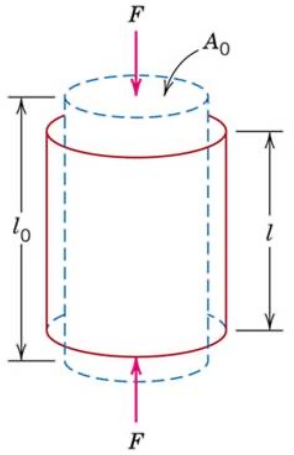
- We can use **Tangent modulus** which is slope of tangent at a particular stress level, or,
- **Secant modulus** which is slope of the line joining origin with some specified stress level.



Compression, Shear and Torsion Tests

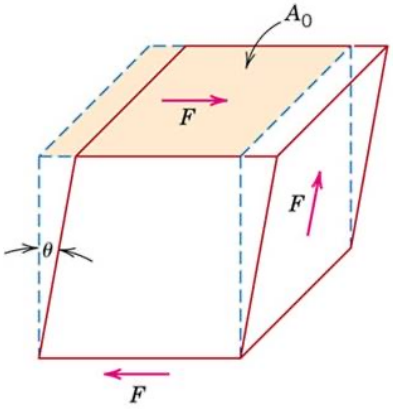
Compression

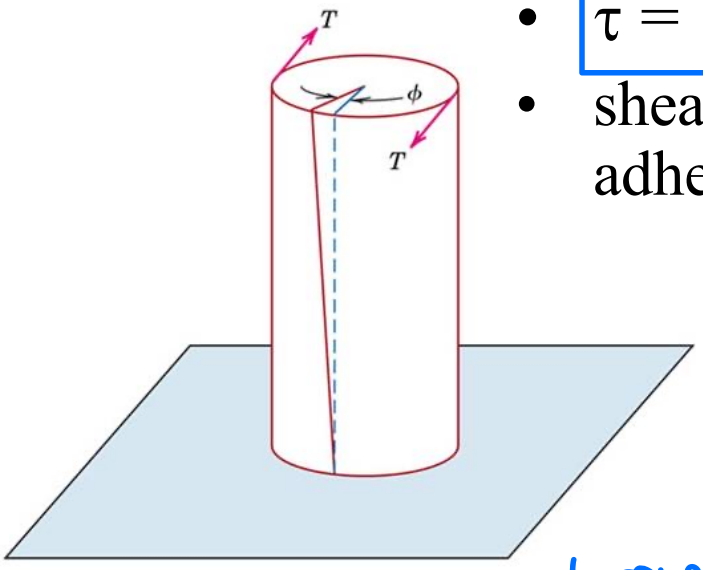
- by convention, stress and strain are negative
- used for measuring strength of brittle materials and for calculating forces required in manufacturing processes which involve compressive deformation



Shear

- shear stress is $\tau = F/A_o$ and γ (shear strain) is tangent of shear angle, θ
- $\tau = G\gamma$, G is shear modulus
- shear tests are often used to measure adhesive bonding, riveted joints etc





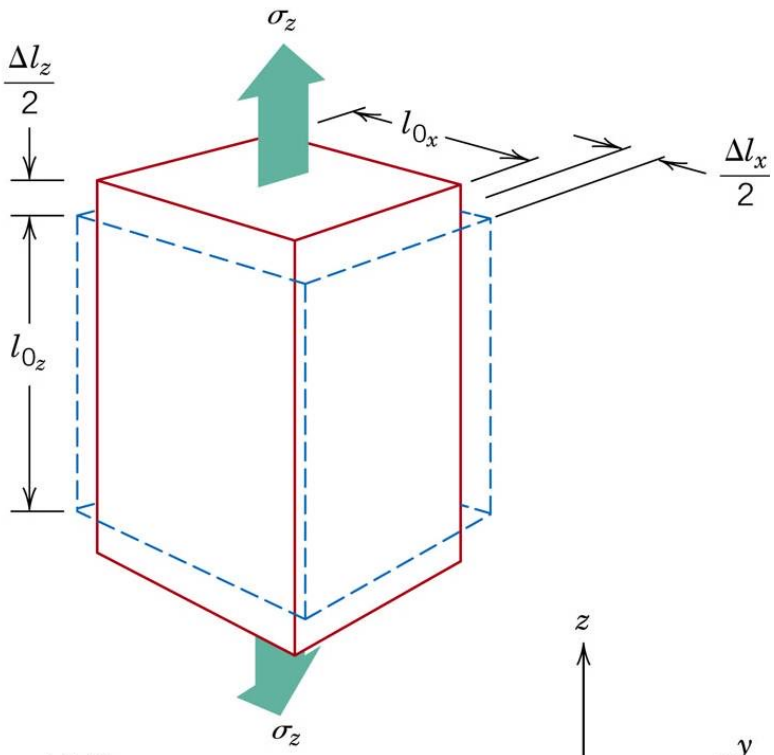
torque

Torsion

- *torsion* is a variation of shear occurring in *machine axles, drive shafts and twist drills*
- $T = f(\tau)$ and $\gamma = f(\phi)$

Poisson's Ratio

- When pulled in tension (along Z), a sample gets longer and thinner, i.e., *a contraction in the width (X) and breadth (Y)*
  - *if compressed gets fatter*



- **Poisson's ratio** defines how much strain occurs in the lateral directions (x & y) when strained in the (z) direction:

$$\upsilon = -\frac{\text{lateral strain}}{\text{longitudinal strain}}$$

$$\nu = -\frac{\varepsilon_x}{\varepsilon_z} = -\frac{\varepsilon_y}{\varepsilon_z}$$

can't be higher than 1

- Typical values = 0.2 to 0.5
- For isotropic materials

$$E = 2G(1+\nu)$$

Some materials are anisotropic so E & G vary with direction (e.g. composite materials and single crystals)

Elastic Deformation: Anelasticity

time dependence of elastic deformation

• So far we have assumed that elastic deformation is time ….independent.. (i.e. applied stress produces instantaneous elastic strain)

some mats are time dependent

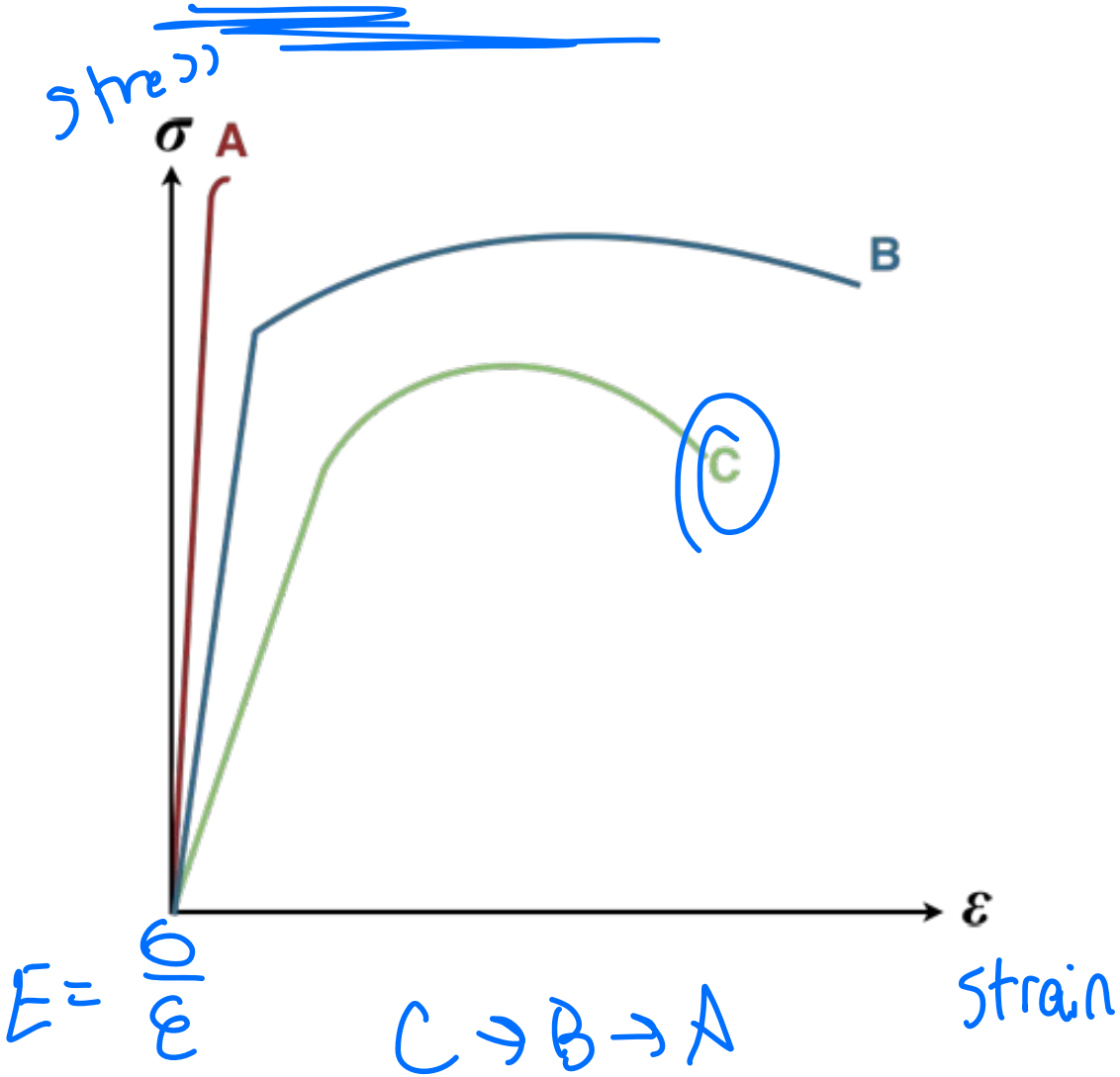
• In some materials, elastic strain may develop gradually after loading, and recovery may take time after unloading. This time-dependent, recoverable deformation is called **anelasticity**.

• The effect is normally small for **metals** but can be significant for **.**polymers**...** (“viscoelastic behavior”).

the load will be varied w/ time

Elastic Behaviour - *Review*

Which tensile engineering stress–strain curve represents the **least** stiff material?



Use the QR code shown in the classroom. This one won't work.

Next topic:

Plastic Deformation

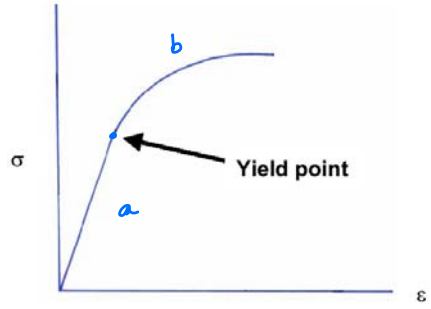
Info: exam
- multiple choice
- might have to draw
- sep by group by family name

Outline

- Yield Strength
- Plastic Deformation
- Mechanical Behavior
- Example
- True Stress and True Strain

Plastic deformation

What happens if we continue to apply tensile loading beyond the elastic limit? *(i.e., stretching atomic bonds to the point of breaking)*

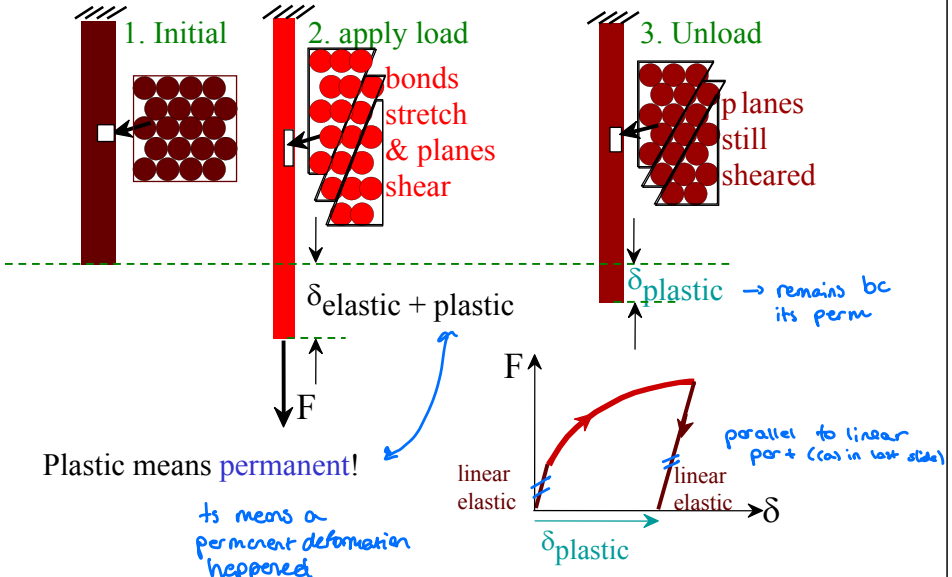


Permanent/

Plastic deformation:

- stress and strain are not proportional
- the deformation is not reversible
- deformation occurs by breaking and re-arrangement of atomic bonds (*in crystalline materials primarily by motion of dislocations, Chapter 7*)

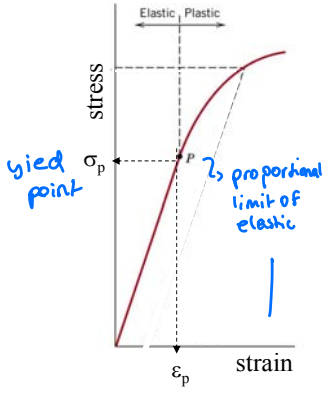
Plastic Deformation: *metals*



Plastic means permanent!

ts means a permanent deformation happened

Proportional Limit:



- Proportional limit is the point where the stress-strain curve becomes nonlinear (the strain deviates from being proportional to the stress).
- The gradient of this portion of the stress-strain curve equals to the elastic modulus of the material.
- The stress and strain values at this point are known as the proportional-limit stress and strain, respectively.

$\sigma = E\varepsilon$

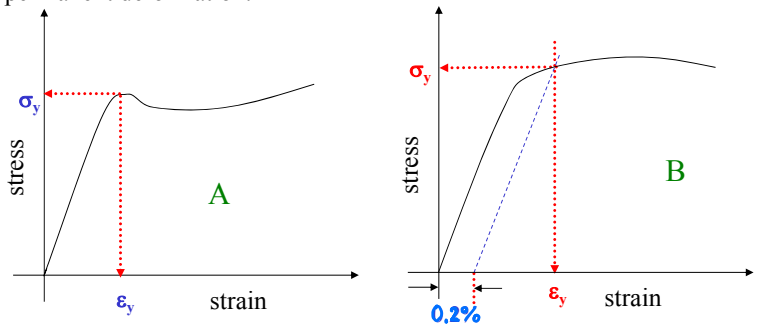
- This is the point beyond which **Hooke's law** can no longer be used to relate stress and strain in axial or shear deformation.

before ts Hooke's law can be applied

$P \neq \sigma_p$

Yield Strength

• The yield point corresponds to the point where the material begins to have permanent deformation.

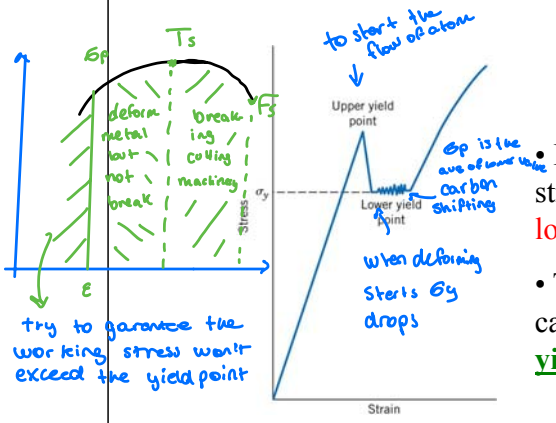


• Some materials have a well-defined yield region (A), others (B) do not.

• In the absence of a distinct yield point, a 0.2% offset is used to obtain an approximate yield point.

• *Although the **yield** and the **proportional** limit points **are close** to each other, they **do not correspond** to the same location on the stress-strain curve.*

Yield Strength

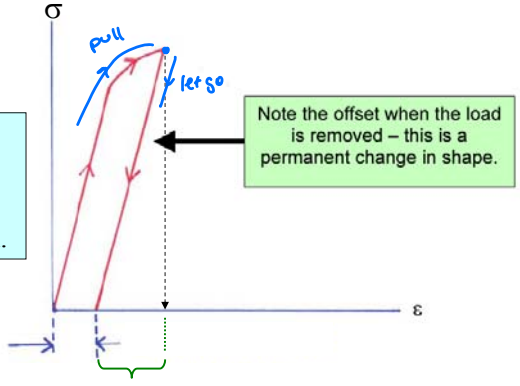


• For a low-carbon steel, the stress vs. strain curve includes both an upper and lower yield point.

• The yield strength is defined in this case as the average stress at the **lower yield** point.

Plastic Deformation

Suppose a tensile load is applied to a specimen and then released after the yield point was reached!

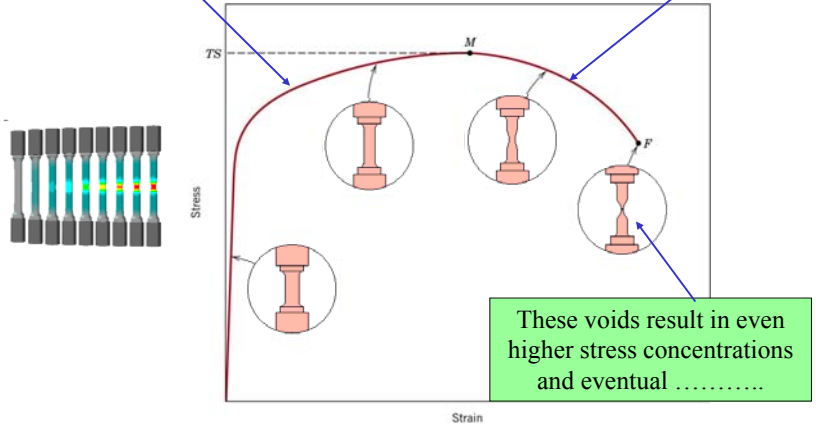


Plastic deformation is Irreversible: when the stress is removed, the material does not return to its original dimension.

Mechanical Behavior

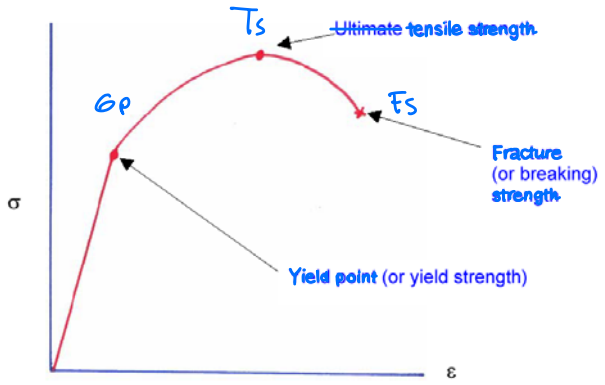
As plastic deformation proceeds, the force increases due to ……..

As more of the stress becomes concentrated in the neck, formation of voids occur



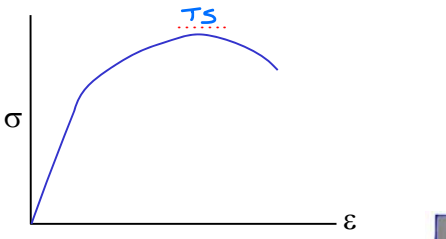
These voids result in even higher stress concentrations and eventual ……..

Mechanical Behavior

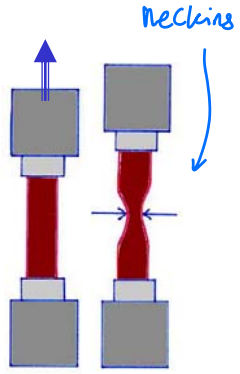


• *For structural applications, the yield stress is usually a more important property than the tensile strength, since once it is passed, the structure has deformed beyond acceptable limits.*

Mechanical Behavior



Question: In ductile metals, the σ - ε curve eventually turns down after reaching the ultimate tensile strength (UTS). Does this mean the specimen is becoming "weaker"? No, stronger (hardening)



we know that the gauge area decreases during plastic deformation due to necking

Recall the definition of stress: "engineering" stress = F/A_o where A_o is the initial cross-sectional area

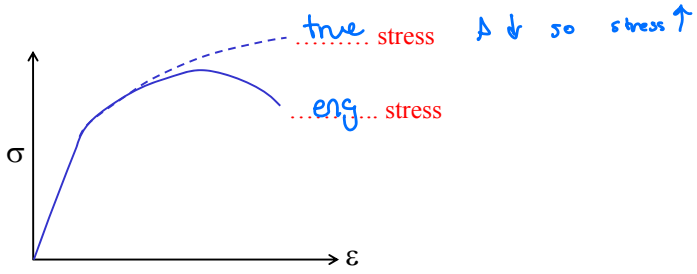
$A_c < A_o$ always

eng stress = $\frac{F}{A_o}$

true stress = $\frac{F}{A_{(after\ deformation)C}}$

Engineering Stress vs. True Stress

Since the actual cross-sectional area is reduced, use of the initial area gives a lower value than the actual one (the ratio is A_o/A_c).



• Even though the true stress-strain curve gives a more accurate picture of the breaking strength of a material, it is difficult to obtain measurements of *the actual area in real-time*.

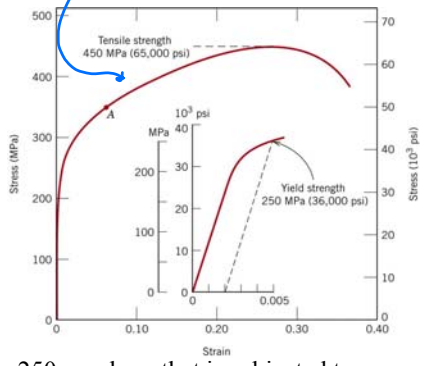
• Usually, the reported values are the engineering stress.

• True fracture strength > tensile strength

✓ *but the engineering σ - ε diagram does not show this*

Example

work hardening



From the tensile σ - ε behaviour for a specimen of brass shown in the figure, determine the following:

a) modulus of elasticity

b) yield strength at a strain offset of 0.002

c) maximum load that can be sustained by a cylindrical specimen having an original diameter of 12.8 mm

d) change in length of a specimen originally 250 mm long that is subjected to a tensile stress of 345 MPa

a) $E = \text{slope} = \frac{\Delta\sigma}{\Delta\varepsilon} = \frac{\sigma_2 - \sigma_1}{\varepsilon_2 - \varepsilon_1} = \frac{(150-0)}{(0.0016-0)} = 93.8\ GPa$

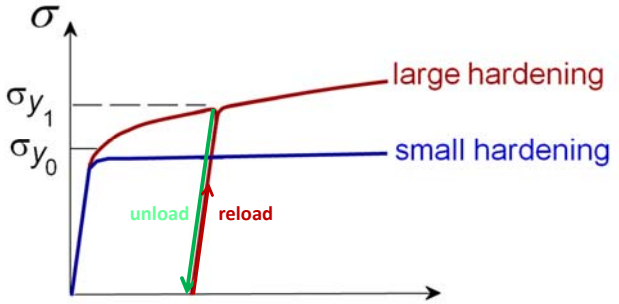
b) 2500 MPa

c) $F = \sigma A_o = \sigma \left(\frac{d_o}{2}\right)^2 \pi = 450\times10^6 \left(\frac{12.8\times10^{-3}}{2}\right)^2 \pi = 57'000N$

d) $345\ MPa > 250\ MPa$ (plastic) $\rightarrow \Delta l = \varepsilon l_o = (0.006)(250mm) = 1.5mm$

Strain Hardening

- An increase in σ_y due to plastic deformation.



- Curve fit to the stress-strain response:

$$\sigma_T = K(\varepsilon_T)^n$$

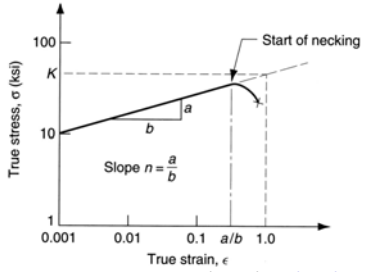
hardening exponent:
n= 0.15 (some steels)
to n= 0.5 (some copper alloys)

"true" stress (F/A)

"true" strain: $\ln(L/L_o)$

constant

True Stress-True Strain Curve



True stress-strain curve plotted on log-log scale

Material	n	K MPa	K psi
Low-carbon steel (annealed)	0.26	530	77,000
Alloy steel (Type 4340, annealed)	0.15	640	93,000
Stainless steel (Type 304, annealed)	0.45	1275	185,000
Aluminum (annealed)	0.20	180	26,000
Aluminum alloy (Type 2024, heat treated)	0.16	690	100,000
Copper (annealed)	0.54	315	46,000
Brass (70Cu–30Zn, annealed)	0.49	895	130,000

Hollomon's eq → $\sigma = K\,\varepsilon^n$

Because it is a straight line in a log-log plot

K: strength coefficient

n: strain-hardening exponent

- The higher the slope the stronger when material is strained

Safety Factors in Design

- Design uncertainties mean we do not push the limit.
- Factor of safety, N

$$\sigma_{working} = \frac{\sigma_y}{N}$$

Often N is between 1.2 and 4

safety factor

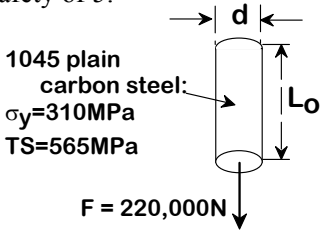
- Example:

Calculate a diameter, d, to ensure that yield does not occur in the 1045 carbon steel rod below. Use a factor of safety of 5.

$\sigma_{working} = \frac{\sigma_y}{N}$

$\frac{220{,}000N}{\pi(d^2/4)}$ ↑ 3

$d \approx 52mm$



Next time:

Toughness, Hardness…